INTERSTATE AVIATION COMMITTEE

AVIATION ACCIDENT INVESTIGATION COMMISSION

FINAL REPORT
BASED ON THE RESULTS OF THE INVESTIGATION OF THE AVIATION ACCIDENT

Type of aviation accident	Catastrophe
Aircraft type	An-24RV aircraft
State and registration identification marks	RA-47366
Owner	JSC "Sakhalin Airlines" Airways»
Operator	JSC Angara Airlines
Aviation Administration	East Siberian Regional Territorial Administration of the Federal Air Transport Agency
Scene of the incident	Russian Federation, Republic of Buryatia, Severobaikalsky area, Nizhneangarsk airfield, coordinates: 55°47ÿ40.4ÿ N, 109°35ÿ00.3ÿ E
Date and time	June 27, 2019, 10:25 a.m. local time (02:25 UTC), day

In accordance with the Standards and Recommended Practices of the International Civil Aviation Organization This report is issued for the sole purpose of preventing aviation accidents.

The investigation carried out within the framework of this report does not intend to establish the share of anyone guilt or responsibility.

The criminal aspects of this incident are presented in a separate criminal case.

List of abbreviations used in this report

2P	- second pilot
a/d	ÿ airfield
ADT	ÿ Automatic fuel pressure control
a/p	ÿ airport
AMGS	ÿ civil aviation meteorological station
AMC	ÿ Aviation Meteorological Center
ANI	ÿ Aeronautical information
ANP	ÿ aeronautical passport
JSC	ÿ joint-stock company
AP	ÿ aviation accident/autopilot (depending on context)
APO	ÿ Aviation Production Association
PRICE	- aircraft repair plant
ASK	ÿ emergency rescue team
ATB	ÿ aviation technical base
AUC	ÿ Aviation training center
BVPP	- concrete runway
B/M	ÿ flight engineer
no education	ÿ without restrictions
v. d.	ÿ east longitude
VV	ÿ propeller
WWAUL	ÿ Higher Military Aviation School of Pilots
VZL.	ÿ takeoff
VKK	ÿ Higher Qualification Commission
VLP	ÿ spring-summer period
VLEK	ÿ medical and flight expert commission
VNA	ÿ input guide vane
runway	ÿ runway
VPR	ÿ decision-making height
VS	ÿ aircraft/East Siberian (according to context)
Armed Forces of Ukraine	ÿ auxiliary power unit/East Siberian Management (by context)
Tue	ÿ air transport

Machine Translated by Google

G.	ÿ city (with names), year (with numbers)
GA	ÿ civil aviation
GAUZ	- state autonomous healthcare institution
State Healthcare Institution	- state budgetary healthcare institution
hot water supply	- civil aircraft
years	ÿ years
GosNII GA	ÿ State Research Institute civil aviation
GP	- state-owned enterprise
fuels and lubricants	ÿ fuels and lubricants
GUDP	- state unitary subsidiary enterprise
PEC	ÿ altitude correction sensor
DPK	ÿ volunteer fire brigade
EDDS	ÿ unified duty dispatch service
JSC	ÿ closed joint-stock company
ZMCB	Zaporizhzhya Machine-Building Design Bureau
Acting	ÿ Acting
IAS	ÿ aviation engineering service
IVPP	- artificial runway
PCM	ÿ torque meter
KBC	ÿ aircraft commander
KDP	ÿ command and control center
Code of Administrative Offenses	ÿ Code of Administrative Offenses
PDA	ÿ advanced training courses
CARP	ÿ Aviation Accident Investigation Commission
KTA	ÿ airfield control point
KTS	ÿ comprehensive aircraft simulator
CFL	ÿ blade feathering button
LO	- flight squad
m/u-iya	ÿ weather conditions
MAC	ÿ Interstate Aviation Committee
MV	ÿ local time
Ministry of Internal Affairs	ÿ Ministry of Internal Affairs

MVL	ÿ local air line
MSTU GA	Moscow State Technical University civil aviation
MK	ÿ magnetic rate
MKpos	ÿ landing magnetic course
MKU	ÿ municipal state institution
WE	ÿ municipal formation
MPU	ÿ magnetic course angle
MRR	ÿ time between repairs
MSRP	ÿ magnetic parameter recording system
MSCH	ÿ medical sanitary unit
MTU	ÿ interregional territorial administration
NBR GA	National Air Accident Investigation Bureau with Ukrainian civil aviation aircraft
BOOK.	ÿ nominal value
rate	- illegible
JSC	ÿ open joint-stock company/united air squadron (according to context)
UAE	- Joint Air Squadron
Department of Internal Affairs	ÿ air traffic services
OG	ÿ task force
OGPS	ÿ State Fire Service detachment
OGBUZ	- regional state budgetary institution healthcare
OIB	ÿ Flight Safety Inspection Department
OOO	ÿ limited liability company
OPLG	ÿ Department of Maintaining Airworthiness
OPRS	ÿ separate homing radio station
ARVD	ÿ air traffic management
Mr.	ÿ point
PVD	ÿ air pressure receiver
PVP	ÿ visual flight rules
PMG	ÿ flight low throttle

POS	ÿ anti-icing system
PPLS	ÿ flight crew training program
PPP	ÿ instrument flight rules
PPR	ÿ after the last renovation
PRT	ÿ temperature limit control and correction system rotation frequency
PSP BPSO	ÿ search and rescue unit of the Baikal search and rescue rescue squad
AKP	ÿ search and rescue operations
PSCH	ÿ fire and rescue unit
PC FPS in the Republic of Belarus	ÿ Fire department of the Federal Fire Service Republic of Buryatia
RV	ÿ elevator
RD	ÿ taxiway
RZGA	Rostov Civil Aviation Plant
RCC	ÿ regional qualification commission
RLE	ÿ flight manual
RPP	ÿ Flight Operations Manual
RU	ÿ regional administration
RUD 1	ÿ left engine control lever
RUD 2	ÿ right engine control lever
RUD	ÿ engine control lever
Russian Federation	ÿ Russian Federation
N	ÿ northern latitude
SAT	Sakhalin Air Routes
PAGE	ÿ mean aerodynamic chord
CIV	ÿ pulse and time counter
SC RF	ÿ Investigative Committee of the Russian Federation
cm.	- look
SME	- forensic medical examination
SPU	- aircraft intercom
MILK	ÿ Investigative Department for Transport
THAT	ÿ technical maintenance

TU ÿ technical conditions

Internal Affairs Directorate ÿ air traffic control

UGAN NOTB SFO - Department of State Aviation Supervision and Oversight
for transport security in the Siberian Federal District
district

UK ÿ Criminal Code

IF ÿ speed corrector amplifier

UPRT ÿ fuel lever position indicator

LONG ÿ temperature controller amplifier

TUE ÿ training detachment

Training Center ÿ training center

FAP-128 ÿ Federal Aviation Regulations "Preparation and execution
flights in civil aviation of the Russian Federation", approved by
order of the Ministry of Transport of Russia dated July 31, 2009, No. 128

FAP-147 ÿ Federal Aviation Regulations “Requirements for members
aircraft crews, technical specialists
aircraft maintenance and flight support personnel (flight dispatchers) of civil
aviation”,
approved by order of the Ministry of Transport of Russia dated September 12, 2008, No. 147

FAP-293 ÿ Federal Aviation Rules "Organization of Air Transport"
movements in the Russian Federation”, approved by order
Ministry of Transport of the Russian Federation dated November 25, 2011 No. 293

FAP ME ÿ FAP "Medical examination of a flight,
air traffic control personnel, flight attendants, cadets and candidates
entering civil aviation educational institutions
aviation”, approved by the order of the Ministry of Transport of Russia dated
22.04.2002 ÿ 50

FAS Federal Aviation Service

FAU - federal autonomous institution

Federal State Budgetary Institution - federal state budgetary institution

Federal State Budgetary Institution ÿ federal state budgetary educational
institution of higher education

Federal State Unitary Enterprise - federal state unitary enterprise

FSB Federal Security Service

FSNST Federal Service for Supervision of Transport

CRB	ÿ central district hospital
CUP	ÿ Mission Control Center
ETD	ÿ operational and technical documentation
Doc 4444	ÿ Rules of air navigation services. Organization Air Traffic Control. Sixteenth Edition, 2016
Doc 8168	ÿ Rules for Air Navigation Services. Production Aircraft operations. Volume I. Flight regulations. Sixth edition, 2018
IN	ÿ echelon flight
Nms	ÿ minimum descent altitude
QFE	ÿ atmospheric pressure at the airfield altitude or at the level runway threshold
QNH	ÿ atmospheric pressure, reduced to mean sea level standard atmosphere
UTC	ÿ Coordinated Universal Time
Vzp	ÿ landing approach speed

General information

June 27, 2019, at 10:25 local time (02:25 UTC) [1], during the day, when landing An An-24RV aircraft crashed at the Nizhneangarsk airfield in the Republic of Buryatia. RA-47366, operated by Angara Airlines JSC. There were 4 crew members on board the aircraft. crew and 43 passengers, including a child under 2 years old. As a result of the accident, 2 members of the crew died. crew, the aircraft burned in a ground fire.

Information about the AP was received by the Interstate Aviation Committee at 05:10 on June 27, 2019.

The investigation by the Presidential Administration was conducted by a commission appointed by the order of the Chairman. KRAP MAK dated 27.06.2019 No. 14/916-r.

In accordance with Appendix 13 of the Convention on International Civil aviation notification about the accident was sent to Ukraine – the state developer and aircraft and engine manufacturer.

The investigation involved specialists from the National Bureau of Civil Aviation (Ukraine) and the Antonov State Enterprise. and the State Enterprise Ivchenko-Progress, as well as specialists from the Federal Autonomous Institution Aviation Register of the Russian Federation.

The investigation began on June 27, 2019.

The investigation was completed on April 13, 2022.

The preliminary investigation was conducted by the Bratsk Special Investigative Department of the Supreme Court of the Russian Federation.

[1] Further in the text, UTC time is indicated; local time corresponds to UTC + 8 hours.

1. Factual information

1.1 Flight history

06/25/2019 for the performance of 06/26/2019 regular commercial passenger flights The following flights were planned for transportation: AGU-99 on the route Irkutsk – Nizhneangarsk; AGU-199 on the route Nizhneangarsk – Ulan-Ude; AGU-200 on the route Ulan-Ude – Nizhneangarsk; AGU-100 on the route Nizhneangarsk – Irkutsk (Fig. 1), – and is assigned the crew of JSC Angara Airlines (hereinafter referred to as the airline) consisting of: the captain, the pilot-inspector, 2P, flight mechanic and flight attendant. In flight assignment #28215, signed the commander of LO No. 1 of the airline, recorded the completion of task 1 of section 2 of the An-24 PPLS, An-26 and item 6.3 FAP-147.

Note: *1. PPLS An-24, An-26 1.2.1:*

Section 2. Preparing the Second Pilot for Commissioning as commander of the armed forces (PIC under supervision).

Task 1. Flight training of the second pilot in production conditions."

2. FAP-147:

"6.3. In addition to the requirements set out in paragraphs 6.1 and 6.2 of these Rules, holder of an air transport pilot's license with a rating with a note about the type of aircraft "airplane":

a) must have at least 1,500 hours of flight time as an aircraft pilot, which is counted as no more than 100 hours of flight time on a simulator. The specified raid includes:

500 hours as a pilot-in-command under supervision; or 250 hours as an aircraft commander; or at least 70 hours as aircraft commander and at least 180 hours as an aircraft commander under supervision"[2] .

In fact, the flights were planned to evaluate the performance of 2P before being put into operation in as the PIC. The performance of the 2P was to be assessed by the airline's inspector pilot, included in the crew.

[2] Here and further, unless otherwise stated, in the responses of organizations, explanatory and other cited Documents highlighted in italics retain the author's version.

Machine Translated by Google

Note: *From the explanatory note of the commander of flight No. 1 of the airline dated July 2, 2019:*

> *"In order to evaluate the performance of the second pilot before entering into service as the aircraft commander was included in the crew of the OIBP pilot-inspector airline (Surname and initials), having a large instructor experience and experience in performing solo flights on An-24 and An-26 aircraft."*

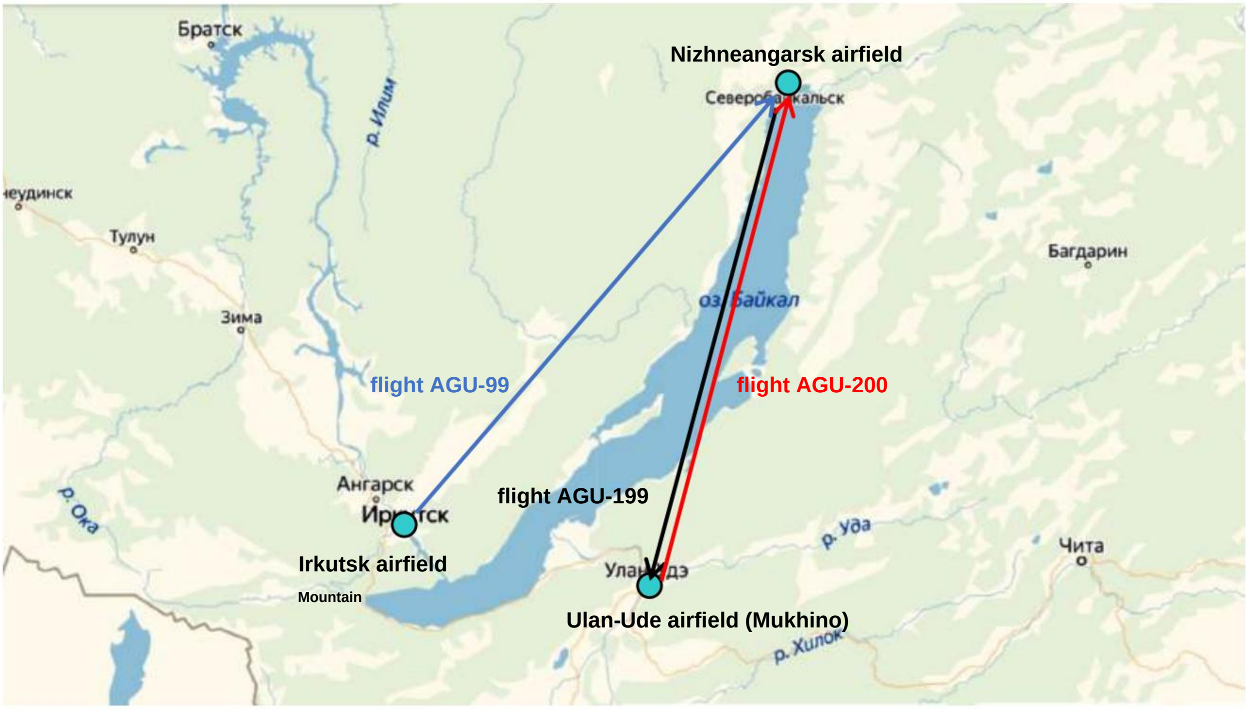


Fig. 1. Flight routes of aircraft on June 26, 2019 and June 27, 2019

On June 26, 2016, at 11:20 p.m., the crew underwent pre-flight medical checkup. medical center of JSC Irkutsk International Airport and began pre-flight preparation.

After passing medical control, approximately 50 minutes before departure, The captain felt a deterioration in his health and reported this to the commander of the 1st wing, who replaced the PIC with an inspector pilot.

Note: *From the explanatory note of the commander of flight No. 1 of the airline dated July 2, 2019:*

> *"…at 07:30 MV I received a telephone message from the captain (last name and initials) about his poor health.*
>
> *In a situation where there is a time crunch (45 minutes before departure), having firm confidence in extensive experience of solo flights (Surname and initials of the pilot-inspector), I have given permission for the replacement Commander of the Aircraft (Surname and initials) as a pilot inspector for*

execution of the planned flight with the recording of his name on

on the title page of the flight assignment in the column "Aircraft Commander 3".

According to the airline's RPP, since the pilot inspector was included in the composition crew and together with the crew underwent training to carry out the task, the commander LO was assigned to him the duties of the captain.

Note: *paragraph 4.1.10. of Chapter 4 of Part "A" of the airline's RPP:*

"Can be used as an additional crew member inspector. An inspector – a pilot by profession – can perform the duties of both the captain and the second pilot.

At 00:20 the crew consisted of: pilot-inspector (hereinafter referred to as PIC), 2P, flight mechanic and flight attendant – took off from Irkutsk airfield. Takeoff was performed by 2P. Flight to Nizhneangarsk airfield with passengers on board passed without incident, landing at at the airfield, he completed 2P at 01:45.

After refueling the aircraft and boarding passengers, at 02:40, 2P took off from the airfield Nizhneangarsk. The flight to Ulan-Ude airfield (Mukhino) with passengers on board was completed. without any comments, the 2P landed at the airfield at 04:00.

Note: *From explanations 2P from 03.07.2019:*

"…Takeoff at Irkutsk airport was carried out from the right, control and radio communication was carried out on the left, I occupied the right seat at all stages of the flight 26 and June 27, 2019. Takeoff and flight on the Irkutsk – Nizhneangarsk route proceeded without any deviations. Landing at Nizhneangarsk airport was carried out on the right, control and radio communication on the left.

…The takeoff and flight on the route Nizhneangarsk – Ulan-Ude took place without incident. deviations. The landing approach to Ulan-Ude airport was carried out from the right, control and maintaining radio communication on the left. Landing without deviations."

According to the schedule, the departure of flight AGU-200 on the route Ulan-Ude – Nizhneangarsk was scheduled for 04:50. Due to weather conditions at the Nizhneangarsk airfield, which were worse than the airfield minimum, the flight departure was initially postponed every hour, and then, In agreement with the airline's Mission Control Center, it was rescheduled for 00:30 on June 27, 2019.

Note: *Actual weather at Nizhneangarsk airfield on June 26, 2019:*

- from 04:00 to 04:58: visibility 10 km, light rain showers, cloudiness significant with a lower boundary height of 420–480 m, significant

[3] In the copy of the flight assignment from June 26, 2019, submitted by law enforcement agencies of the city of Ulan-Ude to the commission, the surname of the pilot-inspector is missing in the "Aircraft Commander" column, and the surname, first name, and patronymic of the captain are crossed out in pencil.

cumulonimbus with a lower boundary of 1500–1620 m, mountains covered by clouds;

- from 04:58 to 05:26: visibility 4000 – 5000 m, light rain showers, haze, significant cloudiness with a lower boundary height of 210–300 m, significant cumulonimbus with a lower boundary height of 990 - 1500 m, the mountains are covered by clouds;

- from 05:26 to 06:00: visibility 2000 m, haze, vertical visibility 80 – 90 m, mountains covered by clouds;

- from 06:00 to 06:55: visibility 2000 - 3000 m, light rain showers, haze, vertical visibility 90 m, mountains covered by clouds;

- from 06:55 to 07:46: visibility 2000 m, moderate rain showers, haze, vertical visibility 90 m, mountains covered by clouds;

- from 07:46 to 10:30: visibility 3000 m, haze, solid cloud cover 120 – 180 m, mountains covered by clouds.

According to the Nizhneangarsk airfield ANP, the minimum for landing approach according to the IPP it is 690x5000 m.

The crew rested at the Polet Hotel, located in close proximity from the airport.

Note: *1. From explanations 2P from 03.07.2019:*

"The departure of flight AGU-200 was delayed due to deteriorating weather conditions at Nizhneangarsk airport. Every hour the captain and I checked the weather conditions Ulan-Ude Airport meteorological service. The weather was below minimum, the crew I was on the plane because we were waiting for the weather conditions to improve... At 18:00 local time, information was received about the flight being postponed. AGU-200 on June 27, 2019, as there was no improvement in the weather. The crew "I went to the Ulan-Ude airport hotel to rest."

2. From the explanation of the head of the airline's Mission Control Center on July 1, 2019:

"On June 26, the flight was unable to depart as scheduled (at 04:50 UTC) weather conditions of Nizhneangarsk airport, until the end of the period weather conditions will not improved. The decision was made to change the departure time to 00:30 UTC. June 27th."

On June 26, 2019, from 11:24 PM to 11:30 PM, the crew underwent a medical examination at the health center. Baikal Airport LLC has begun pre-flight preparations.

At 23:36 in the premises of the Ulan-Ude AMGS the crew got acquainted with the meteorological information and received a package with cartographic information along the flight route.

At 01:03:00 on June 27, 2019, the PIC took off from the Ulan-Ude airfield (Mukhino). The flight was carried out at flight level 170 (5200 m) with the autopilot engaged.

At 01:53:44, the 2P established contact with the air traffic control tower of the Nizhneangarsk airfield. (hereinafter referred to as the control tower dispatcher), reported the estimated time of arrival and requested weather conditions.

At 01:53:52 the control tower dispatcher transmitted the weather conditions at the airfield to the crew and determined the approach through the NRS with further visual maneuvering to runway 05. 2P confirmed receipt of the information and transmitted the aircraft loading and the required information to the control tower dispatcher refueling.

At 02:07:57 the crew contacted the control tower dispatcher, reported the flight conditions and requested an approach on heading 225° via the NRS.

At 02:08:19 the control tower dispatcher confirmed receipt of the information and transmitted the instruction crew proceed at FL170, descend as calculated, approach the landing at FL170 FL100 (3050 m).

At 02:08:51 the control tower controller cleared the approach to landing on runway 23 (MKpos = 225°) and transmitted meteorological conditions at the airfield.

At 02:09:25 the crew began descending to FL100.

At 02:11:35 during descent the left engine shut down spontaneously feathering of the explosives. The crew reported the engine's spontaneous shutdown to the control tower dispatcher.

In response to the control tower dispatcher's request for a landing decision, the crew responded that they would sit down.

At 02:13 the control tower controller announced the “Alarm” signal at the airfield.

At 02:14:35 the crew reported reaching FL100.

At 02:14:45 the control tower controller cleared the approach using the NPR and descent to the QFE altitude. 2100 m.

At 02:16:06 the crew reported setting the pressure and reaching a QFE altitude of 2100 m.

At 02:16:16 the control tower controller confirmed the flight of the NRS and allowed the descent to height of QFE 1500 m.

At 02:18:29 the control tower controller cleared a left turn to the landing zone and a descent to altitude QFE 640 m.

At 02:22:18 the crew reported reaching an altitude of 640 m and visual visibility of the runway.

At 02:22:23 the control tower controller cleared the crew to approach runway 23 using visual maneuvering.

At 02:23:49 the crew reported readiness for landing.

At 02:23:52 the control tower controller cleared the crew to land.

At 02:24:35 the crew reported landing.

During the landing, the aircraft began to deviate to the right, left the runway, and ran along the ground, broke through the airfield fence and collided with the wastewater treatment plant building.

A fire broke out on the aircraft after it stopped. The flight attendant evacuated passengers through luggage door. 2P, according to him, was unable to evacuate the PIC and flight engineer, who were trapped in the burning cockpit of the aircraft and were unconscious. 2P left the plane. through the left front emergency hatch.

1.2. Bodily injury

Bodily injury	Crew	Passengers	Other persons
Fatal	2	0	0
Serious	1	9	0
Minor/absent	1/0	23/11	2[4]/0

1.3. Aircraft damage

The aircraft burned in a ground fire.

1.4. Other damages

At the site of the accident, the following were damaged: a building, technological equipment of the treatment plant structures, KAMAZ type vehicle, GAZ type vehicle, diesel generator installation and fencing of treatment facilities and the airfield.

1.5. Information on personnel

Inspector pilot

Job title	Pilot-inspector of the An-24, An-26 aircraft (order of the AO) Angara Airlines dated May 14, 2005, No. 174/l)
Floor	Male
Age	58 years old
Education	Omsk Flight Technical School of Civil Aviation, 1982 specialty - flight operation of aircraft, awarded the qualification of civil pilot aviation, diploma GT-I No. 211522, issued on June 29, 1982. In 1991, retraining in Kirovograd Higher Flight School of Civil Aviation *"according to the program pilot of the An-24 and An-26 aircraft,*

[4] Workers who were in the wastewater treatment plant building.

	certificate issued on July 25, 1991, Registration number 4254. In 2012, training at the Federal State Unitary Enterprise “VS UTC” on program *"Initial flight training instructor staff"*
Certificate	Airline Transport Pilot Certificate II P No. 000473, issued by the RSC VS RU FAS Russia on 11.11.1996, validity period actions - indefinitely. Qualification marks: *"The An-24 multi-engine land-based aircraft, "Co-pilot"* from 10.10.1991; *"The An-26 multi-engine land-based aircraft, "Co-pilot"* from 03/22/2006; *Multi-engine land-based aircraft An-24 "KVC-trainee"* from 10.06.2008; *"Multi-engine land-based aircraft An-24. Aircraft commander, Captain An-24"* from 26.09.2008; *"The An-24 multi-engine land-based aircraft, An-26-100. Aircraft commander, Captain An-24, An-26-100»* ÿÿ 24.09.2009
Medical report	VLEK MCh JSC "International Airport" Irkutsk, medical certificate of the 1st class VT No. 074602 dated 05/22/2019: *“Recognized as fit for work as an air line pilot”,* is valid until 22.05.2020
Minimum weather conditions for the An-24 aircraft, An-26	Landing: lower cloud limit – 50 m, visibility – 700 m; takeoff: visibility – 300 m (order of the director of JSC Angara Airlines dated 03.10.2018 No. 1034)
Raid on 06/27/2019:	
- general	15167 h 15 min (An-2, An-24, An-26)
– on An-24, An-26	10667 hours 51 minutes
– on the An-24 as a KAB	3468 hours 15 minutes
– on the An-26 as a KAB	2092 hours 25 minutes

– for June in total - for June as a captain – for June on the An-24RV – for June on the An-26 - over the last three days - on the day of the incident	28 hours 10 minutes 28 hours 10 minutes 23:05 05:05 04:40 01:22
Working hours (before the AP)5	02:55
Exercise machine	05/20/2019, KTS An-24 (An-26) of the center for training of aviation specialists Irkutsk branch of MSTU GA under the leadership simulator instructor-examiner. Conclusion: *Airline Pilot Qualifications corresponds"*
Date of last equipment inspection piloting	November 6, 2018, as a pilot-instructor-examiner JSC Angara Airlines. Conclusion: *“Maybe perform flights on the An-24 aircraft as a pilot instructors in weather conditions: takeoff – no x 300 m; landing – 50 x 700 m. Civil Air Transport Pilot Qualifications corresponds"*
Date of last check aircraft navigation	05.04.2019, senior navigator of LO No. 1, instructor-examiner of JSC "Airline" "Angara". Overall rating *"Five"* (conclusion (the inspector is not listed in the flight log)
Admission to instructor work6	12.04.2012, order No. 120/k of the Armed Forces of the Ministry of Transport of the Russian Federation: *"To be allowed to fly as a pilot- An-24, An-26, An-26-100 aircraft instructors"*
Permission to fly without a navigator	Order 7 of the commander of the 2nd Irkutsk LO joint aviation detachment of the Ukrainian Civil Aviation 31.10.1991
Clearance for flights in unattached areas crew composition	Order of JSC Angara Airlines from 15.06.2010 ÿ 774

5 For all aircraft crew members, working hours are calculated from the time of medical examination.

6 The approval is recorded in the Special Remarks section of the Airline Transport Pilot's Certificate.

7 There is no order in the flight file; the airline pilot's certificate and the flight log contain an entry about permission to fly without a navigator; the order number is not indicated.

Preparation for VLP	Approved for flights in the spring-summer period 2019, order of the commander of LO No. 1 JSC Angara Airlines dated April 23, 2019 ÿ 0411
PDA	03.12.2018 – 07.12.2018, ATC FGBOUVO "St. St. Petersburg State University of Civil Aviation, certificate issued on 07.12.2018, registration number 47627
Preliminary preparation	05/20/2019, under the leadership of the commander of LO No. 1 JSC Angara Airlines
Pre-flight preparation	At Ulan-Ude airport before departure
Pre-flight rest	In a hotel setting, at least 12 hours
Pre-flight medical examination by a medical worker at the health center	LLC "Airport Baikal" (Ulan-Ude)
Incidents and AP in the past	No

After graduating from the Omsk Civil Aviation Flight School in 1982, he worked in Kirensk joint air detachment in the position of 2P Armed Forces An-2. In October 1990, he was appointed to position of commander of the An-2 aircraft.

In April 1991 he was transferred to the Irkutsk United Aviation Detachment of the Ukrainian Civil Aviation Armed Forces position 2P VS An-24 second flight detachment.

In April 1992, he was transferred to JSC Baikal Airlines to the position of 2P VS An-24. In June 2002, dismissed due to staff reductions.

From June 2002 to May 2005 he worked in organizations not related to the aviation industry activities.

In May 2005, he was hired by JSC Irkutsk Aircraft Repair Plant No. 403 position of 2P aircraft An-24.

In September 2005, he was transferred to JSC Angara Airlines to the position of 2P VS An-24. In September 2008, he was appointed to the position of Captain of the An-24 Armed Forces. In January 2013, he was transferred to the Flight Safety Inspectorate for the position of pilot-inspector for An-24 and An-26 aircraft.

2P

Job title	Second pilot of An-24, An-26 (order of the director of JSC Angara Airlines dated July 4, 2018, No. 155/l)
Floor	Male
Age	43 years old

Education	Balashov Higher Military Aviation School of Pilots, 1998, awarded qualification – pilot engineer by specialty *"Command Tactical Aviation, Operation air transport",* diploma BVS 0826635 issued on June 11, 1998. In 2013-2014, training at the Federal State Unitary Enterprise "VS UTC" in program *"Retraining of pilots for aircraft An-24, 26, 26-100,* certificate No. 92-14
Certificate	Commercial Pilot License No. 0026700, issued by the Armed Forces of the Ministry of Transport of the Russian Federation on January 26, 2017, validity period actions - indefinitely, qualification marks: *"airplane AN24 Co-pilot, An-2/ AN2, An-26/ AN26 Co-pilot; Instrument Flight Rules (IFR) flights airplane (instrument airplane)»*
Medical report	VLEK MCh JSC "International Airport" Irkutsk, medical certificate of the 1st class VT No. 075330 dated 02/19/2019: *"Recognized as fit for work as an air line pilot",* is valid until 19.02.2020
Raid on 06/27/2019:	
- general	6012 h 24 min (L-410, An-2, An-24, An-26)
– on the An-24	1325 hours 45 minutes
– on the An-26	538 hours 40 minutes
– for June	27 hours 55 minutes
- over the last three days	04:40
- on the day of the incident	01:22
Working hours (before the AP)	02:57
Exercise machine	11.04.2019, KTS An-24 (An-26) of the center for training of aviation specialists Irkutsk branch of MSTU GA under the leadership simulator instructor-examiner. Conclusion Commander of LO No. 1: *"Cleared for flights"*
Date of last equipment inspection piloting	November 21, 2018, as a pilot-instructor-examiner

	JSC Angara Airlines. Conclusion: *“Maybe to perform flights on the An-24 aircraft as second pilot with the right to take off and land. Commercial Civil Aviation Pilot Qualifications corresponds"*
Date of last check aircraft navigation	20.02.2019, senior navigator of LO No. 1, instructor-examiner of JSC "Airline" "Angara". Overall rating *"Five".* (Conclusion (the inspector is not listed in the flight log)
Permission to fly without a navigator	Report of the acting commander of the LO No. 1 AO Angara Airlines, dated 05.04.2016, No. 0122
Clearance for flights in unattached areas crew composition	Report of the acting commander of the LO No. 1 AO Angara Airlines, July 21, 2017, No. 0721
Preparation for VLP	Approved for flights in the spring-summer period 2019, order of the commander of LO No. 1 JSC Angara Airlines dated April 23, 2019 ÿ 0411
PDA	03.12.2018 – 07.12.2018, ATC FGBOUVO "St. St. Petersburg State University of Civil Aviation, certificate issued on 07.12.2018, registration number 47629
Preliminary preparation	03/22/2019 under the leadership of the deputy commander of LO No. 1 of JSC Angara Airlines
Pre-flight preparation	In the airport of Ulan-Ude under the leadership of KVS
Pre-flight rest	In a hotel setting, at least 12 hours
Pre-flight medical examination by a medical	worker at the health center LLC "Airport Baikal" (Ulan-Ude)
Incidents and AP in the past	No

After graduating from the Balashov Higher Military Pilot School 2P, he retired from the Russian Armed Forces.

In October 1999, he was hired by Samara-Avia LLC as a 2P An-2 aircraft pilot.

In May 2012, he resigned of his own free will and was hired JSC Angara Airlines for the position of 2P An-2 aircraft. In July 2015, he was transferred to position of Captain of the An-2 Aircraft. In October 2016, after training and completing the program retraining, transferred to the position of 2P aircraft An-24, An-26.

In March 2018, he resigned of his own free will and was hired in April 2018. work at JSC Baikal Airlines as a 2P An-24 aircraft pilot.

In July 2018, he resigned and was hired by JSC Angara Airlines position 2P of the An-24, An-26 aircraft.

Flight engineer

Job title	Flight mechanic An-24, An-26 (order of the director of JSC Angara Airlines dated 03.06.2013 No. 130/l)
Floor	Male
Age	57 years old
Education	Yegoryevsk Aviation Technical School, 1982, *“assigned the profession – aviation mechanic 2nd class",* certificate No. 316, issued 18.01.1982. Moscow Institute of Civil Engineers aviation, 1989, specialty - *"operation aircraft and engines”,* awarded qualification - *"mechanical engineer",* diploma TV No. 161103, issued on June 16, 1989, registration ÿ 3-6738. In 1989, retraining in Kirovograd Higher Flight School of Civil Aviation *"according to the program flight mechanic of An-24 and An-26 aircraft,* certificate issued on 27.12.1989, registration number 1965
Certificate	Flight engineer (flight mechanic) certificate No. 0026451, issued by the Armed Forces of the Ministry of Transport of the Russian Federation on January 25, 2018, validity period - indefinite, qualification marks: *Airplane AN24, AN26*
Medical report	VLEK MCh JSC "International Airport" Irkutsk, medical certificate of the 1st class VT No. 075181 dated 12/19/2018: *“Recognized as fit for work as a flight mechanic”,* really until 19.12.2019

Raid on 06/27/2019: - general – on the An-24 – on the An-26 – for June - over the last three days - on the day of the incident	 12853 h 19 min (An-24, An-26, Tu-154, Il-76) 7149 h 00 min 1616 hours 45 minutes 30 hours 10 minutes 04:40 01:22
Working hours (before the AP) 02 h 59 min	
Date of last check practical work	06/18/2019, flight mechanic-instructor-examiner of JSC Angara Airlines. Conclusion: *"Can perform flights as Flight mechanic. Civil aviation flight mechanic qualifications corresponds"*
Clearance to fly in unassigned crew composition	Order of the Director of JSC Angara Airlines dated 12.01.2011 ÿ 3
Preparation for VLP	Approved for flights in the spring-summer period 2019, order of the commander of LO No. 1 JSC Angara Airlines dated April 23, 2019, No. 0411
PDA	04.02.2019 – 08.02.2019, ATC FGBOUVO "St. St. Petersburg State University of Civil Aviation, certificate issued on 08.02.2019, No. 49094
Preliminary preparation	04/17/2019 under the leadership of the deputy commander of LO No. 1 of JSC Angara Airlines
Pre-flight preparation	In the airport of Ulan-Ude under the leadership of KVS
Pre-flight rest	In a hotel setting, at least 12 hours
Pre-flight medical inspection	Medical worker at the health center LLC "Airport Baikal" (Ulan-Ude)
Incidents and AP in the past	No

After graduating from the Yegoryevsk Aviation School of Civil Aviation, the flight mechanic worked in Nadym United Aviation Detachment of the Tyumen Civil Aviation Directorate in the position of 3rd-category aircraft technician. In July 1982, he was transferred to the position of aircraft technician. 3 classes.

In February 1984, he was transferred by the Irkutsk JSC to the ATB as a technician. In October In 1989 he was transferred to the position of flight mechanic for the An-24 aircraft in the second flight detachment.

In April 1992, he was transferred to JSC Baikal Airlines to the position of flight mechanic of the An-24 aircraft. In April 1996, he was transferred to the position of flight engineer of the aircraft. Il-76.

In May 2000, he was dismissed and hired by Tesis Airlines CJSC. (Moscow) for the position of flight engineer for the Il-76 aircraft.

In May 2002, he was transferred to the Irkutsk branch of Tesis Airlines CJSC. position of flight engineer for the Il-76 aircraft. In January 2003, he resigned of his own accord. desire.

In February 2003, he was hired by Siberia Airlines OJSC for the position of Flight engineer of the Tu-154 aircraft in flight detachment No. 2 (Irkutsk).

In August 2004, he resigned of his own free will and was hired at Irkutsk branch of JSC "Tesis Airlines" for the position of flight engineer 2nd class Tu-154 aircraft.

From July 2005 to April 2006 he worked in organizations not related to aviation activities.

In April 2006, he was hired by Siberia Airlines OJSC for the position of flight engineer of the Tu-154 aircraft in the UAE (Omsk). In September 2006, he was transferred to the position flight engineer of the Tu-154 aircraft of Flight Detachment No. 3 (Moscow). In October 2008, he was dismissed for reduction of the organization's staff.

In October 2008, he was hired by Angara Airlines CJSC for the position of flight mechanics VS An-24, An-26-100.

In March 2011, he resigned of his own free will and was hired at UTair-Express LLC for the position of flight mechanic for the A-24 aircraft in the aviation unit An-24 (Irkutsk). In December 2011, he resigned voluntarily.

In February 2012, he was hired by Volga-Dnepr Airlines LLC. (Ulyanovsk) for the position of flight engineer of the Il-76 aircraft. In May 2012, he resigned due to at one's own request.

In June 2013, he was hired by Angara Airlines OJSC for the position of flight mechanics VS An-24, An-26.

Flight attendant

Job title	Flight attendant An-24, An-26 (director's order) JSC Angara Airlines dated September 30, 2005 No. 32/l)
Floor	Female
Age	40 years old

Education	2003, training at the State Enterprise Training Center of the Armed Forces of the Russian Federation (Irkutsk) under the program *"Preparation flight attendants to perform and ensure flights on domestic airlines on An-24 aircraft",* certificate No. 61-01
Certificate	Flight attendant certificate No. 0026481, issued Armed Forces of the Federal Air Transport Agency 03.03.2018, validity period – indefinitely, qualification marks: *«Instructor»*
Medical report	VLEK MCh JSC "International Airport" Irkutsk, medical report RA No. 207272 from 03/16/2016: *"Recognized as fit for work flight attendant",* valid until March 16, 2021
Raid on 06/27/2019: - general – on the An-24 – for June - on the day of the incident	 10373 h 27 min (An-24, An-26, An-148, Tu-154) 5457 h 05 min 76 hours 35 minutes 01:22
Working hours	03:01
Preparation for VLP	Approved for flights in the spring-summer period 2019, order of the head of the service flight attendants of JSC Angara Airlines from 19.04.2019 No. 01/04/p
PDA	04.02.2019 in the ATC of the Federal State Budgetary Educational Institution of Higher Education "St. St. Petersburg State University of Civil Aviation (Irkutsk) under the program: *"Annual emergency rescue training for cabin crew members An-24 aircraft crew",* certificate dated 04.02.2019, registration number 17260
Date of last check practical work	October 15, 2018, flight attendant instructor JSC Angara Airlines. Conclusion: *"Flight Attendant of the Russian Civil Aviation Authority" qualifications compliant. Can continue to perform flights as a flight attendant on the An-24 aircraft"*

Pre-flight rest	In a hotel setting, at least 12 hours
Pre-flight medical examination by a medical worker at the health center	LLC "Airport Baikal" (Ulan-Ude)
Incidents and AP in the past	No

In January 2003, the flight attendant was hired by JSC Irkutsk Aircraft Repair Plant No. 403" for the position of flight attendant for the An-24 aircraft.

In September 2005, she was transferred to work at Angara Airlines CJSC. position of flight attendant An-24, An-26.

The commission believes that the level of professional training of the crew members corresponded to the flight mission.

Dispatcher KDP MVL

Job title	The dispatcher who carries out direct air traffic control vessels of the 2nd class (order of the director of the State Department of the Ulan- (Udeaeronavigation of 03.03.2003 No. 69/k)
Floor	Male
Age	61 years old
Education	Vyborg Aviation Technical School GA, 1978, diploma No. 055140 issued 06/15/1978, specialty *"technical operation of aircraft and aircraft engines",* awarded the qualification of *"mechanical technician".* UTO-15 of the East Siberian Civil Aviation Directorate, completed courses from 01.10.1981 to 30.05.1982 retraining under the "Air Traffic Controller" program, Air Traffic Controller's Certificate issued on May 30, 1982
Certificate	Air Traffic Controller Certificate motion SD No. 005438, issued on May 22, 2003 by the FAS Russia, valid until September 25, 2020
Class	Air Traffic Controller, Class 1 (Order of the Supreme Control Commission of the Federal Air Transport Agency) dated 31.05.2011 No. 5)
Admission to work as a dispatcher	Dispatcher of the combined control tower of the international air route, Nizhneangarsk (order of the director of the State Unitary Enterprise "Ulan- (Udeaeronavigation of May 22, 2003, No. 170/k)

Medical report	VLEK MSC of the Air Navigation branch Central Siberia" Federal State Unitary Enterprise "State Corporation for OrVD" (Krasnoyarsk), medical report Class III, series BT No. 054569 dated 09.11.2018: *"Recognized as fit to work as an air traffic controller"* valid until 09.11.2019
PDA	09.15.2016, Siberian branch of the non-profit additional educational institution professional education "Institute air navigation" (Krasnoyarsk), program *"Air traffic management (for air traffic controllers),* certificate No. 014838, cattle registration number No. 3718.14838
Date of last check	02.03.2019 senior structural dispatcher units of Nizhneangarsk Ulan-Ude ATS center of the Eastern Air Navigation branch Siberia" Federal State Unitary Enterprise "State Air Traffic Management Corporation", *"Overall rating - 5"* (conclusion after checking with (Not included in the dispatcher's book)
Pre-shift rest	At home, at least 12 hours
Pre-shift medical examination	According to the dispatcher, self-monitoring
Incidents and AP in the past	No

It should be noted that, in accordance with paragraph 1.11 of Appendix 14 of the Federal Aviation Regulations of the Ministry of Civil Aviation, *“When on duty shifts of limited number (up to 12 people) in remote areas pre-shift medical examination is not carried out in the main base areas. List such shifts are determined by the head of the regional air department transport of the Ministry of Transport of Russia based on the report of the chief specialist of the department for Aviation Medicine. Decision on admission to work in air traffic control The flight director (senior dispatcher) accepts* the order . Interregional Directorate of the Federal Air Transport Agency for ATM and AKPS of the Siberian Federal The district approved the list of divisions of the Eastern Siberia Air Navigation branch Federal State Unitary Enterprise “State Air Traffic Management Corporation” (outgoing No. 2-21/753 dated 08/19/2011), in which the pre-shift It is permitted not to conduct a medical examination of air traffic controllers, and the decision on admission to work on air traffic control to accept the flight director (senior dispatcher). This list includes the structural subdivision of the airfield air traffic control Nizhneangarsk.

The Commission believes that the level of professional training of the air traffic controller The MVL met the established requirements.

1.6. Aircraft details

Fig. 2. Aircraft An-24RV RA-47366 before the accident

1.6.1. Airframe

Type of aircraft	An-24RV aircraft
Serial (factory) number	77310804
Manufacturer, production date	Kiev APO (Ukraine), September 16, 1977
State and registration identification marks	RA-47366
Certificate of registration	No. 3440, issued by the Inspectorate Directorate Flight Safety FSNST 03.04.2006
Owner of the aircraft	JSC "SAT Airlines"
Operator	JSC Angara Airlines
Certificate of Airworthiness	No. 20222180029, issued by the Armed Forces of the Ministry of Transport of the Russian Federation May 23, 2018, valid until July 28, 2019 (within the MRR 15,060 hours, 8,660 flights)
Designated resource, service life	40,000 hours/25,000 landings/41 years 11 months before 08/16/2019 (decision of the Federal State Unitary Enterprise GosNII GA dated

	08.05.2018 No. R1.24.3-18/18, approved (Rosaviatsia, May 21, 2018)
Operating time since the start of operation 26.06.2019	38,014 hours/18,584 landings
Remaining assigned resource/term services as of June 26, 2019	1986 hours/6416 landings/51 days
Number of repairs	5, last repair 04/28/2000 in Irkutsk APPLICATION No. 403
Service life between repairs	15,060 hours/8,660 landings/19 years 3 months before July 28, 2019, with maximum takeoff weight 22.5 tons (decision of the Federal State Unitary Enterprise GosNII GA from 08.05.2018 No. R1.24.3-18/18, approved (Rosaviatsia, May 21, 2018)
Maintenance schedule as of June 26, 2019	14,237 hours 40 minutes/8,383 landings
Remaining service life between repairs/period services as of June 26, 2019	822 h 20 min/277 landings/31 days
Amount of fuel on board in last flight	3000 kg
Last periodic maintenance	12/26/2018, according to form F-28 + preparation for flights, performed by IAS JSC "Angara Airlines", work order card from 26.12.2018 ÿ 375
Operational maintenance	June 27, 2019, completed according to form A-1+OV IAS JSC "Angara Airlines", map-order from June 27, 2019 No. 240/U

1.6.2. Engine data

Engine type	AI-24 2 series	
Power plant number	1 (left)	2 (right)
Engine serial number	H47622024	H4722058
Manufacturer	Zaporizhzhya Motor-Building production association (Ukraine)	
Release date	17.05.1976	30.06.1967
Assigned resource	25000 hours	22000 hours

Operating time since the start of operation as of June 26, 2019	21213 h 14 min	9671 hours
The remainder of the assigned resource as of June 26, 2019	3786 hours 46 minutes	12329 h
Number of repairs	8	4
The company that produced the last one repair	JSC "RZGA No. 412" (Rostov-on-Don)	
Date of last repair	29.07.2018	28.04.2016
Date of installation on the aircraft	19.11.2018	16.12.2016
Time between repairs	4000 hours	3000 hours
Service life between repairs	7 years	7 years
Operating time since last repair as of June 26, 2019, hours/cycles	307 h 34 min/190	1863 h 52 min/not indicated
Remaining service life between repairs as of June 26, 2019	3692 hours 26 minutes	1136 h 08 min
Remaining service life between repairs as of June 26, 2019	6 years 3 days until 29.07.2025	3 years 2 days until 28.04.2023
Last periodic maintenance	12/26/2018, according to form F-28 + preparation for flights, performed by IAS JSC "Airline" "Angara", work order card dated December 26, 2018, No. 375	

1.6.3. Data on the Armed Forces of Ukraine

Engine type	RU19A-300
Serial number	R615400AR5-1
Manufacturer	Tyumen Motor Plant
Release date	24.02.1986
Assigned resource	4288 hours 52 minutes
Operating time since the start of operation 26.06.2019	1628 h 22 min/9574 cycles
The remainder of the assigned resource on 26.06.2019	2660 h 30 min
Number of repairs	1
Date and place of last repair	August 9, 2001, Tyumen Motor Plant
Date of installation on the aircraft after last renovation	08.10.2010

Time between repairs	1500 h/10000 cycles
Operating time since last repair 26.06.2019	1417 h 14 min/6618 cycles
Remaining service life between repairs/period services as of June 26, 2019	82 h 46 min/3382 cycles
Last periodic maintenance	12/26/2018, according to form F-28 + preparation for flights, performed by IAS JSC "Angara Airlines", work order card from 26.12.2018 ÿ 375

1.6.4. Propeller data

Type BB	AV-72	
Place	left	right
Serial number	S14L2768	S13L2255
Release date	31.12.1981	23.07.1981
Assigned resource	14000 hours	
Operating time since the start of operation 26.06.2019	13456 hours	13403 h
The remainder of the assigned resource	544 hours	597 h
Number of repairs	3	4
Date of last repair	29.09.2017	30.10.2014
Date of installation on the aircraft after last renovation	19.11.2018	11.09.2017
Interrepair life/interrepair life service life	2000 h/6 years	
Operating time since last repair 26.06.2019	1158 h	1956 h
Remaining service life between repairs as of June 26, 2019	842 hours	44 hours
Remaining service life between repairs as of June 26, 2019	4 years 3 months	1 year 4 months

The aircraft was operated:

ÿ from the beginning of operation until November 1991 - in the Khabarovsk JSC Far Eastern Civil Aviation Management;

ÿ from 28.11.1991 to 11.11.1994 – at the “Ministry of Education and Science “Start”[8] (Yuzhno-Sakhalinsk);

ÿ from 11.11.1994 to 01.03.2013 – at Sakhalin Air Routes Airlines OJSC (Yuzhno-Sakhalinsk);

ÿ in accordance with the agreement dated 01.03.2013 No. A-22/02-22.5/2013-0081 “For rent An-24RV aircraft" the aircraft was transferred to Kontaero LLC;

ÿ in accordance with the agreement dated 01.05.2013 No. A/11-13/01 for the sublease of air The An-24RV aircraft was transferred to Angara Airlines;

ÿ in accordance with the agreement of 26.03.2018 No. A/3-18/03 for the sublease of air The An-24RV aircraft was transferred to JSC Baikal Airlines;

ÿ in accordance with the agreement of 14.05.2018 “On early termination According to the An-24RV aircraft sublease agreement No. A/3-18/03 dated March 28, 2018, the aircraft was transferred to JSC Angara Airlines.

The aircraft maintenance was carried out by the IAS JSC "Airline" "Angara" in accordance with the current maintenance regulations for An-24, An-26 aircraft, publications 1991, Moscow.

JSC Angara Airlines has a certificate of organization for maintenance No. 285-17-085, issued by Rosaviatsia on June 22, 2017, for the right to perform aircraft maintenance. Certificate indefinite.

After completing the F-28 maintenance + flight preparation on December 26, 2018, the aircraft carried out flights until 20.04.2019.

From 20.04.2019 to 22.06.2019, research work was carried out on the aircraft technical condition of the aircraft for continued operation. The work was carried out program No. 1.24.1.3-19/29, approved by the chief designer of the Federal State Unitary Enterprise GosNII GA 29.03.2019.

After the work on investigating the technical condition of the aircraft, it performed 2 flights, The third flight was an emergency.

Having analyzed the technical documentation presented, the commission came to the following conclusion: conclusion that the technical operation of the aircraft was carried out in accordance with established requirements, with the exception of the aircraft's left engine (see conclusion of the State Enterprise “ZMKB Ivchenko-Progress” in section 2.2 of this report).

1.7. Meteorological information

On June 27, 2019, weather conditions in the Nizhneangarsk airfield area were determined as follows:

[8] In the entry in the Supreme Court form, the commission was unable to decipher the abbreviation “MSPTA”.

at the earth's surface - a low-gradient baric field of reduced pressure;

at altitudes – the influence of the northern periphery of the cyclone with the center over the city of Ulaanbaatar, with direction of the leading flow 110° and speed of 30 km/h, cold trough, barotropic atmosphere.

Weather forecast for Nizhneangarsk airfield from 00:00 to 06:00 on June 27, 2019:

Surface wind 200°–3 m/s, gusts 8 m/s; visibility 10 km; cloud cover: significant cumulonimbus with a baseline of 1200 m above the ground; from 00:00 to 04:00: significant cloudiness with a lower boundary height of 270 m, continuous cumulonimbus with the height of the lower boundary is 1200 m from the earth's surface.

Actual weather conditions at Nizhneangarsk airfield on June 27, 2019:

1. Measurement by the “Alarm” signal at 02:19 (MKpos 225°):

ÿ ground wind 050°– 3 m/s;

ÿ visibility 10 km;

ÿ cloudiness: several (1-2 oktas) cumulonimbus with a lower altitude boundaries 1500 m from the earth's surface;

ÿ air temperature + 20 °C;

ÿ dew point temperature + 12 °C;

ÿ atmospheric pressure: QFE 713 mmHg/951 hPa; QNH 1006 hPa.

2. Control measurement of wind parameters at the dispatcher’s request at 02:22: wind at land: 060°– 4 m/s.

3. Weather conditions at 02:30 (local regular weather report):

ÿ ground wind: 080°– 2 m/s;

ÿ visibility: 10 km;

ÿ cloudiness: several (1-2 oktas) cumulonimbus with a lower altitude boundaries 1500 m from the earth's surface;

ÿ air temperature: + 20 °ÿ.

ÿ dew point temperature: + 12 °ÿ.

ÿ pressure: QFE 713 mmHg/951 hPa; QNH 1006 hPa.

1.8. Navigation, landing and air traffic control aids

During the approach and landing of aircraft at the airfield, the following were in operation: radio-technical means of navigation, landing and air traffic control:

ÿ separate automatic homing radio station PAR-10S (APRM 250);

ÿ automatic radio direction finder RDF-734;

- airfield surveillance radar (DRL-7SM);

ÿ radar information display equipment (APM "Corinth");

ÿ radio stations "PHASAN-19P5" No. 1 and No. 2;

ÿ satellite communication earth station;

ÿ Aviton dispatcher conversation registration system;

- runway occupancy warning equipment.

1.9. Communication means

During the approach and landing, the aircraft crew maintained radio communication with the air traffic control tower. International air route of Nizhneangarsk airfield (call sign "Nizhneangarsk tower") on a frequency of 132.0 MHz. Radio communication during the flight was stable.

1.10. Airfield details

Nizhneangarsk airfield is located in the Republic of Buryatia, 3.5 km north-east of the village of Nizhneangarsk, 477 km north of the city of Ulan-Ude, refers to airfields class "D" and is a civil airfield.

Certificate of state registration of the aerodrome No. 9/45 issued on October 20, 2016 Armed Forces of the Federal Air Transport Agency.

Information about the airfield is published in ANI Collection No. 122.

The airfield's ANP was registered on 06.04.2018 No. SP1-594 SV MTU Rosaviatsiya.

KTA coordinates: 55º48ÿ07ÿ N, 109º45ÿ45ÿ E, mountain airfield, elevation KTA above sea level +473.9 m, magnetic declination – minus 7.2°. By the airfield operator is OJSC “Airports of local air lines of Buryatia” (Ulan-Ude).

The surface of the airfield is flat, the soil is loamy with stones inclusions, subject to soaking during heavy precipitation. The runway size 1800 x 150 m.

The airfield has one runway: a runway measuring 1503 x 32 m; surface: airfield slabs PAG-14 (6 x 2 m). The runway slopes downwards from the southwest to the north-east, no transverse slopes. Runway threshold heights: Runway 23 – H = 469 m, Runway 05 – H = 474 m. At threshold 05 there is a 150 m long stopping strip.

Magnetic take-off track angles: MPU = 045° (RWY 05) and MPU = 225° (RWY 23).

There are no alternate runways at the airfield.

There is no lighting equipment at the airfield. The airfield is used in Daylight hours. Permitted flights: IFR/VFR.

The level of required fire protection (hereinafter referred to as RLFP), according to the Collection ANI No. 122 and ANP of the airfield – 3rd category.

Note: *Order of the Ministry of Transport of the Russian Federation dated March 28, 1991 No. 65 "On approval of the management for search and rescue support of flights Civil Aviation of the USSR" (SPASOP GA-91), Appendix 13: "Determination of the required fire protection level (RFP) airfield."*

The category of the artificial landing strip according to the UTPZ is determined according to Table 1 depending on dimensions of the largest (by fuselage length) aircraft (AC), using the artificial landing strip.

Table 1

The length of the fuselage of the largest aircraft, m	Maximum width of the fuselage of the largest aircraft, not more than m 2	Category of IVPP by UTPZ
from 0 to 9		1
from 9 to 12	2	2
from 12 to 18	3	3
from 18 to 24	4	4
from 24 to 28	4	5
from 28 to 39	5	6
from 39 to 49	5	7
from 49 to 61	7	8
from 61 to 76	7	9

The category of the artificial runway according to the UTPZ may be downgraded by one level relative to the value determined by the length and maximum width fuselage, if at the airfield, the number of movements is greatest for the aircraft's runway is less than 700.

The number of movements is determined for the three most intense flights of the months of the year. One movement is considered to be a takeoff or landing VS.

The number of fire engines (FA) in combat readiness, fire extinguishing compositions located at these PA, and the total the productivity of supplying the compositions, ensuring the required level fire protection for artificial runways must be no less than those given in Table 2.

Table 2

Category of UTPZ IVPP	Quantity PA, pcs.	Quantity fire extinguishing compositions, kg	In that number foaming caller, kg	Total production feed rate, kg/sec
1	1	800	55	6
2	1	1700	120	14
3	1	2600	180	20
4	2	8000	500	64
5	2	12000	840	80
6	3	15200	1060	100
7	3	24000	1680	133
8	4	32500	2160	180
9	5	41000	2870	226

The total number of PA, fire extinguishing agents and the total the productivity of their supply at the airfield must ensure the UTPZ for each runway.

3. Deployment time at any point of each runway of the first firefighter vehicles (from the quantity established in Table 2) should not exceed 3 minutes, and subsequent ones - 4 minutes from the moment of announcement fire and rescue teams to give an alarm signal before the start supply of fire extinguishing agent.

According to the UTPZ, the An-24 aircraft has category 4 and is the largest aircraft, approved for operation at this airfield. Moreover, the number of its takeoffs and landings for the three busiest months of the year do not exceed 700, i.e. category 3 according to the UTPZ for both runways at the Nizhneangarsk airfield, it was established in accordance with current regulatory documents.

The following requirements are defined for category 3 UTPZ:

ÿ number of fire trucks – 1;

ÿ quantity of fire extinguishing agents – 2600 kg, including

foaming agent – 180 kg;

ÿ total feed capacity – 20 kg/s.

In fact, there was one KAMAZ 43118 fire truck at the airfield. AA-8.0, state registration plate C606KS03RUS. The vehicle has: a 7500 liter water tank and a 500 liter foam tank, water capacity in nominal mode is 60 l/s, used foaming agent – PO-RZA (6%). Fire truck, its fire tanks and the performance of fire pumps meets the requirements airfields with UTPZ 3 category.

On the day of the AP, the AIP of Russia provides detailed information, including air navigation there were no diagrams for the Nizhneangarsk airfield (only four-letter code and coordinates of the CTA). This report provides diagrams from AUV of the airfield and aeronautical collection prepared by SZ RCAI LLC, which the crew used during the preparation and execution of the flight.

Aerodrome map showing aerodrome minima and available distances shown in Fig. 3.

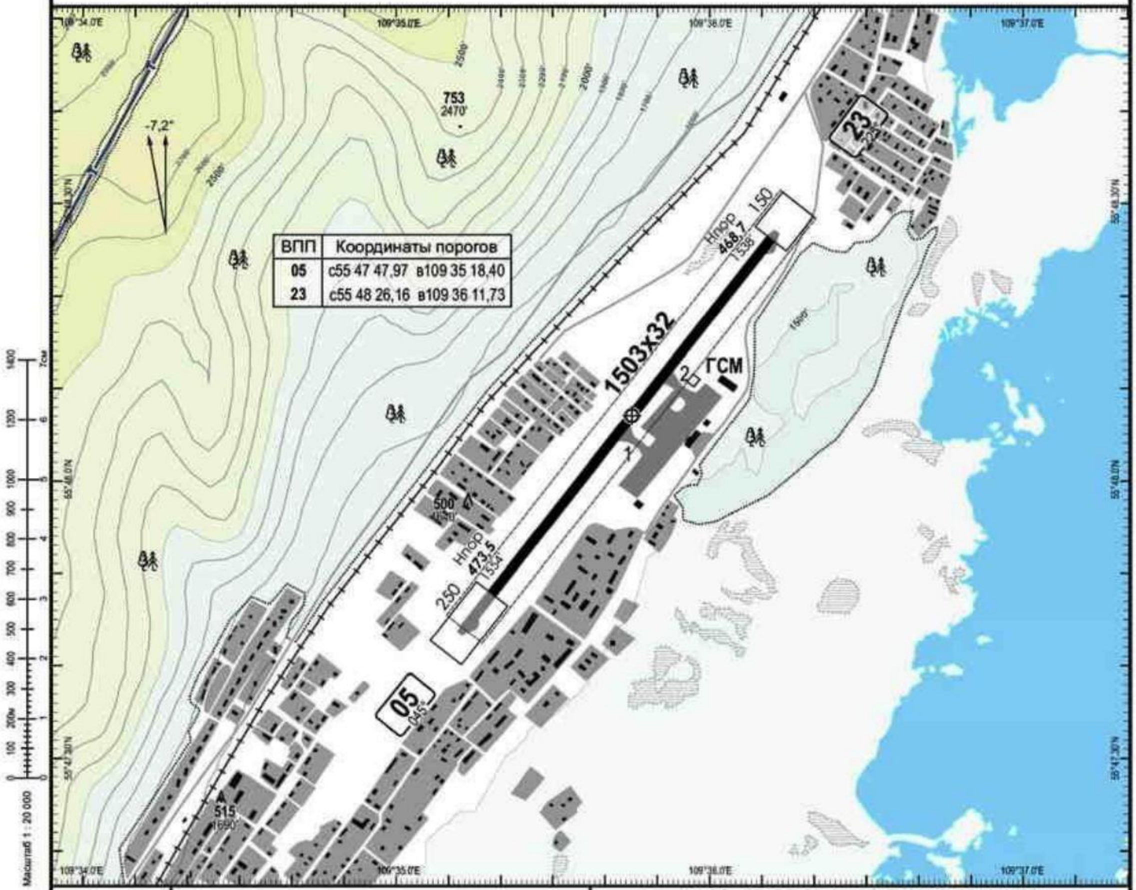
Приложение 2.1

КАРТА АЭРОДРОМА

НИЖНЕАНГАРСК, РОССИЯ
НИЖНЕАНГАРСК

НИЖНЕАНГАРСК Вышка 132.000

КТА: с55 48 07,1 в109 35 45,1 На/д 473,5м / 1554'



ВПП	СВЕТООБОРУДОВАНИЕ	РАСПОЛАГАЕМЫЕ ДИСТАНЦИИ РДР	РДВ	РДПВ	РДВ
05	отсутствует	1503	1653	1503	1503
23	отсутствует	1503	1753	1503	1503

Размеры летной полосы (ЛП) 1803x150.

Кат. ВС	МИНИМУМЫ ДЛЯ ВЗЛЕТА (ППП) ВПП05 без огней оси ВПП день	ВПП05 ночь	ВПП23 без огней оси ВПП день	ВПП23 ночь	МИНИМАЛЬНЫЕ МЕТЕОУСЛОВИЯ (ПВП) ДЕНЬ Ннго	ДЕНЬ Lвид	НОЧЬ Ннго	НОЧЬ Lвид
Верт.	*4120'* (780) x5000	-	*2410'* (260) x1900	-	(600) *3550'*	2000	-	-
A					-	-		
B								
C								
D	-		-					

Аэронавигационный паспорт аэродрома НИЖНЕАНГАРСК 16.04.18г.

Fig. 3. Airfield map from the Nizhneangarsk airfield ANP

It should be noted that in the airfield map, in the column "Available distances", no landing distance available (LDA), but twice (the second time in error) The available takeoff distance (ARTD) is indicated, which is equal to the RPD.

The diagram of standard routes of arrival at the airfield from the AUV is shown in Fig. 4, and from the collection of OOO SZ RCAI - in Fig. 5.

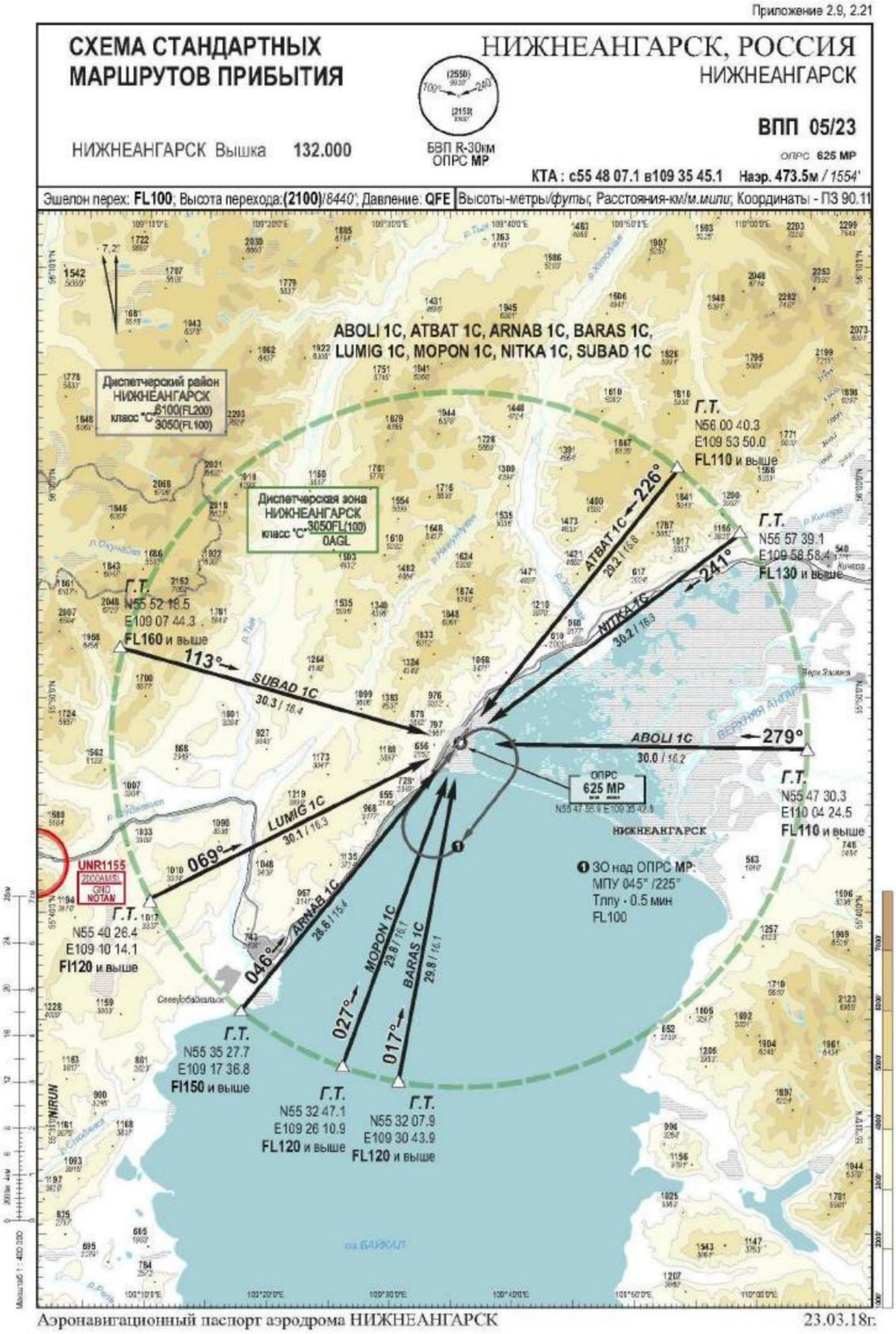


Fig. 4. Scheme of standard arrival routes from the AUV

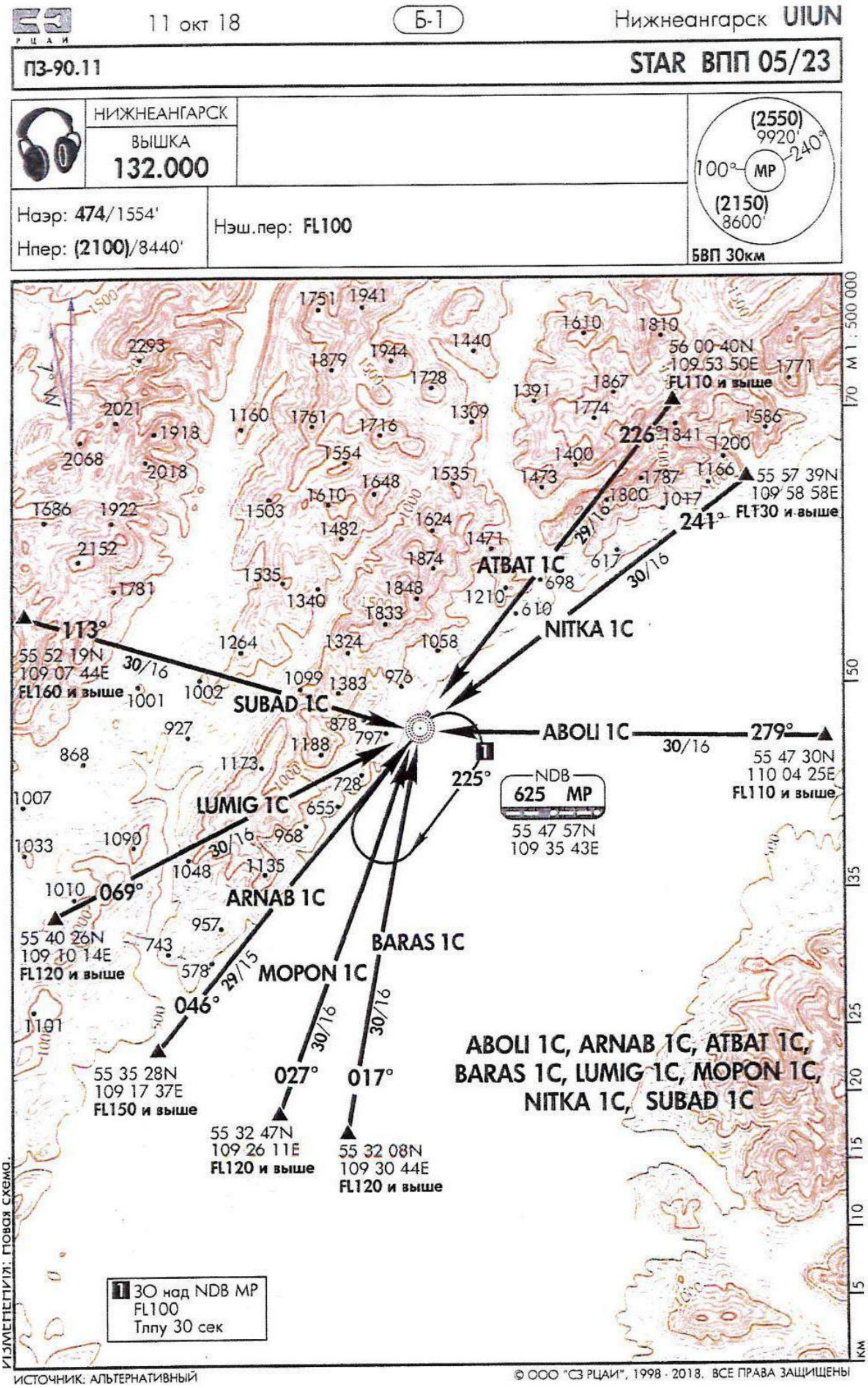


Fig. 5. Scheme of standard arrival routes from the collection of LLC "SZ RCAI"

According to the diagrams, arrival from Ulan-Ude to the Nizhneangarsk airfield is carried out along the standard route MOPON 1C with a descent from the 27° MPU to the NRS airfield. The altitude or flight level when entering the approach is not indicated on the diagram, however After reaching the OPRS, you must make a right turn to enter the waiting area above it (holding area parameters: FL 100, MPU 225°/045°) and descend to an altitude 2100 m by QFE pressure. The specified procedure does not depend on the selected landing course crew (45° or 225°, Fig. 6-9).

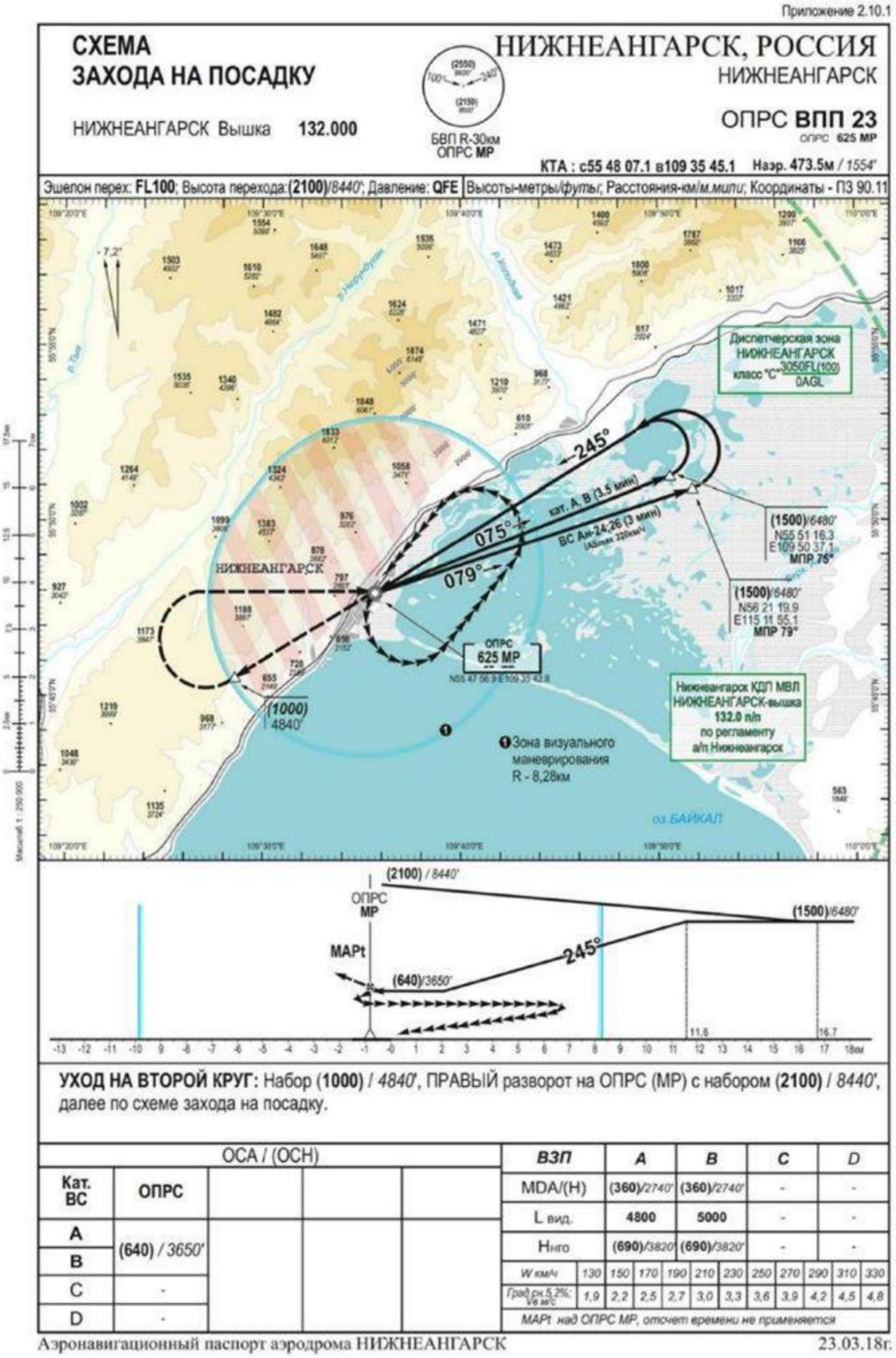


Fig. 6. Approach diagram with a heading of 225° from an AUV

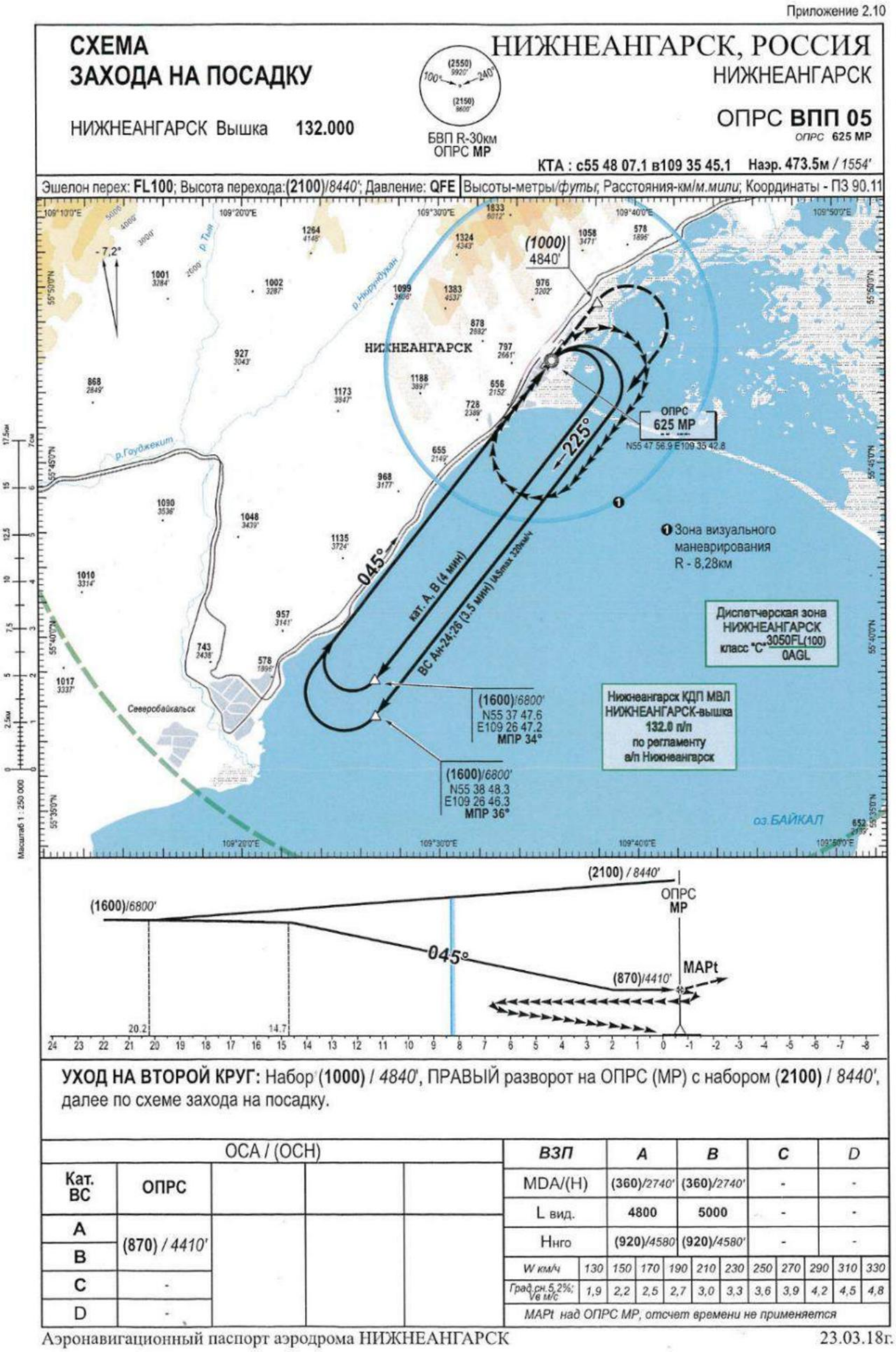
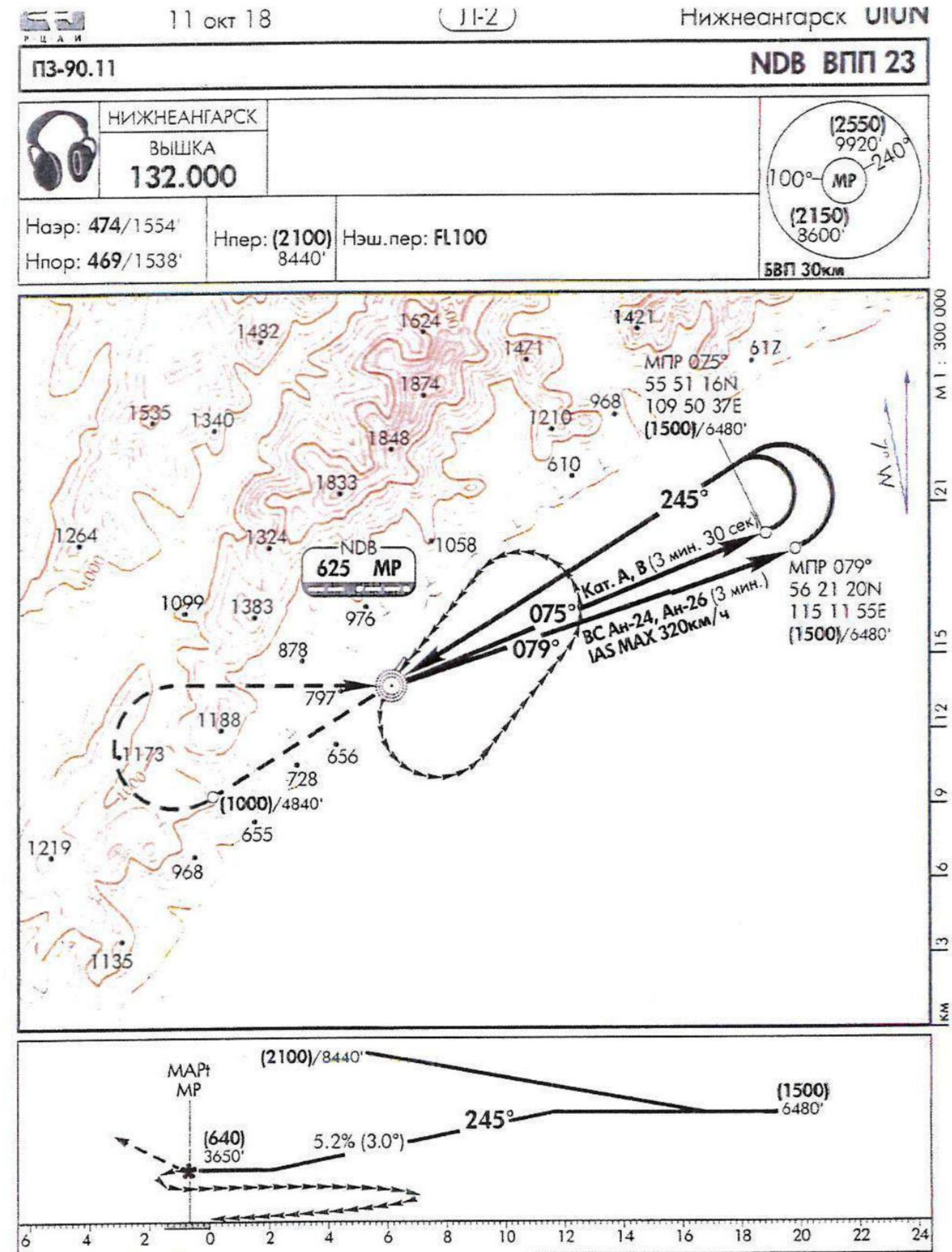


Fig. 7. Approach diagram with a heading of 45° from the AUV

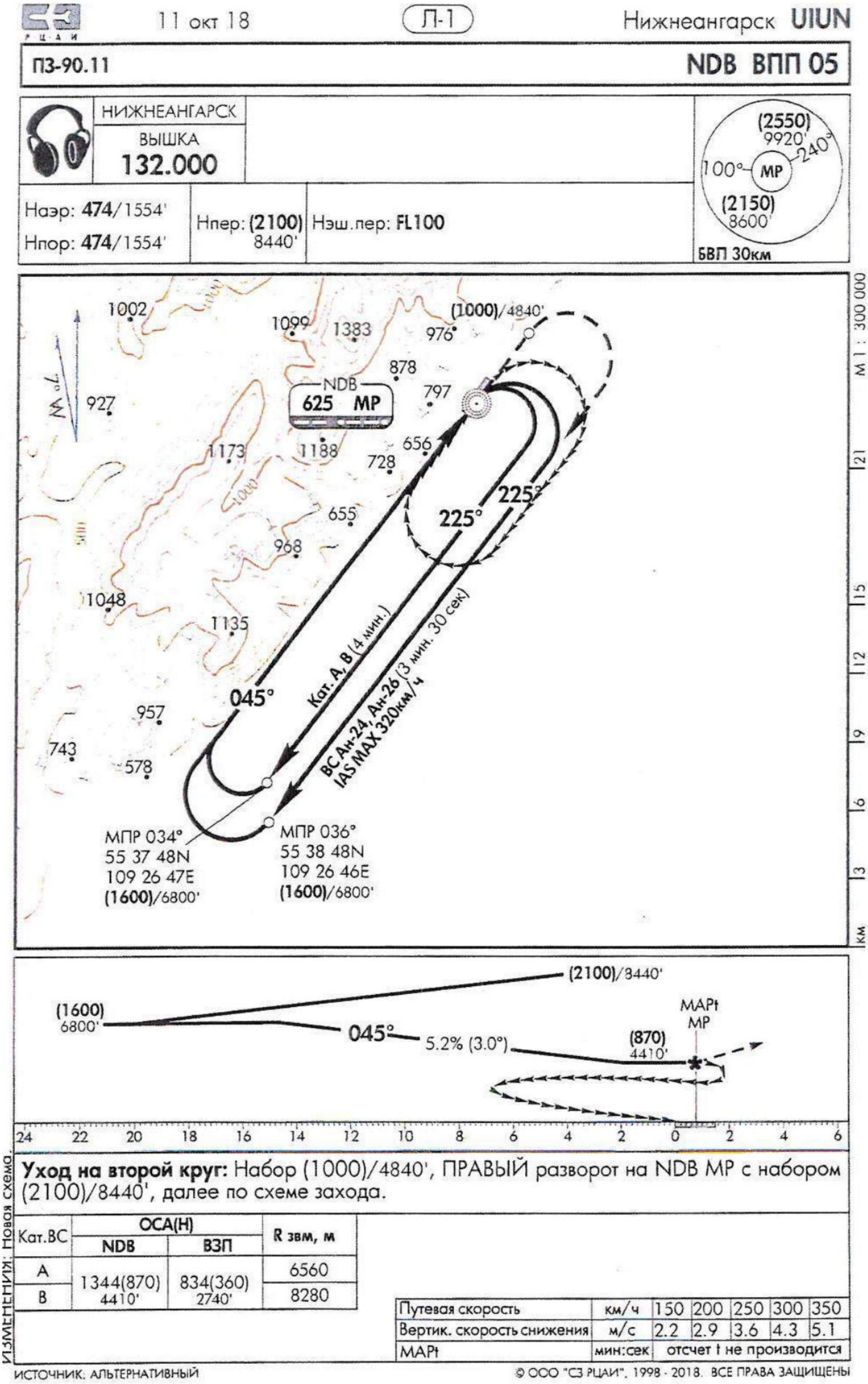


Кат.ВС	OCA(H) NDB	OCA(H) ВЗП	R звм, м
–	1114(640) 3650'	834(360) 2740'	6560
			8280

Путевая скорость	км/ч	150	200	250	300	350
Вертик. скорость снижения	м/с	2.2	2.9	3.6	4.3	5.1
МАРt	мин:сек	отсчет t не производится				

ИСТОЧНИК: АЛЬТЕРНАТИВНЫЙ

Fig. 8. Approach diagram with a course of 225° from the collection of SZ RCAI LLC



Кат.ВС	OCA(H) NDB	OCA(H) ВЗП	R звм, м
A	1344(870) 4410'	834(360) 2740'	6560
B			8280

Путевая скорость	км/ч	150	200	250	300	350
Вертик. скорость снижения	м/с	2.2	2.9	3.6	4.3	5.1
MAPt	мин:сек	отсчет t не производится				

Fig. 9. Approach diagram with a course of 45° from the collection of SZ RCAI LLC

Note: *The QFE pressure altitude values are given in brackets on the diagrams.*

To carry out landing with a course of 225°, it is necessary, after completing the zone waiting with a descent to the QFE altitude of 2100 m, perform a turn to the MPU 79° and on the indicated airspeed is not more than 320 km/h and follow this course for 3 minutes descend to a QFE altitude of 1500 m, then perform a left paired (merged) turn at MPU 245°, maintaining an altitude of 1500 m. After turning at MPU 245°, proceed to descend to the NRS and take an altitude of 640 m. The NRS flight is carried out at an altitude of 640 m, then perform visual maneuvering in the designated landing approach area.

To carry out landing on a course of 45°, it is necessary after flying over the NRS at an altitude of 2100 m, make a right turn to a course of 225°, at a speed of no more than 320 km/h follow for 3.5 minutes descending to QFE altitude 1600 m, then execute right paired turn to course 045°, maintaining altitude 1600 m. After turning to course 045° begin descent to the NRS to an altitude of 870 m. After flying over the NRS at the specified at the designated altitude, perform visual maneuvering in the designated landing approach area.

Part C of the airline's RPP provides the following operating minima: for Nizhneangarsk airfield:

ВПП	Тип ВС	Минимумы для захода на посадку				
		H п.р. x L вид. (m)		H мс (m)	H нго (m)	Lвид. (m)
		ОПРС (РЛК)	ОПРС (без РЛК)	Визуальный заход		
05	Ан-24 Ан-26	870/5000	-	360	2100	5000
23	Ан-24 Ан-26	640/5000	-	360	2100	5000

1.11. Flight recorders

The aircraft was equipped with flight data recorders. information K3-63 and MSRP-12-96, sound recorder MS-61. Removed from aircraft recorders, with the exception of the K3-63 recorder, which was burned in a ground fire and it was not possible to read the information from it, they were transferred for research to the MAK laboratory.

The recordings of the accident flight were available on both recorders, the results The transcripts were used to establish the circumstances and causes of the accident.

Machine Translated by Google

1.12. Information on the condition of aircraft components and their location at the scene of the incident

The AP area is a mountainous area with elevations of up to 1800 m the northern shore of Lake Baikal, covered with coniferous and deciduous trees. Airfield It is located in the foothills at a distance of 2.5 km from the shoreline of Lake Baikal.

The accident site is located 400 m southwest of the threshold of runway 05 (Fig. 10). terrain at the AP location is 472 m.

Fig. 10. AP region in a photo from space

The aircraft landed with attitude control (MC) = 225° (RWY 23). The aircraft touched down along the axis The first touchdown occurred with the left main landing gear at a distance of 530 m. from the entrance end. After approximately 130 m on the runway there are traces of the beginning of tire destruction right main landing gear (Fig. 11). At a distance of approximately 1430 m from the runway entrance The aircraft went off to the right onto the ground (Fig. 12) and continued moving along the ground. On the tracks of movement The aircraft detected the outer wheel tire of the right main landing gear on the ground (Fig. 13), and also a fragment of the inner wheel (Fig. 14). 470 m after the aircraft left the runway broke through the airfield fence (Fig. 15) (at the same time, the folding right main landing gear brace), "jumped" across the road, knocked down a wooden a pole on a concrete base collided with the left engine with the sewage treatment plant building structures, overturning a GAZ-type vehicle, and stopped. Upon collision with the left side of the crew cabin was deformed by the sewage treatment plant building and The fuel tanks were destroyed and a fire started. The fire destroyed the aircraft, with the exception of the tail section of the fuselage and part of the right wing console, which separated during collision with a wooden pole on a concrete base and a diesel generator installation.

Machine Translated by Google

Fig. 11. Traces of the beginning of destruction of the tires of the right main landing gear

Machine Translated by Google

Fig. 12. Tire marks on the right main landing gear when leaving the runway and landing on the ground

Fig. 13. Outer tire of the right main landing gear

Fig. 14. Fragments of the inner and outer tires of the right main landing gear

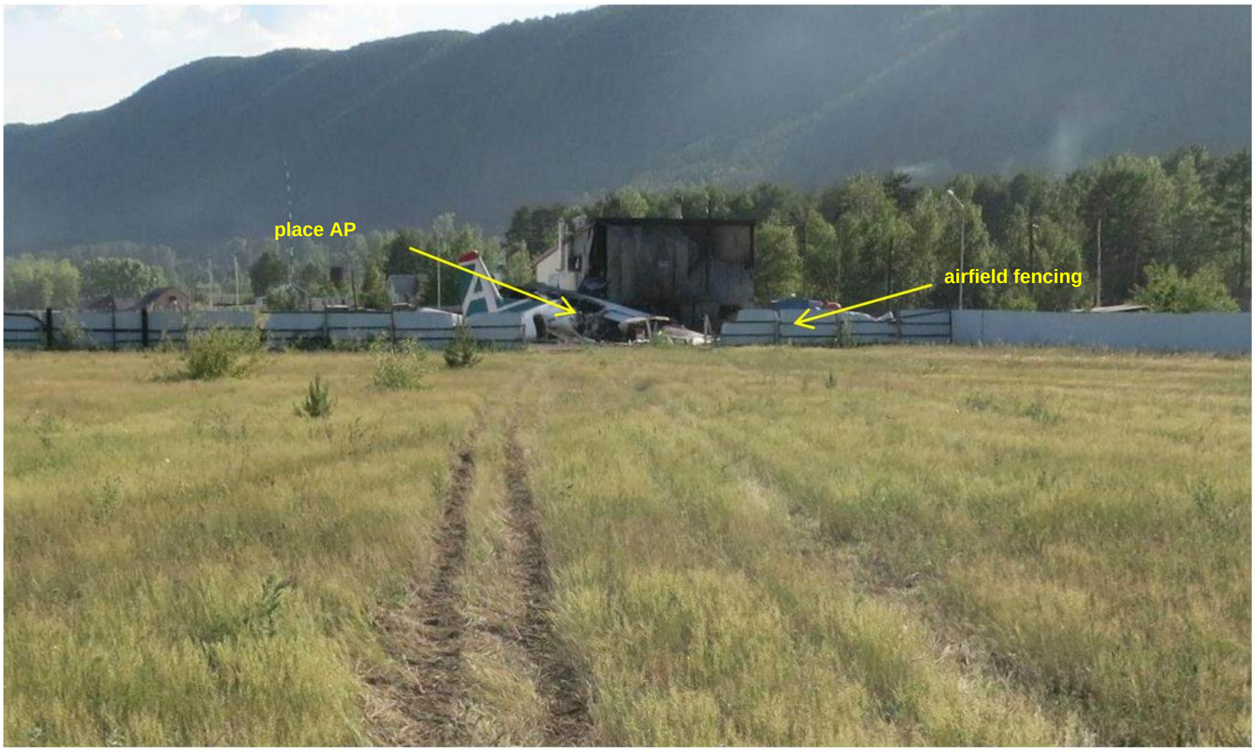


Fig. 15. Destroyed airfield fence

At the site of the AP, the aircraft is located with MK = 276° (Fig. 16). General view of the aircraft and its parts after AP is shown in Fig. 16 – Fig. 23.

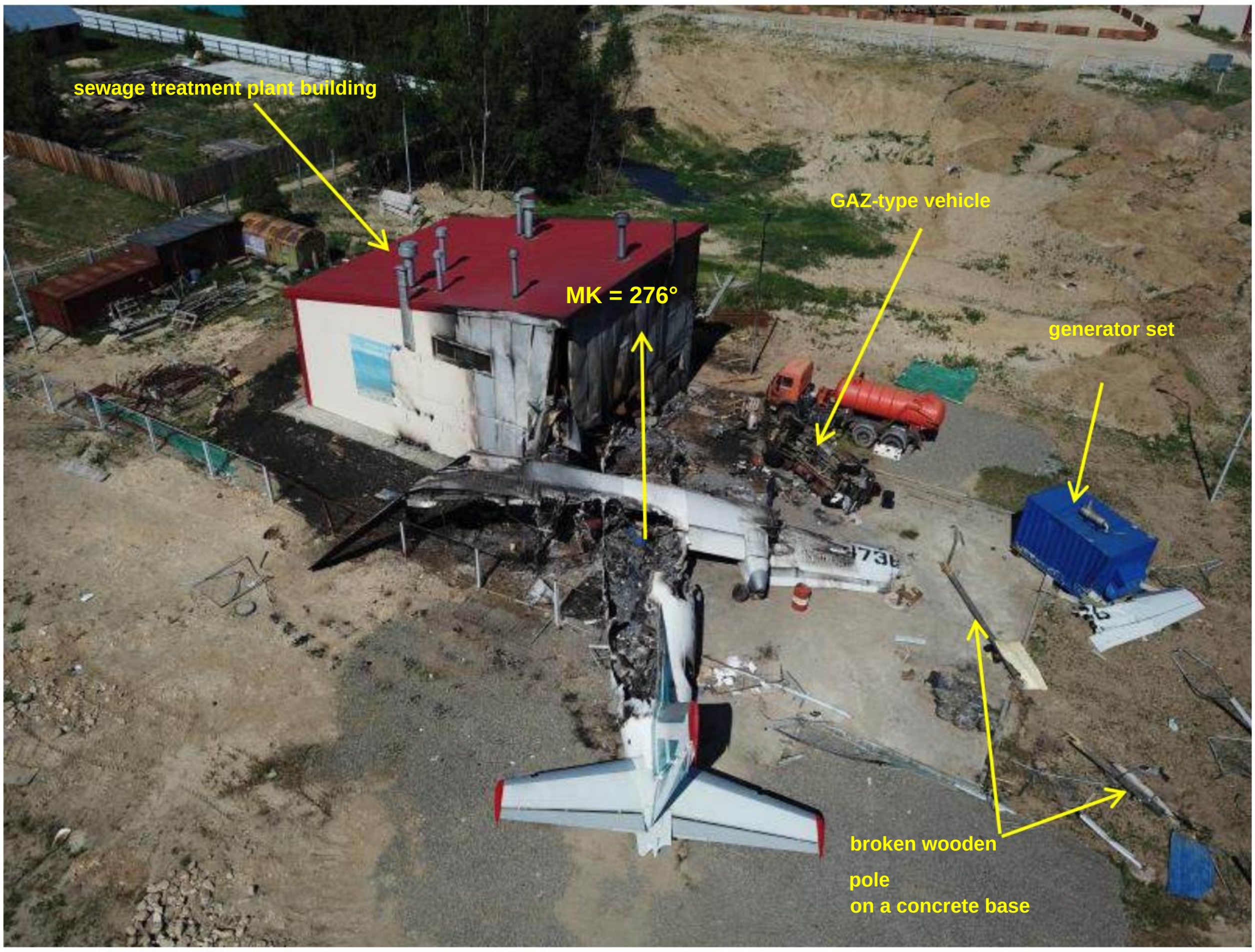


Fig. 16. Airplane at the accident site. Top view.

Fig. 17. Airplane at the accident site. Front view

Machine Translated by Google

Fig. 18. Airplane at the accident site. Right side view.

Fig. 19. Airplane at the accident site. Rear view.

Fig. 20. Airplane at the accident site. Left side view.

Fig. 21. Airplane at the accident site. View of the passenger cabin.

Fig. 22. Aircraft at the accident site. Left main landing gear

Fig. 23. Aircraft at the accident site. Right main landing gear

The sketches of the AP location are shown in Fig. 24.

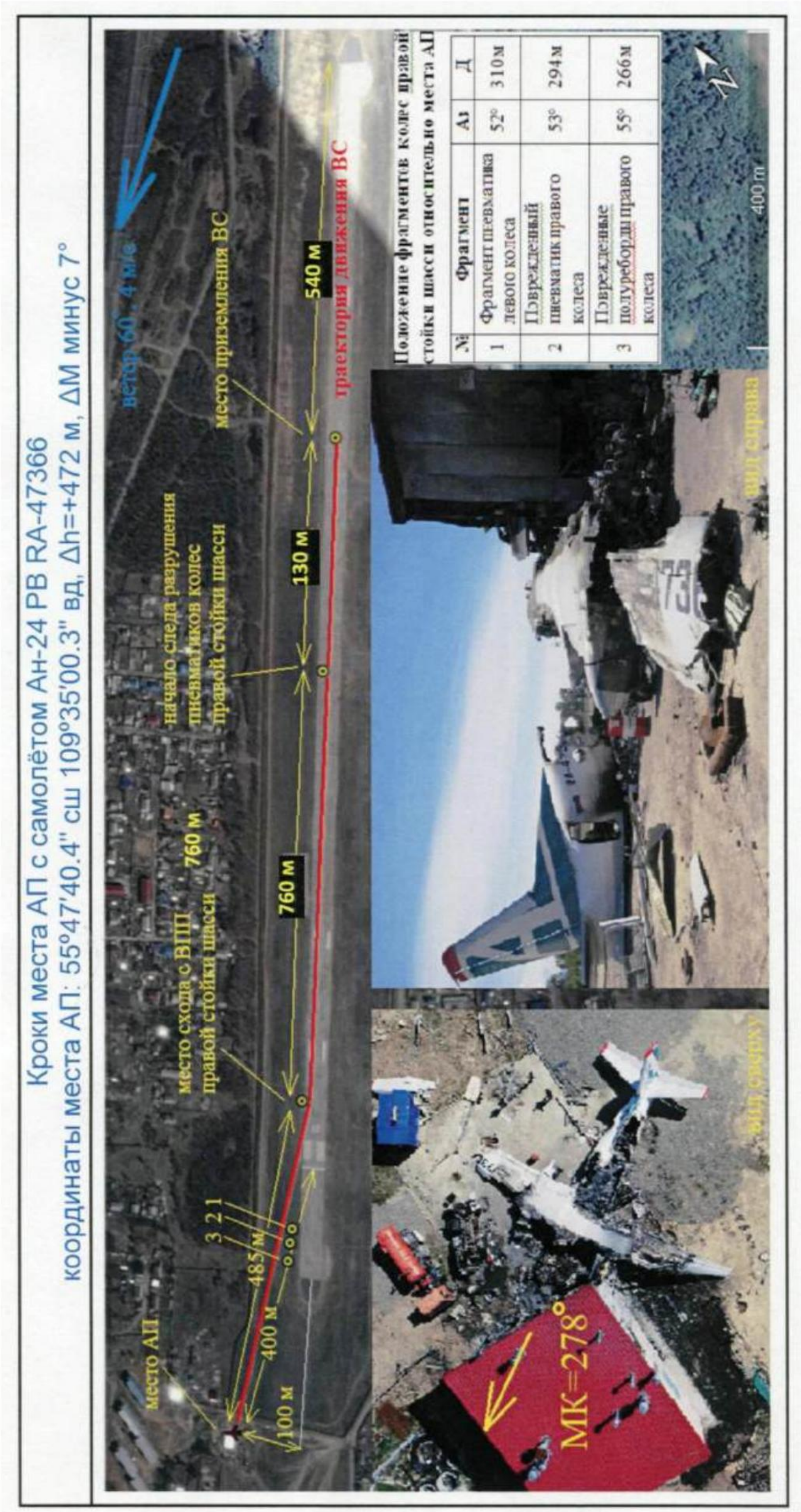


Fig. 24. Steps of the AP site

Machine Translated by Google

1.13. Medical information and summary of pathological results anatomical studies

On June 27, 2019, a blood sample was taken from 2P at the Nizhneangarsk Central Regional Hospital. On July 4, 2019, State Autonomous Healthcare Institution "Republican Narcological Dispensary" conducted chemical Toxicological examination of a blood sample. According to the results report studies, alcohol was not detected in the blood.

On June 28, 2019, the Irkutsk City Clinical Hospital No. 3 conducted Medical examination of a flight attendant for alcohol intoxication. According to the test results, no alcohol was detected.

A forensic medical examination of the bodies of the captain and flight engineer was conducted at the State Budgetary Healthcare Institution Irkutsk Regional Bureau of Forensic Medical Examination. During forensic chemical analysis of blood and urine No ethyl alcohol was detected in the bodies of the captain and flight engineer.

1.14. Data on the survival of passengers, crew members and other persons in an aviation accident

When landing, the flight crew was seated in the aircraft cabin in their standard workplaces. Everyone was wearing seat belts.

The flight attendant and passengers were in the aircraft cabin. Everyone was wearing seat belts. seat belts. Accommodation of the crew and passengers on the aircraft and The injuries sustained during the accident are shown in Fig. 25.

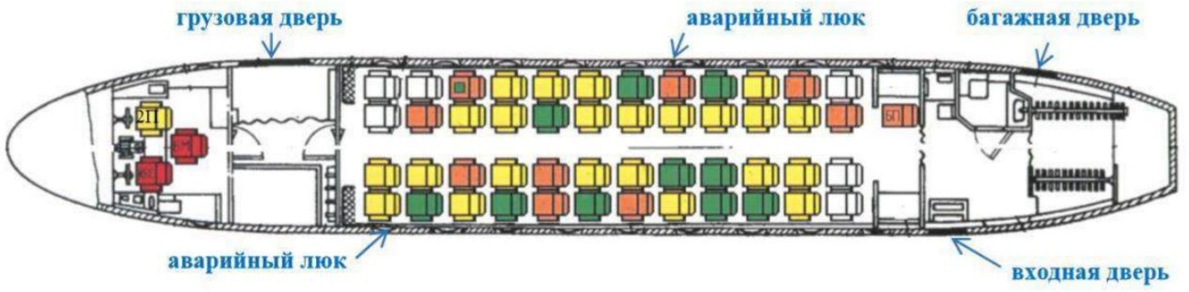
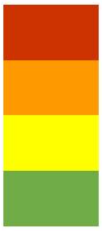


Fig. 25. Layout of the crew and passengers on the aircraft:



- bodily injury resulting in death;

- serious bodily harm;

- minor bodily injuries;

- there are no bodily injuries.

After the plane stopped, the flight attendant tried to open the front door, but failed. could, as it was deformed during the aircraft's collision with obstacles. The flight attendant opened the baggage door and, on his command, the passengers exited the aircraft.

Note: *From the flight attendant's explanation on July 5, 2019:*

“…Smoke instantly appeared in the front of the aircraft, and I ran towards emergency exits in the tail. The tail was partially destroyed, there were cracks fuselage, the ramp was torn out and the entrance door was deformed, it I tried to open it, but the male passenger and I couldn’t open it. Then I rushed to the tailgate...it didn't work right away, but I managed open...There was panic, of course, but only from fear, because the plane filled with acrid black smoke. The last passenger got out (crawled out), because only below could you still breathe, and he said that in the cabin there was more "there's no one here..."

2P was unable to evacuate the PIC and flight engineer, who were trapped in burning cabin of the aircraft and remained motionless.

Note: *From explanations 2P from 03.07.2019:*

“…When I raised my head I saw that smoke was coming from behind the front panel and The top panel (visors) caught fire. The flames spread quickly. I tried to bring the crew to their senses, since the captain and flight engineer were unconscious. Attempts were unsuccessful. I began to leave the work place. I unbuckled myself and climbed out to the door. I tried to bring it back again. feeling the crew and pull out, attempts were unsuccessful, because they were “fastened and clamped…”.

Forensic examination of the bodies of the captain and flight engineer revealed that *"The death of (the last name and initials of the captain and flight engineer) was caused by acute poisoning carbon monoxide."*

Having left the aircraft cabin into the passenger cabin, 2P evacuated one passenger through left front emergency hatch, after which he evacuated himself.

After the evacuation, all passengers were able to move independently. The injured The passengers, 2P and the flight attendant were taken to the Nizhneangarsk Central Regional Hospital. The remaining passengers were taken to the airport terminal building.

There were no design flaws in the aircraft that could have contributed to the severity of the consequences. revealed.

1.15. Actions of emergency rescue and fire brigades

At 02:13, after the aircraft crew reported the failure of the left engine, the control tower dispatcher announced the alarm signal.

After the alarm signal was given, in the period from 02:13 to 02:23, the following were notified: MO "Severobaikalsky District", DPK rural settlement "Nizhneangarskoye" and the station emergency medical care of the State Budgetary Healthcare Institution "Nizhneangarsk Central Regional Hospital".

The plane collided with the sewage treatment plant building after it rolled out runway limits occurred at 02:24:49.

At 02:27, a fire and rescue team from Nizhneangarsk Airport arrived at the scene of the accident. 1 fire truck and 5 specialists, as well as a fire department crew from Nizhneangarsk Airport consisting of 1 UAZ vehicle and 17 specialists. Then a detachment of PSCh-50 OGPS-12 arrived in consisting of 3 vehicles (two of which are fire trucks) and 12 specialists. Calculations of the PSR and ASC for Upon arrival at the scene of the emergency, they immediately began extinguishing/cooling the fire with water.

Note: *From the protocol of additional interrogation of the head of the departmental Fire department of Nizhneangarsk airport from July 10, 2019:*

"…After the collision, the plane immediately caught fire. Having approached it on distance no more than 15 meters, (name of firefighter) climbed onto the gun carriage and began to extinguish the fire with a jet of water for cooling. At this time the evacuation of passengers took place. After all passengers left the aircraft cabin, (firefighter's name) switched the shutter gun carriage and began to extinguish the fire with foam."

At 02:29, after the evacuation of passengers from the aircraft, the Nizhneangarsk airport emergency response team began extinguishing the fire using foam concentrate PO-6R3 type A.

The remaining units, after arriving at the scene of the accident, began extinguishing the fire. water.

The following arrived at the scene of the accident:

ÿ at 02:47 the 1st group of rescuers of the Severobaikalsk PSP BPSO consisting of 1 car and 5 specialists;

ÿ at 02:50 the DPK of the rural settlement "Nizhneangarskoye" consisting of 1 vehicle and 2 specialists;

ÿ at 03:00 the OG of the Federal State Budgetary Institution "9th Fire Department of the Federal Fire Service of the Republic of Bashkortostan" consisting of 1 fire truck and 4 specialists and the fire and rescue garrison of the Severobaikalsky district, consisting of 1 fire truck and 2 specialists;

ÿ at 03:10 the 2nd group of rescuers of the Severobaikalsk PSP BPSO consisting of 1 car and 11 specialists.

At 03:20 the fire was localized.

At 03:53 the open fire at the site of the emergency was extinguished.

At 05:15 the fire was completely extinguished.

About 250 liters of foaming agent and 400 m³ of water were used.

At 06:30, rescuers from the Severobaikalsk PSP BPSO from the burned-out plane The bodies of the deceased crew members were recovered and handed over to the Ministry of Defense personnel "Severobaikalsky" Ministry of Internal Affairs of Russia.

Deficiencies in emergency rescue operations that affected the severity consequences were not identified.

1.16. Testing and Research

1.16.1. Research of fuels and lubricants

Analysis of fuel samples from the aircraft's fuel tanks and oil from the left engine conducted in the fuel and lubricants testing laboratory of JSC Irkutsk International Airport.

As a result of fuel testing it was established that *“the submitted sample No. 2, taken from the drain cock of the fuel tanks of the second group of the right half-wing of the aircraft An-24RV RA-47366 is a TS-1 aviation fuel.*

Changes in the quality indicators of aviation fuel sample TS-1 No. 2, taken from the drain cock of the fuel tanks of the second group of the right half-wing of the aircraft An-24RV RA-47366, compared with the standards and quality indicators of standard samples “No aviation fuels were identified that were being refueled into aircraft.”

As a result of oil testing it was established that *“the presented sample No. 1, taken from the drain cock of the front crankcase of the left engine of the An-24RV aircraft RA-47366 is an CM-4.5 aviation oil blend that has been exposed to high temperature, "burned" ...".*

1.16.2. Aircraft engine research

The research of the left engine of the aircraft was carried out by specialists of the Federal Autonomous Institution "Aviation" Register of the Russian Federation" based on JSC "RZGA No. 412".

Conclusion on the results of the study: *“AI-24 engine series 2 No. H47622024 An-24RV RA-47366 aircraft, its parts and assemblies are in satisfactory condition technical condition, with the exception of parts that were damaged during the process development of an aviation accident and subsequent fire.*

Signs indicating in-flight engine shutdown due to damage or there is no damage to its parts."

1.16.3. Research of braking systems

Research of the main landing gear wheel brakes and the shuttle valves of the system The braking was carried out by the Federal Autonomous Institution "Aviation Register of the Russian Federation".

Conclusion on the results of the study: *“The brake devices of the wheels of the left and the right landing gear of the An-24RV RA-47366 aircraft and their parts have no damage and* *are in satisfactory technical condition, signs of failure are absent.*

The shuttle valves of the An-24RV RA-47366 aircraft are in the extreme positions, which indicates that brake pressure is entering the brake devices from main braking system.

Anti-skid devices of the right landing gear of the An-24RV aircraft RA-47366 are in working order."

1.16.4. Examination of the right main landing gear tires

The FAA conducted a study of the fragments of the aircraft tires on the right main landing gear. Aviation Register of the Russian Federation.

Conclusion of the research results: *"Destruction of internal and external aircraft tires 900 x 300 models 8A No. Ya I 1734869 and No. YaXIII 1733733 right main landing gear of the An-24RV RA-47366 aircraft was the result of the aircraft landing with "with locked wheels."*

1.16.5. Results of mathematical modeling

In order to evaluate the operation of the aircraft braking system and the actions of the crew during During the landing, specialists from the Antonov State Enterprise carried out a mathematical analysis Simulation of aircraft motion during takeoff. The simulation results are presented in Fig. 26 – 30.

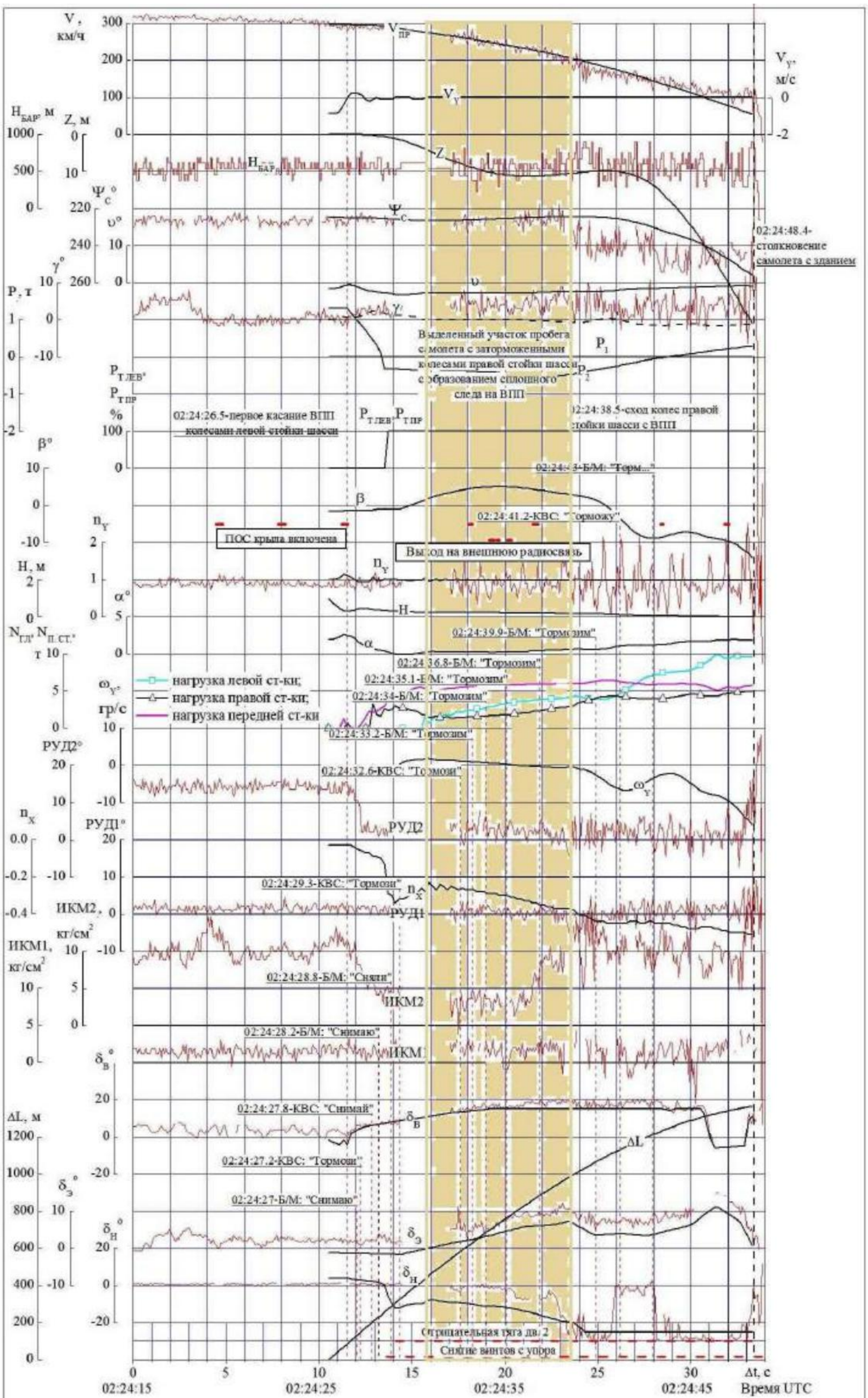


Рис. Самолет Ан-24РВ RA-47366. Параметры движения при выполнении посадки в а/п Нижнеангарск: V_{REF}=290 км/ч ИН; W=3 м/с; $\psi_{ВПП}$=225 (№23); ψ_W=50°; δ_3=15°; G=20850 кг; t=20°; $H_{АЭР}$ ~ 470 м; R_0=ПОТР; БВПП; μ≥0.7; $\bar{X}_T$=0.233

Параметры из МСРП-12-96;
Расчет

Fig. 26. Results of mathematical modeling of landing and movement of aircraft along the runway in an emergency flight

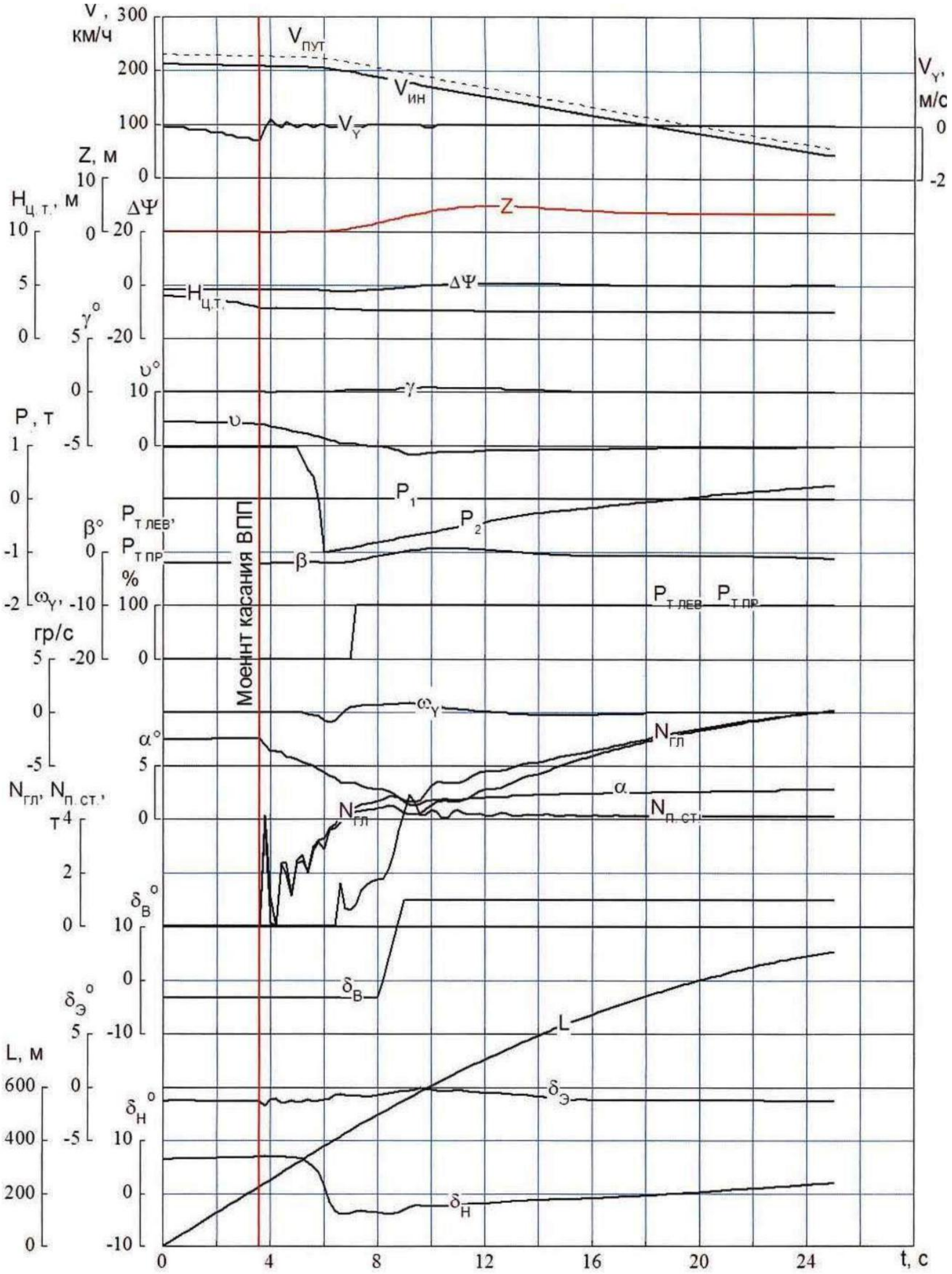


Рис. 1. Самолет Ан-24РВ RA-47366. Параметры движения при выполнении посадки в а/п Нижнеангарск: V_{REF}=211.3 км/ч ИН; W_z=3 м/с; $\psi_{впп}$=225 (№23); ψ_W=50°; δ_3=15°; G=20850 кг; t=+ 20 °C; $H_{АЭР}$ ≈ 470 м; R =ПОТР: БВПП: μ>0.6: $X_{..}$=0.233

Fig. 27. Results of mathematical modeling of aircraft movement along the runway under emergency flight conditions during landing at the speed recommended by the flight manual and with intact tires

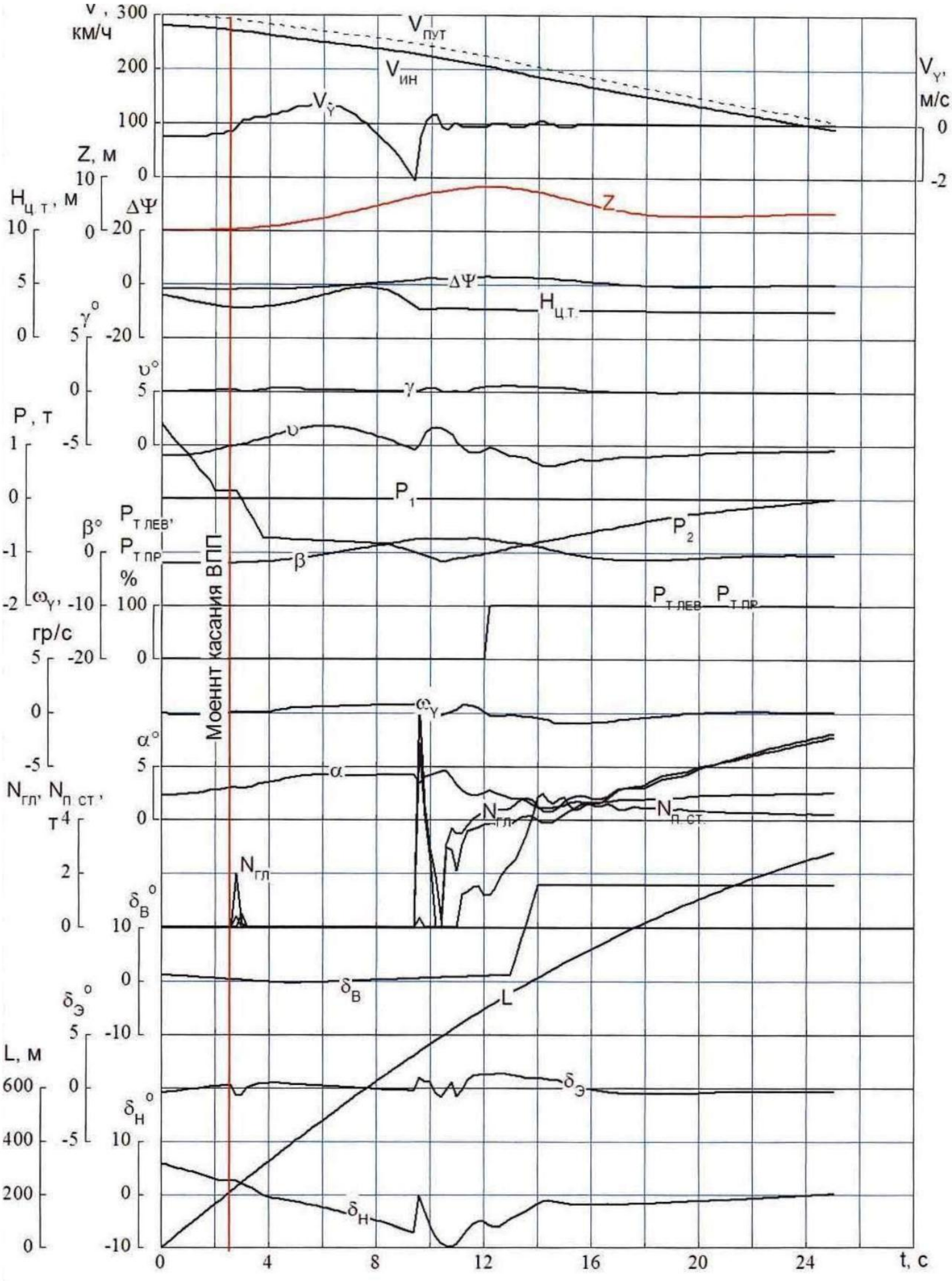


Рис. 2. Самолет Ан-24РВ RA-47366. Параметры движения при выполнении посадки в а/п Нижнеангарск: V_{REF}=290 км/ч ИН; W_z=3 м/с; $\psi_{ВПП}$=225 (№23); ψ_W=50°; $\delta_з$=15°; G=20850 кг; t=+ 20 °C; $H_{АЭР} \approx$ 470 м; R_0=ПОТР; БВПП; $\mu \geq 0.6$; $\bar{X}_т$=0.233

Fig. 28. Results of mathematical modeling of aircraft movement under emergency flight conditions, confirming the possibility of aircraft separation from the runway

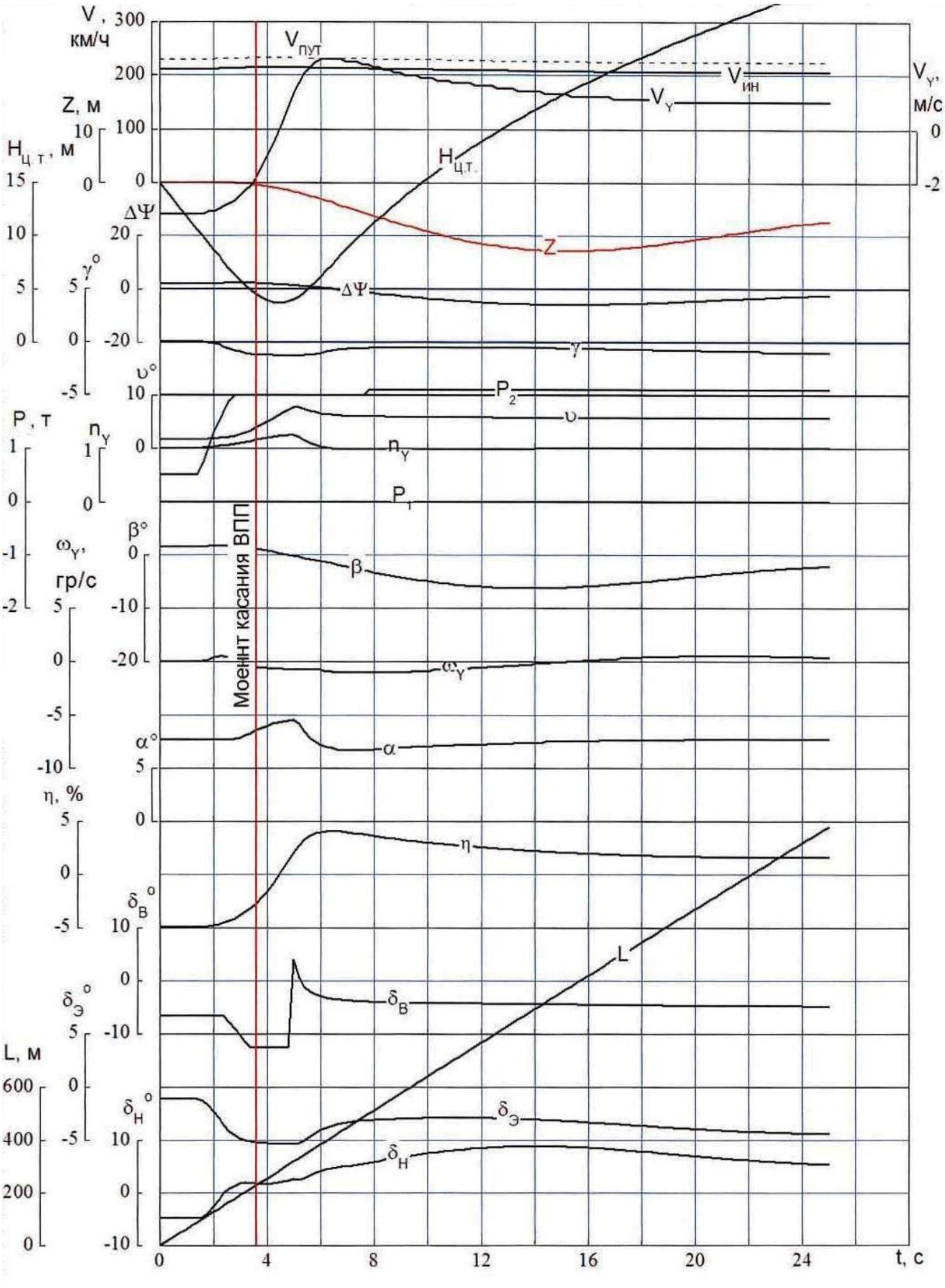


Рис. 3. Самолет Ан-24РВ RA-47366. Параметры движения при выполнении ухода на второй круг в а/п Нижнеангарск:
V_{REF}=211.3 км/ч ИН; W_z=3 м/с; $\psi_{ВПП}$=225 (№23); ψ_W=50°; $\delta_з$=15°; G=20850 кг;
t=+ 20 °C; $H_{аэп}$ ≈ 470 м; R_0=ПОТР; БВПП; μ≥0.6; $\bar{X}_т$=0.233

Fig. 29. Results of mathematical modeling of an aircraft go-around in emergency flight conditions from a height of 10 m

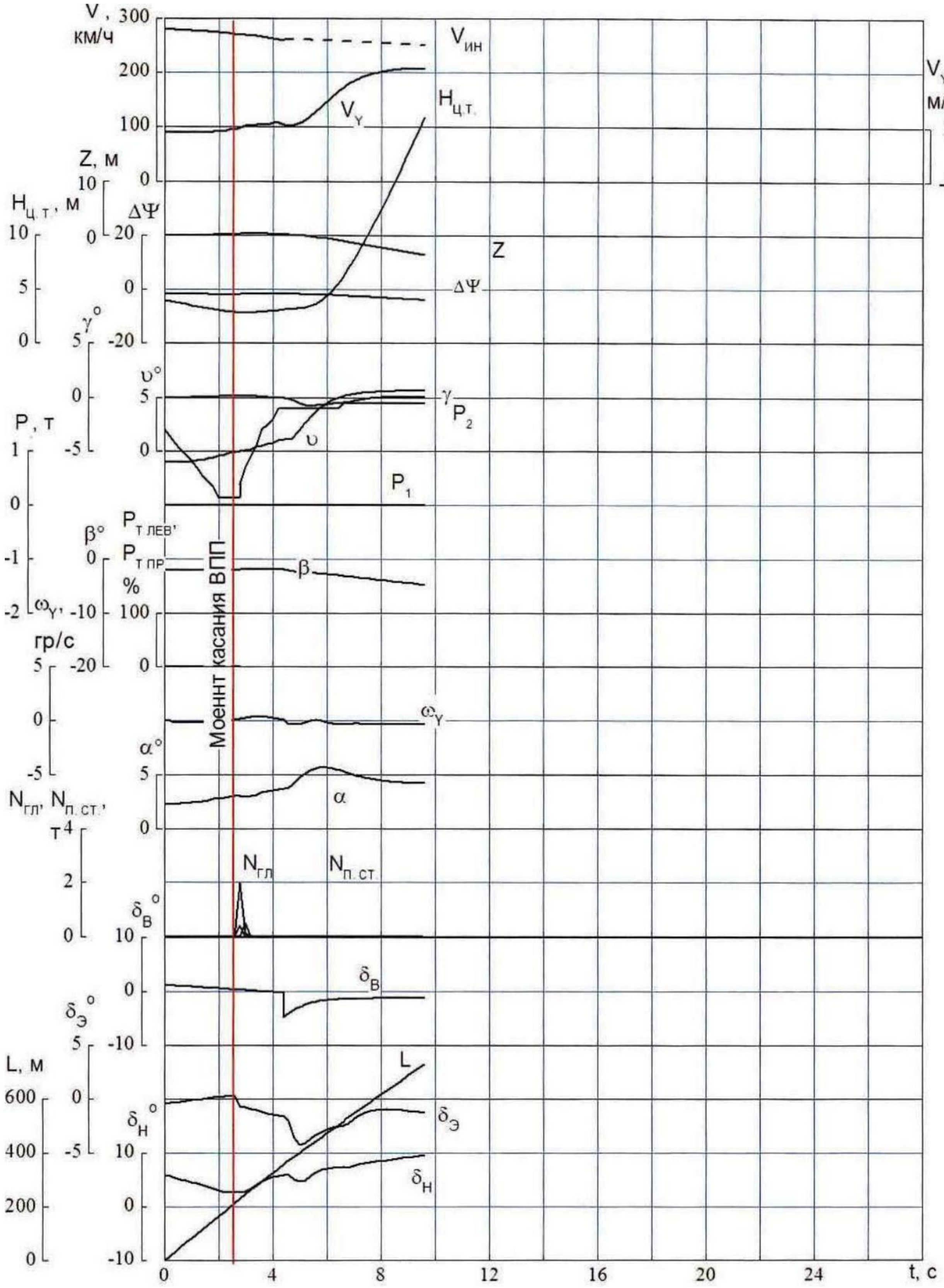


Рис. 4. Самолет Ан-24РВ RA-47366. Параметры движения при выполнении ухода на второй круг после отскока от ВПП в а/п Нижнеангарск:
V_{REF}=290 км/ч ИН; W_z=3 м/с; $\psi_{ВПП}$=225 (№23); ψ_w=50°; δ_3=15°;
G=20850 кг; t=t_{CA}; $H_{АЭР}$ ≈ 470 м; R_0=ПОТР; БВПП; $\mu \geq 0.6$; $\bar{X}_T$=0.233

Fig. 30. Results of mathematical modeling of an aircraft go-around in emergency flight conditions after a bounce from the runway

As a result of the mathematical modeling, it was established:

1. Maximum available total directional moment from tires wheels of the front landing gear and from the full deflection of the rudder it becomes insufficient to maintain a given direction of movement along the runway at speeds below 200 km/h (Fig. 26). Total required directional moment from tires braked wheels of the right landing gear and from the right propeller, removed from the stop, exceeds maximum available total rudder yaw moment and nose landing gear tires. Aileron consumption does not exceed ÿ 50%, elevator fully deflected to press the front support.

2. When performing landing in accordance with the provisions of the Flight Manual with one with the engine off and the right wheels intact in emergency conditions the available rudder deflections are sufficient to prevent the aircraft from drifting to the right and its exit from the runway (Fig. 27).

3. Landing at an increased speed of 290 km/h instead of 220 – 240 km/h, recommended in the flight manual, leads to the aircraft “bounce” (separation) from the runway (Fig. 28).

4. Aerodynamic characteristics of the An-24RV aircraft and its the thrust-to-weight ratio with one engine inoperative allows for a go-ahead the second circle from a height of 10 m (Fig. 29), as well as after the first bounce from the runway when the occurrence of a high-speed "goat" (Fig. 30). However, according to the provisions of the Aircraft Manual (section 5.1.8), A safe go-around with one engine out is only possible from a height 50 m.

5. The maximum permissible weight of the aircraft to ensure the standard gradient during go-around, according to the flight manual, for actual airfield conditions Nizhneangarsk is 20.75 tons.

6. The required landing distance for an An-24RV type aircraft with a mass 20.85 tons with one engine inoperative, with flaps at 15°, at recommended in the RLE, the gliding and landing speeds are 240 – 220 km/h with a course of 225° at actual airfield conditions that existed on the day of the accident (with associated components wind), is equal to 2160 m.

1.16.6. Results of the analysis of psychological testing data crew members[9]

This section presents the results of the analysis of psychological data testing of the KVS. The study was conducted by an experienced psychologist with experience in

[9]

in the field of psychodiagnostics (primarily clinical) for more than 25 years, expert level, with teaching experience and supervision experience, as well as experience in areas of forensic examination and personnel assessment.

[10]

10

Текст данного раздела не публикуется для обеспечения соответствия положениям Главы 2.6 ПРАПИ.

Текст данного раздела не публикуется для обеспечения соответствия положениям Главы 2.6 ПРАПИ.

1.17. Information on organizations and administrative activities, related to the incident

The owner of the aircraft is JSC SAT Airlines (certificate of aircraft registration No. 3440, issued by Rosaviatsia on 03.04.2006).

Postal address: 693023, Russian Federation, Yuzhno-Sakhalinsk, Gorky Street, Building 50A.

06.11.2013 on the basis of two airlines JSC "Airline" SAT "and JSC Vladivostok Avia was formed by JSC Aurora Airlines. Postal address: 693023, Russian Federation, Yuzhno-Sakhalinsk, Gorky street, house 50A.

The aircraft operator is JSC Angara Airlines (certificate operator No. 477, issued by Rosaviatsia on 20.12.2016). According to the operational specifications, JSC Angara Airlines has the right to operate the An-24RV RA-47366 aircraft to carry out commercial air transportation of passengers and cargo. Postal address: 664009, Russian Federation, Irkutsk, Shiryamova street, house 2.

The operator of the Nizhneangarsk airfield is JSC “Local Airports” Air Lines of Buryatia" (abbreviated name - OJSC "Airports of Buryatia"). OJSC Airports of Buryatia has a certificate as a civil aviation aerodrome operator. No. FAVT.OA.09-007, issued on October 20, 2017 by the Armed Forces of the Moscow Territorial Administration of the Federal Air Transport Agency. Postal address: 670018, Russian Federation, Republic of Buryatia, g. Ulan-Ude, village Airport, house 10.

Control (supervision) over the implementation of requirements by supervisory entities in the field of civil aviation is carried out by the Department of State Aviation Supervision and Supervision of ensuring transport security in the Siberian Federal District Rostransnadzor. Postal address: 630091, Russian Federation, Novosibirsk, Krasny Prospekt, Building 44.

1.18. Additional information

1.18.1. Judicial issues regarding the wastewater treatment plant building

Based on the order of the UGAN NOTB SFO Rostransnadzor (hereinafter referred to as the Management) on March 27, 2017, during a scheduled inspection and inspection of the aerodrome territory of the Nizhneangarsk airfield by the Chief State Inspector of the Directorate It was established that the construction company Inter-Comfort LLC was the general contractor in the construction of treatment facilities and external sewerage networks up to drop points in the village of Nizhneangarsk within the aerodrome's aerodrome area Nizhneangarsk. The customer of construction under contract No. F.2016.335199 dated November 21, 2016 was the Municipal Public Institution "Committee for Municipal Economy Management" of the North-Baikalsky district of the Republic of Buryatia, which has a permit for the construction of the facility in the absence of agreement with the owner of the airfield.

Based on the circumstances of the violation identified by the official of the Administration On April 18, 2017, administrative offense report No. 4171540013-01 was drawn up, provided for in Part 2 of Article 11.4 of the Code of Administrative Offences of the Russian Federation.

Based on the results of the examination of the case on an administrative offence based on the report of the Chief State Inspector, Deputy Chief

On May 2, 2017, the Directorate issued Resolution No. 4171540013-03, by which the MCU The Committee for Municipal Economy Management has been brought to administrative responsibility liability under Part 2 of Article 11.4 of the Code of Administrative Offences of the Russian Federation with the appointment of an administrative punishment in the form of a fine.

Considering that the said resolution does not comply with the law, the Municipal Institution “Committee on Municipal Economy Management" appealed to the Arbitration Court of the Republic Buryatia filed a petition to invalidate the decree of May 2, 2017. ÿ 4171540013-03.

On September 12, 2017, the Arbitration Court of the Republic of Buryatia, having considered the case in open court meeting case No. A10-2461/2017 on the application of the Municipal Institution "Committee on Management municipal economy" to the UGAN NOTB SFO Rostransnadzor on recognition illegal resolution of 02.05.2017 No. 4171540013-03 on the involvement of the MUK "Committee for the management of municipal economy" to administrative responsibility for Part 2 of Article 11.4 of the Code of Administrative Offences of the Russian Federation, issued a decision:

“Guided by Articles 167–170 , 211 of the Arbitration Procedure Code Russian Federation, DECIDED:

The stated requirement is satisfied.

Resolution of the State Aviation Supervision and Supervision Authority ensuring transport security in the Siberian Federal District Federal Service for Supervision of Transport dated 02.05.2017 No. 4171540013-03 involving the Municipal State Institution "Management Committee" municipal economy" (OGRN 1080317000543, INN 0317009080) to administrative liability under Part 2 of Article 11.4 of the Code of Administrative Offenses of the Russian Federation shall be recognized as illegal and "cancel completely."

No appeal was filed against this decision of the Siberian Federal District Department of Transport Supervision.

On June 27, 2019, after the AP considered the case, the investigator of the Bratsk Investigative Committee The East Siberian Transport Department of the Investigative Committee of the Russian Federation opened a criminal case No. 11902009304970707 on the grounds of a crime under Part 3 of Article 263 Criminal Code of the Russian Federation. A copy of the decision to initiate criminal proceedings was sent To the Baikal transport prosecutor.

The Baikal transport prosecutor filed a lawsuit in the interests of an unspecified a circle of persons with a claim against the administration of the Severo-Baikalsky District, in which they requested:

“- to recognize the permission of the administration of the Severo-Baikalsky municipal district as illegal district" for the commissioning of a wastewater treatment facility and an external network sewerage to the discharge point in the village of Nizhneangarsk dated 02/28/2019 No. 04517102-01-2018;

- prohibit the administration of the Severo-Baikalsky District from exploiting treatment facilities and external sewerage network located on land plot with cadastral number 03:17:000000:5995;

- oblige the administration of the Severo-Baikalsky District to carry out the demolition treatment facilities and external sewerage network located on land plot with cadastral number 03:17:000000:5995."

November 26, 2019, Severobaikalsk City Court of the Republic of Buryatia (city of Severobaikalsk), having examined in open court a civil case on the said statement of claim, decided:

"The claims of the Baikal Transport Prosecutor are satisfied. To recognize as illegal the permission of the administration of the Severo-Baikalsky District commissioning of the treatment plant and external sewerage network facility discharge points in Nizhneangarsk settlement dated February 28, 2019 No. 04517102-01-2018.

Prohibit the Administration of the MO "Severo-Baikalsky District" from exploitation treatment facilities and external sewerage networks located on land plot with cadastral number 03:17:000000:5995.

Obligate the Administration of the Severo-Baikalsky District to carry out demolition treatment facilities and external sewerage networks located on land plot with cadastral number 03:17:000000:5995.

The decision can be appealed to the Supreme Court of the Republic of Buryatia by filing an appeal through the Severobaikalsk City Court of the Republic Buryatia within a month from the date of its adoption in its final form."

08.06.2020 Judicial Collegium for Civil Cases of the Supreme Court of the Republic Buryatia (Ulan-Ude) considered in an open court session a civil case on the statement of claim of the Baikal Transport Prosecutor in the interests of an unspecified circle of persons to the administration of the MO "Severo-Baikalsky District" on recognition of illegality permission to put the facility into operation, prohibition of operation of the facility, obligation to demolish the facility based on the appeals of the head of the Severo-Baikalsky Municipal District district", a representative of the administration of the MO GP "Nizhneangarsk settlement", a representative Ministry of Construction and Modernization of Housing and Communal Services of the Republic of Bashkortostan Decision of the Severobaikalsk City Court of the Republic of Buryatia dated November 26, 2019. Guided by Articles 328 and 329 of the Civil Procedure Code of the Russian Federation, the judicial The panel determined: *"The decision of the Severobaikalsk City Court of the Republic of Buryatia from November 26, 2019 leave unchanged, appeals* **-** *without satisfaction."*

According to the information available to the investigation commission, at the time of preparation The demolition of the treatment facilities and the external sewerage network has not been completed as per this report. was.

1.18.2. About visual maneuvering (circle-to-land) and visual approach for boarding

Provisions of international and domestic documents on issues of circular maneuvering (circle-to-land)

Doc 8168 ICAO

Section 5 "Approach Procedures", Chapter 6 "Visual Maneuvering (Flight by circle)»

6.1 PURPOSE

6.1.1 The term "visual maneuvering" (circling) is used for descriptions of the flight phase performed after completion of the instrument approach. At this stage, the aircraft is brought into a landing position relative to the runway, the location of which is not suitable for a straight-line landing approach, i.e. in relation to which the criteria for straightening of direction or gradient cannot be met reduction.

6.2 VISUAL FLIGHT MANEUVER

6.2.1 A circling approach is a visual maneuver in flight. The conditions in each circuit flight are different because they depend on variables such as factors such as runway location, final approach flight path, wind speed and meteorological conditions. Therefore, it is impossible to develop a single a scheme that would take into account any conditions in which the landing approach is carried out in a circle.

6.2.2 After initial visual contact, you should, while maintaining air the vessel is within the visual maneuvering zone and not lower than MDA/ H of the flight in a circle, keep the runway environment in view at all times. The runway environment includes such characteristic landmarks such as the runway threshold or approach lighting, or other runway identification markings.

6.3 PROTECTION

6.3.1 Visual maneuvering zone

The visual maneuvering zone during a circling approach is determined by drawing arcs with centers at the location of each runway threshold and tangents, *connecting these arcs[11] .*

6.3.2 Obstacle clearance

After establishing a visual maneuvering zone (circling flight) The OSA/ H is determined for each category of aircraft[12] .

6.3.3 Minimum Descent Altitude/ Height (MDA/ H)

Once the OSA/ H is established, the MDA/ H is also determined to ensure taking into account operational aspects. Reduction below MDA/ H is not carried out until Bye:

a) the required visual contact will not be established and maintained reference points throughout the maneuver;

b) the pilot will not see the runway threshold;

c) the required clearance above obstacles will not be maintained and the aircraft will not be in a suitable position to land using normal descent rates and bank angles.

6.3.4 Exceptions within the visual maneuvering zone (circling)

6.3.4.1 A sector of the circling flight zone in which there is a protruding obstacle may not be taken into account if it is located outside the final approach and go-around zones. This the sector is limited by the dimensions of the approach surfaces devices of Volume I Appendix 14.

6.3.4.2 If this option is used, the published scheme prohibits the execution of a flight in a circle within the entire sector where there is this obstacle.

FAP 128: Terms and Definitions[13]

Visual maneuvering (circle-to-land maneuver) - continuation of the procedure instrument approach, which involves making turns within the limits visual maneuvering zones for bringing the aircraft into a landing position relative to the runway, the location of which in relation to the final stage trajectory The instrument approach does not allow a straight-in landing.

***Visual maneuvering zone** - the zone within which one should take into account obstacle clearance for aircraft approaching from*

[11] The radius of these arcs depends on the aircraft category.

[12] The document contains a note that the established OSA/N should not be interpreted as an operational minimum.

13 There are no provisions regarding the procedure for performing visual maneuvering (circle-to-land maneuver) in FAP-128.

using visual maneuvering (circle-to-land maneuver);

Minimum Descent Altitude (MDA) or Minimum relative descent height (MDH) *- the established absolute or relative altitude during a non-precision approach or during an approach landing using visual maneuvering (circle-to-land maneuver), descent below which it is prohibited if the required visual contact with ground references to continue the landing approach. MDA is measured from the mean sea level, and MDH is measured from the aerodrome elevation or exceeding the runway threshold if its excess is more than 2 m below the excess airfield. In cases where both concepts are used, for convenience it is possible use the form "minimum absolute (relative) descent altitude" (MDA/ H).*

Provisions of international and domestic documents on visual issues landing approach

Doc 4444 ICAO

Chapter 1 «Definitions»

Visual approach - an approach to landing when flying under IFR, when the procedure instrument approach is partially or completely not observed and the approach performed using visual ground references.

Chapter 6 "Separation in the Airfield Area"

6.5.3 Visual approach

6.5.3.1 Subject to the conditions set out in paragraph 6.5.3.3, permission an aircraft operating under IFR to perform a visual approach to landing can be requested by the flight crew or issued by the controller. In the latter In this case, agreement with the flight crew is required.

6.5.3.2 Controllers shall exercise caution when providing visual landing approach if there is reason to believe that the relevant flight crew unfamiliar with the airfield and its surroundings. When initiating visual approaches to landing controllers must also take into account the prevailing traffic and weather conditions.

6.5.3.3 An aircraft operating an IFR flight may be issued clearance to perform a visual approach provided that the pilot has the ability to maintain visual contact with ground landmarks and:

a) the reported cloud base corresponds to the level at which the initial landing approach of an aircraft that has received such a designation begins *resolution, or exceeds this level, or*

b) the pilot reports being at the level at which the initial section begins approach, or at any time during the flight according to the instrument approach procedure, that meteorological conditions allow us to confidently assume that visual approach and landing can be accomplished.

FAP-293

Terms and definitions

Visual approach - an approach to landing when flying under IFR, when the procedure instrument approach is partially or completely not observed and the approach performed using visual ground references.

Chapter V. Arriving and departing aircraft, aerodrome control service

5.6. Permission for an aircraft operating an IFR flight to perform visual approach is requested by the aircraft crew or initiated by the ATS authority. In the latter case, approval from the crew is required.

5.6.1. When initiating visual approaches, the ATS unit must take into account the air situation and meteorological conditions.

5.6.2. The ATS authority issues permission to carry out a visual approach to landing of an aircraft operating an IFR flight, provided:

a) the crew has the ability to maintain visual contact with the runway or its landmarks;

b) the reported cloud base is equal to or greater than the altitude at which which marks the beginning of the initial landing approach of an aircraft that has received such permission, or

c) the crew reports that meteorological conditions allow the flight to be carried out visual approach and landing.

An-24RV Flight Manual, Section 4.6.2a "Piloting Features during a Visual Approach" [14]

(1) Visual approach - an approach carried out in accordance with instrument flight rules (IFR), when part of the procedure or the entire approach procedure according to instruments, the landing approach is not completed and the landing approach is carried out with visual contact with runway and/ or its landmarks.

(2) Entry into the aerodrome zone (area) is carried out by the PIC or 2P according to established schemes (STAR) or along the trajectories specified by the ATC service. Descent and approach to

[14] The procedure for performing visual maneuvering (circle-to-land maneuver) is not described in the An-24RV flight manual.

landing under IFR should be carried out with the help of radio-technical landing aids and navigation - RMS, RSP, OSP, OPS (DPRS, BPRS), VOR, VOR/ DME to the established altitude of the visual approach point (VAP).

(3) Before reaching the visual approach point, there must be The chassis and wing mechanization were released to an intermediate position.

(4) As a rule, a rigid visual approach procedure is not established. In general, visual flight in the visual maneuvering zone is carried out with performing a circular maneuver at a flight altitude in a circle (Hkr.vzp) of at least Nms a specific airfield.

(5) At the altitude of the visual approach point, if not established visual contact with the runway or its landmarks, the aircraft should be transferred to level flight until reliable visual contact with the runway or its landmarks.

(6) When reliable visual contact is established, the PIC must report to the dispatcher: "I see the lane," and receive permission (confirmation) to execute visual approach.

Piloting during a visual approach to landing must be performed by the pilot-in-command. aircraft in constant visual contact with the runway or its landmarks. If when approaching the runway, visual contact is not established or is subsequently lost, a turn towards the runway must be made with a climb and exit to established missed approach procedure using instruments for subsequent approach to landing according to IFR.

7) Maneuvering during a visual approach to landing should not be carried out with rolls more than 30°.

8) Before starting the turn in the direction of the runway of the intended landing at an altitude of not below the minimum descent altitude it is necessary:

- release the wing mechanization into the landing position,

- set the speed Vÿÿ according to table 4.6 or 4.8;

- perform control operations according to the Control Check Card, corresponding to the Map "After giving the aircraft the landing configuration".

Turn to landing course while maintaining the speed Vÿÿ descent at a vertical speed not exceeding 5 m/ s to the glide path entry altitude.

The recommended bank angle when turning to the landing course is 20°, but not more than 30°. Height the glide path must be at least 150 m.

CAUTION! WHEN MAKING A TURN TO LANDING COURSE

IT IS POSSIBLE AND ALLOWED FOR THE ALARM TO BE TRIGGERED AT LIMITS KRENOV.

(9) After entering the landing course, the PIC must assess the position aircraft relative to the runway. If the aircraft is in landing position, set Approach speed Vzp and glide path descent mode (-3°). PIC to report inform the landing controller of your readiness for landing and obtain landing clearance.

10) From the point of commencement of the visual approach, piloting is carried out Only the PIC controls the flight using instruments, paying special attention to maintaining the minimum descent altitude established for a given aerodrome, speed and bank angles. When turning onto the landing course with the indicator light on Maximum bank angle alarm - 2P informs the PIC that the bank angle has reached 30°.

The navigator controls the altitude and speed of the flight and, if possible, the position aircraft relative to the runway.

RPP Part B, *paragraph* 6.6.5. "Visual approach"[15]

The aircraft crew is allowed to make a visual approach to landing when the weather is not below the minimum for a visual approach.

With this approach method, the crew makes a visual approach to landing according to the scheme in accordance with the aeronautical information publication (aerodrome schemes) or instructions for flight operations at a given aerodrome.

If the airfield is equipped with landing approach systems (KGS or OSP), then these systems are backup and control the aircraft's visual approach.

In the absence of ATC and RTS points at the landing aerodrome, approach to the aerodrome destination and landing approach is carried out according to the scheme approved for this airfield.

Before passing the visual approach point when establishing reliable visual contact, the aircraft commander informs the crew: "I see the strip, "continue approach," and the second pilot, who is monitoring the flight, reports to the dispatcher ATC: "I see the runway," and receives clearance to perform a visual approach.

During a visual approach, the visual approach start point must be the chassis and wing mechanization were released into an intermediate position, and before the start turn towards the runway at an altitude not lower than the minimum descent altitude The aircraft's landing equipment must be prepared. When turning on landing course vertical descent rate should be no more than 5 m/ s. Angle

[15] The procedure for performing visual maneuvering (circle-to-land maneuver) is not described in the airline's flight manual.

The descent trajectory slope is maintained visually. Glide path entry altitude must be at least 150 m.

Maximum permissible bank angle during visual approach:

- at altitudes above 80 m .... 30°

- at altitudes of 80 m and less ..... 15°.

Attention: When performing a turn to the landing course, it is possible and permitted activation of the maximum tilt alarm.

The VPR is set equal to the personal minimum height of the aircraft commander cloudiness.

The interaction of crew members and the distribution of responsibilities are similar interaction when using equipment of landing approach systems.

Appendix 2 to the Order of the Department of Air Transport dated September 20, 1993 No. DV-123/3816 Supplement to the “Programs for training flight personnel on air transport aircraft of the Civil Aviation Authority" dated June 22, 1992

Introduction. General Provisions

Visual flight was once the primary mode of flight and many pilots, especially older, have practical experience of such flights, accumulated on an airplane Li-2, Il-14, and others. However, the ICAO definition of a visual approach should be considered as a special category of visual flight requiring carrying out work on organizing such flights at the airfield, and on preparing the flight composition.

...

Visual approach - an approach carried out in accordance with instrument flight rules, when part of the procedure or the entire approach procedure is instruments have not been completed and the landing approach is carried out with visual contact with runway and/ or its landmarks.

A visual approach may involve performing a visual circling the airfield before landing.

...

In general, visual approach patterns are not are established and if published in aeronautical information documents, then are advisory in nature, as a type of maneuver. The absence of a rigid scheme visual approach means that the pilot is given more freedom

[16] On the introduction of visual approaches into civil aviation practice for aircraft of classes 1-3.

actions to maneuver during a visual approach, but only on it and bears full responsibility for flight safety. In all cases, the maneuver must be performed only strictly within a defined visual maneuvering zone.

1. Organization of a visual landing approach at an aerodrome

1.1 Visual maneuvering zone in the aerodrome area

Visual maneuvering zone - the zone within which one should take into account obstacle clearance for aircraft performing a visual approach landing (flight in a circle).

The visual maneuvering zone (circling flight) is established on a specific aerodrome to take obstacles into account when determining the minimum safe obstacle clearance heights.

The visual maneuvering zone (circling flight) is limited by arcs, drawn from the center of the threshold of each airfield runway, which can be used for landing aircraft of this category, connected by tangents to these arcs.

1.18.3. On the performance of landing approaches to runway 23 of the aerodrome Nizhneangarsk by airline crews

The commission conducted an analysis of five randomly selected approaches to runway 25, performed by the airline's crews in August-September 2021. In all During the flights considered, visibility was more than 10 km, the height of significant cloud cover was not less than 1200 m above runway level. Approach paths with indication of characteristic moments of approach (release of landing gear, flaps, etc.) are shown in Fig. 31-35.

In the figures given, the first stage refers to the flight stage with the exit to OPR and flight in the holding area, under the second stage – execution of the approach pattern up to re-entry to the landing strip or before landing (with a visual approach), under the third – performing a visual maneuver (circle-to-land maneuver) with landing.

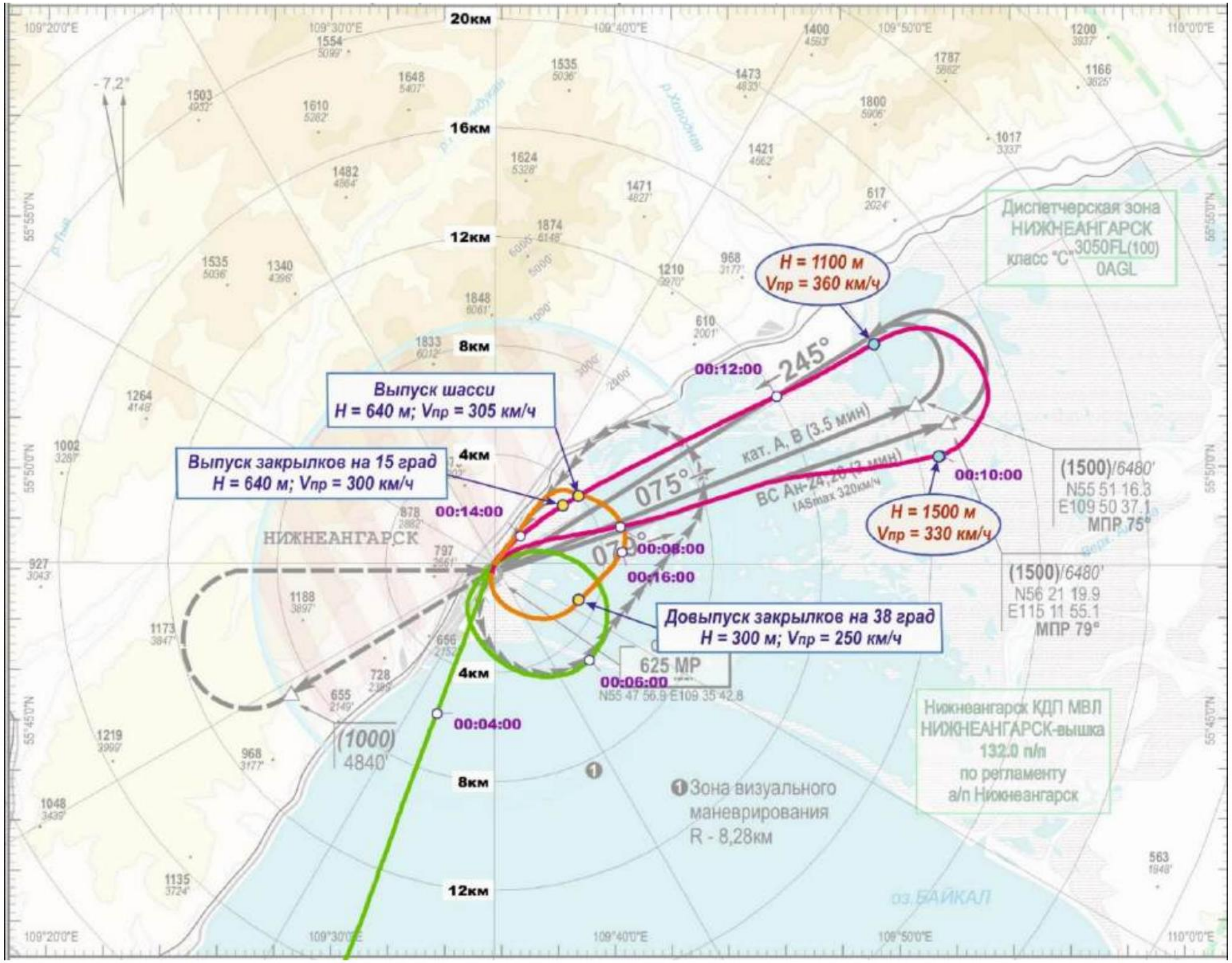


Fig. 31. Landing approach of An-24RV RA-47355, flight 31.08.2021 on the route: Irkutsk - Nizhneangarsk (1st stage of approach - green; 2nd stage of approach - red; 3rd stage of approach - orange)

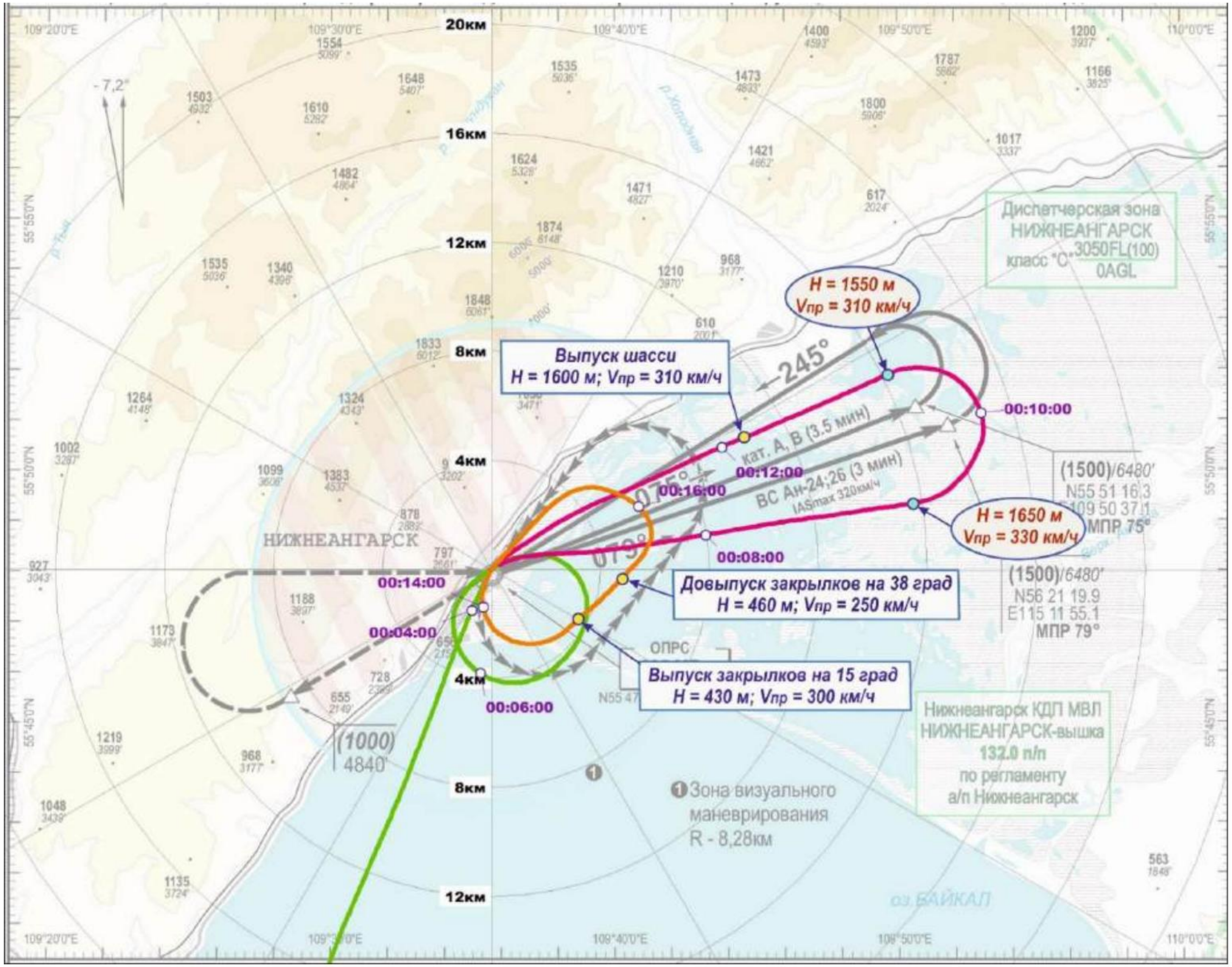


Fig. 32. Landing approach of the An-24RV RA-47355 aircraft, flight 08/31/2021 on the route: Ulan-Ude - Nizhneangarsk (1st stage of approach - green; 2nd stage of approach - red; 3rd stage of approach - orange

Machine Translated by Google

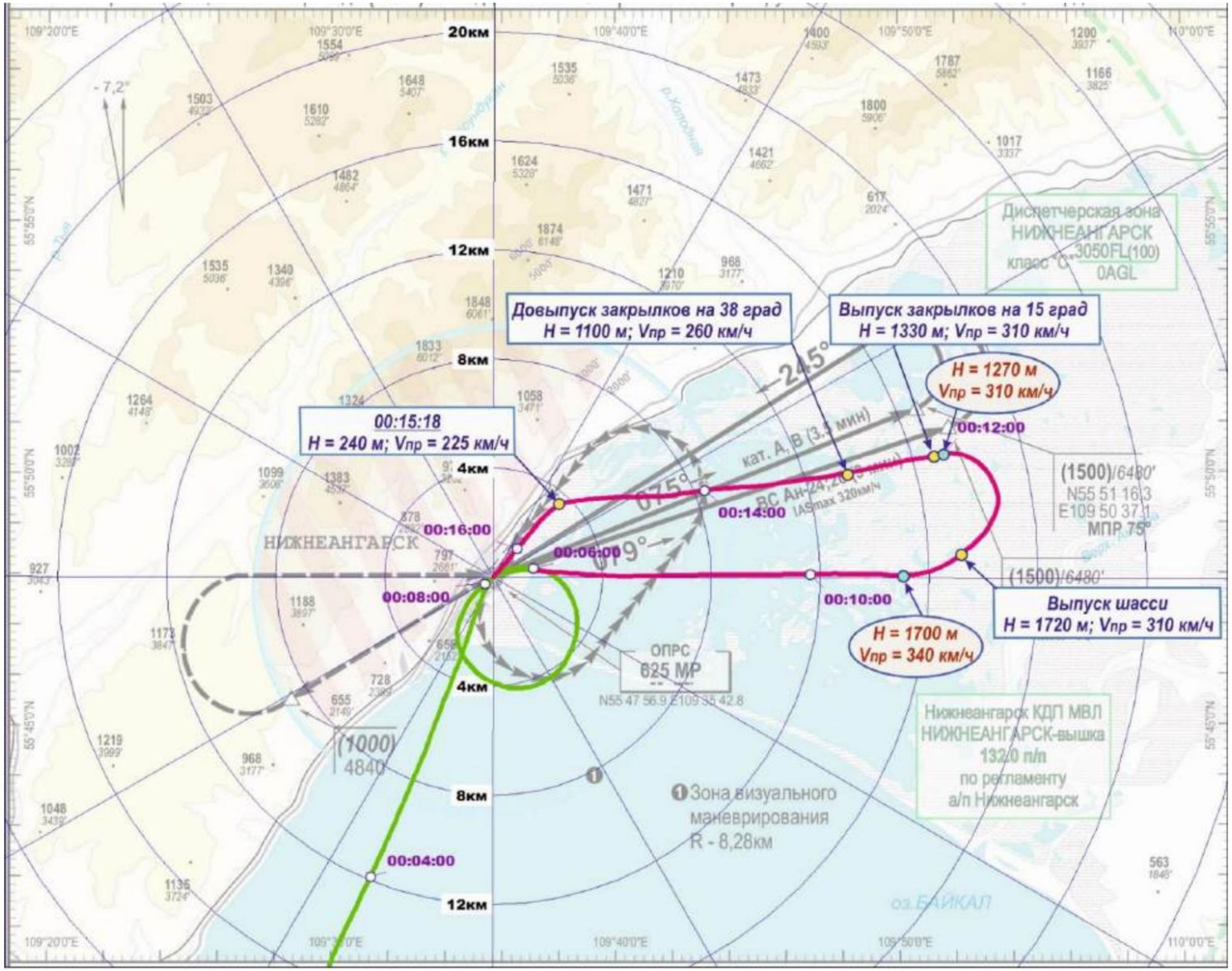


Fig. 33. Landing approach of An-24RV RA-47355, flight 09/03/2021 on the route: Ulan-Ude - Nizhneangarsk (1st stage of approach - green; 2nd stage of approach - red)

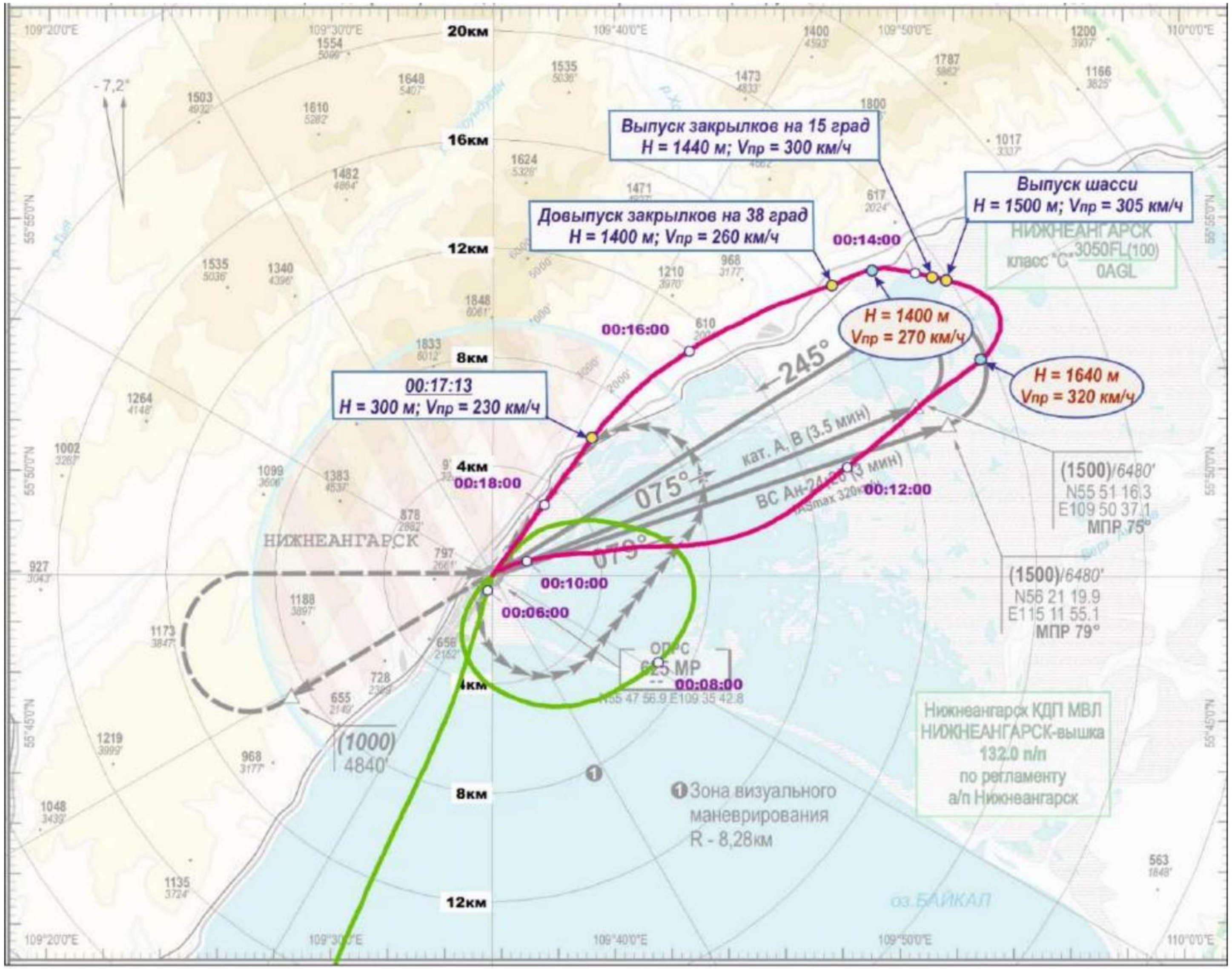


Fig. 34. Landing approach of An-24RV RA-47355, flight 10.09.2021 on the route: Irkutsk - Nizhneangarsk (1st stage of approach - green; 2nd stage of approach - red)

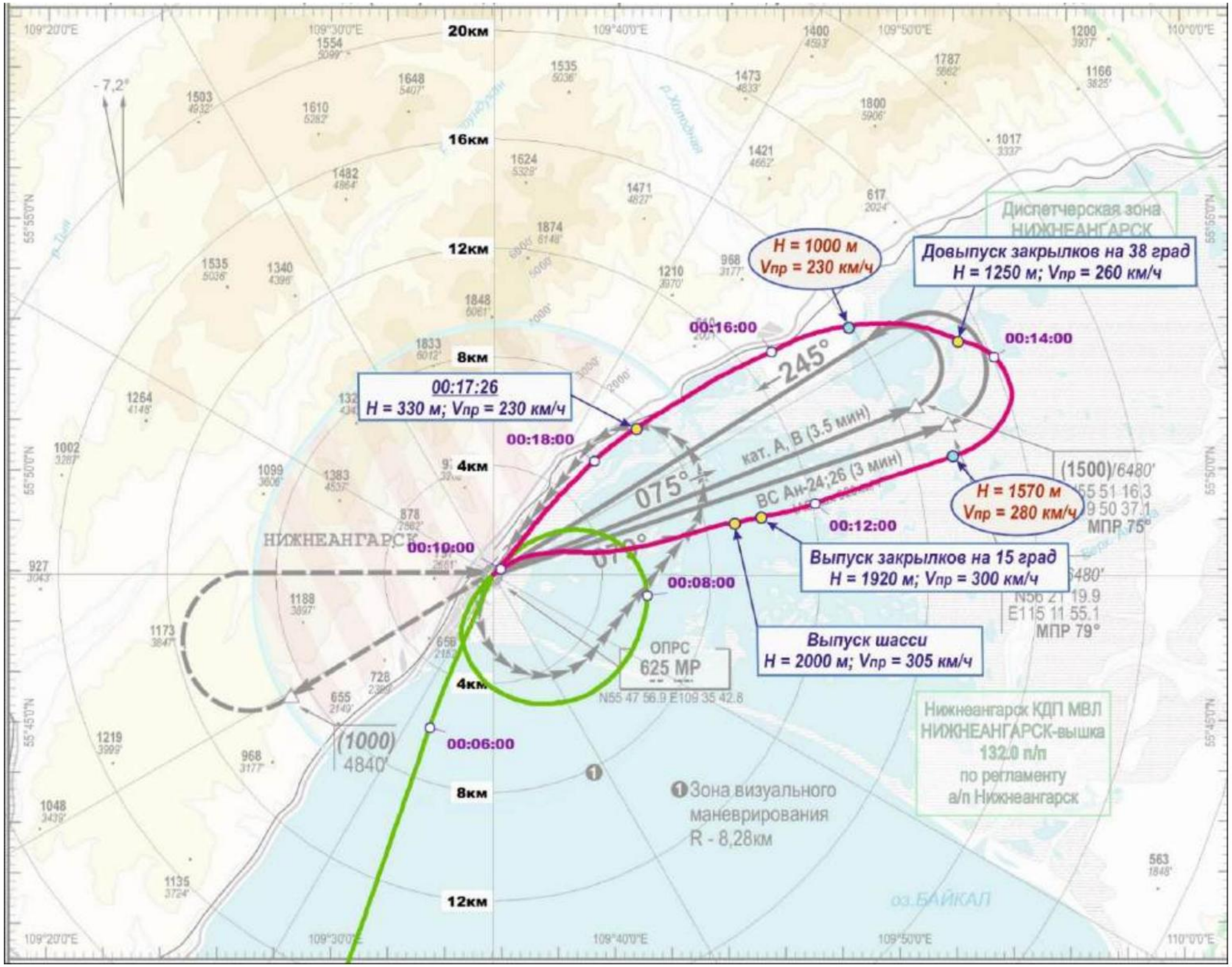


Fig. 35. Landing approach of An-24RV RA-47355, flight 10.09.2021 on the route: Ulan-Ude – Nizhneangarsk (1st stage of approach - green; 2nd stage of approach - red)

Based on the analysis results, the following main conclusions were made:

- in all flights, the first stage of the landing approach (the first overflight of the NRS and the flight in the waiting area) was carried out in accordance with the approach plan;
- In flights on 31.08.2021, further approaches were carried out in general in accordance with the approach pattern, including visual maneuvering (circle-to-land maneuver). In one of the flights, a paired turn was not carried out at a constant altitude of 1500 m, but with a descent to an altitude of 1100 m;
- in flights 09/03/21 and 09/10/21 the exit to the area of the third turn was carried out with a descent to altitudes of 1700...1570 m. Paired turns were performed not at a constant altitude of 1500 m, but with a decrease to altitudes of 1390...960 m exit to a course of 260° … 270°, and not 245°, as provided for in the diagram. After exiting the paired turns, the crews performed a maneuver visual approach, rather than re-entering the landing zone at altitude 640 m. The landing gear and flaps were extended at 15° before or during the paired turn, and the additional extension of the flaps to 38° was performed during the paired turn or after exiting it.

2. Analysis

2.1. Flight description and crew actions analysis

Description of the preparation of the crew and aircraft for flights and the execution of previous flights (flights AGU-99, AGU-199) are given in section 1.1. of this report.

This section analyzes the actions of the crew during the flight.

AGU-200 27.06.2019 on the route: Ulan-Ude - Nizhneangarsk, ending AP.

At 23:36 on June 26, 2019, in the premises of the Ulan-Ude AMGS, the crew familiarized themselves with meteorological information and received a package with cartographic information on flight route. Forecast and actual weather along the flight route, alternate airfields and at the destination were suitable for carrying out the flight mission.

The captain made a justified decision to take off.

There were 43 passengers on board the aircraft, including one child under 2 years old in her arms. The total weight of cargo, baggage and mail was 734 kg. Fuel load was 3000 kg, takeoff weight – 22405 kg, center of gravity – 25.0% MAC, which did not exceed the limits, installed RLEs of the An-24 (An-24RV) aircraft.

Note: *Flight manual of the An-24RV aircraft:*

Section 2.1. "Takeoff weight restrictions"

The maximum permissible take-off weight of the aircraft is 22,500 kg.

Section 2.2. "Limitations on centering"

Operational centering: maximum forward centering – 15% MAC

maximum rear center of gravity – 33% MAC

The parameters of the entire accident flight are shown in Fig. 36. At 01:03 on 27.06.2019 there was The aircraft began takeoff from the Ulan-Ude (Mukhino) airfield. The takeoff was performed in takeoff mode. operation of the engines and the nominal mode of the RU19A-300, with the flaps extended to 15°, which corresponded to the provisions of the flight manual of the An-24RV aircraft. Active piloting on The PIC carried out the entire flight. After takeoff, the mechanization and landing gear were removed. At a barometric altitude of approximately 4,000 m, the crew engaged the autopilot. roll and pitch. The climb was accomplished with the RU19A-300 engine running. The RU19A-300 engine was turned off after reaching flight level 170 (5200 m), which was busy at 01:20:50. The flight along the route (at flight level) was carried out with the autopilot engaged. at indicated airspeed (hereinafter referred to as speed) ÿ 340 km/h. When flying at the flight level of one-time commands and analog parameter values indicating abnormal operation, or There are no registered reports of aircraft failures.

[17] From here on, altitude values are given at standard pressure.

Machine Translated by Google

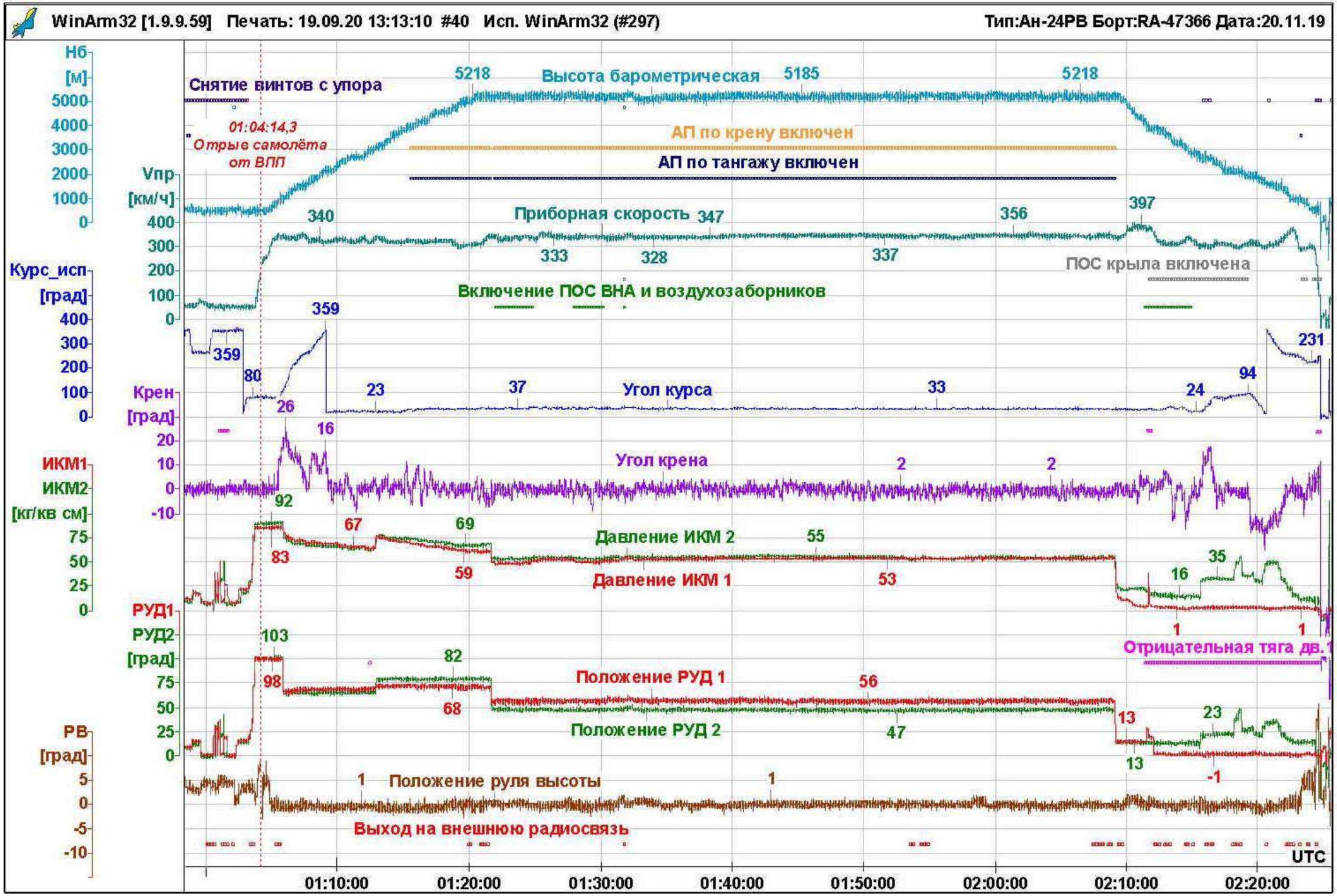


Fig. 36. Parameters of the emergency flight of the An-24RV RA-47366 aircraft

In the time interval from 01:22:10 to 01:30:16, while flying at cruising altitude, the crew twice included an anti-icing system for the inlet guide vanes and engine air intakes (duration: 2 min 45 sec and 2 min 06 sec accordingly). During the flight, the crew reports at the flight level The MS-61 onboard radio recorder did not register any icing. In the explanatory In the note, the second pilot noted that during the flight, *"visual icing was* observed *"It was not observed, the alarm did not go off, and the POS was activated when entering cloud cover."*

At 01:53:44, the 2P established contact with the air traffic control tower of the Nizhneangarsk airfield. (hereinafter referred to as the control tower dispatcher), reported the estimated time of arrival and requested weather conditions: *"Good morning, 366, arrival at your circuit in 16 minutes, please, "your weather."*

At 01:53:52 the control tower dispatcher transmitted the weather conditions at the airfield and the diagram to the crew sunset: *"47366, good morning, understood, Nizhneangarsk weather in one hour thirty: wind seventy degrees, two, visibility more than 10, clouds insignificant cumulonimbus- Rain at 1500, temperature 20 degrees, QFE 713 mm, strip 05 dry, grip 0.6. Calculate approach to runway 05 using the NDS, then visual maneuver to runway 05. "Report entering the zone."*

2P confirmed receipt of the information and transmitted the aircraft loading and required refueling to the control tower dispatcher: *"We have 42 passengers[18] on board, 428 kilograms of baggage, 218 cargo, 88 mail, the maximum refueling required is 4600."* The control tower dispatcher confirmed obtaining information.

At 01:55:12 the captain said: *"So, how are we going to approach according to the plan, in…"* and *"225 or 05"?* There was no response from the remaining crew members to the on-board tape recorder recordings. registered.

Note: *The on-board tape recorder records the conversations of the crew members, which are conducted with the help of the SPU. Registration of negotiations without The use of SPU is not provided.*

At 01:58:48 the flight engineer reported to the crew: *"Fuel consumption per hour was 1100 kilograms, instantaneous fuel consumption (seven hundred) thirty, remaining fuel two tons. The fuel system has been checked and is in good working order. The electrical system has been checked and is in good working order. "Objective control means are operational."*

[18] The crew reported the number of passengers without taking into account a child under 2 years of age.

At 01:59:40 the PIC gave the command to the crew to conduct pre-landing preparations: *"The crew is to begin pre-landing preparations, landing airport Nizhneangarsk, Landing course 45 degrees."* Thus, the crew was preparing to land on runway 05.

At 01:59:48 the flight engineer reported that *"… the automatic braking system included…".*

Note: *An-24RV Aircraft Flight Manual. Section 7.6.3. "Landing Gear Operation in Flight"*

p. 2. The brake wheels are equipped with inertial sensors, which together with the automatic braking system are intended to prevent wheel skidding when braking. Turning on the system is performed by the "AUTO. WHEEL BRAKING" switch, located on the central instrument panel (at the bottom). There There are also two yellow warning lights installed, which flash indicates the activation of inertial sensors.

The "AUTOMATIC WHEEL BRAKING" switch is turned on before taxiing, and turning off - after the aircraft has taxied to parking lot.

At 02:00:00 the PIC informed the crew: *"Crew, attention, the actual weather at The Nizhneangarsk landing airfield is above minimum. The temperature is plus 20, the pressure is 7-1-3, At the alternate airfield in Irkutsk, the actual weather is above the minimum, landing approach OPRS, minimum 640 to 219, (illegible) after visual maneuvering... Fuel on VPR on two hours. The approach with flaps will be 38 degrees. The descent rate is 210, in case (increased shift) 220. Throttle lever latches – first position 12 degrees. In case go-around on a straight line, climb to 1000 meters, right turn to drive with climb 2100, then according to the approach scheme.*

Landing approach. On concrete. Clean, dry, friction coefficient 0.6, brakes on. During landing, the commander uses it, after lowering the front strut, remove the screws from the stop, with I'm braking with information to the crew. Active control on the left, controlling, leading radio communications on the right, flight engineer switches radio communication channels. Other features "No. The airfield is in the mountains. We strictly adhere to the plan. Control by map."

Next, the crew completed the section of the checklist "BEFORE "DESCENT FROM FLIGHT LEVELS."

It should be noted that the on-board tape recorder did not record any reports from crew members on the calculation of the required landing distance for runway 05.

[19] As stated in section 1.10, the minimum established by the airline's RPP for this type of approach is 640 x 5000 m.

According to nomogram 6.41 of the An-24 aircraft flight manual, the required landing distance with the aircraft landing weight of 20,850 kg, headwind of 3 m/s, with flaps extended position 38°, was 1240 m (Fig. 37), that is, it did not exceed the available landing distance (1503 m).

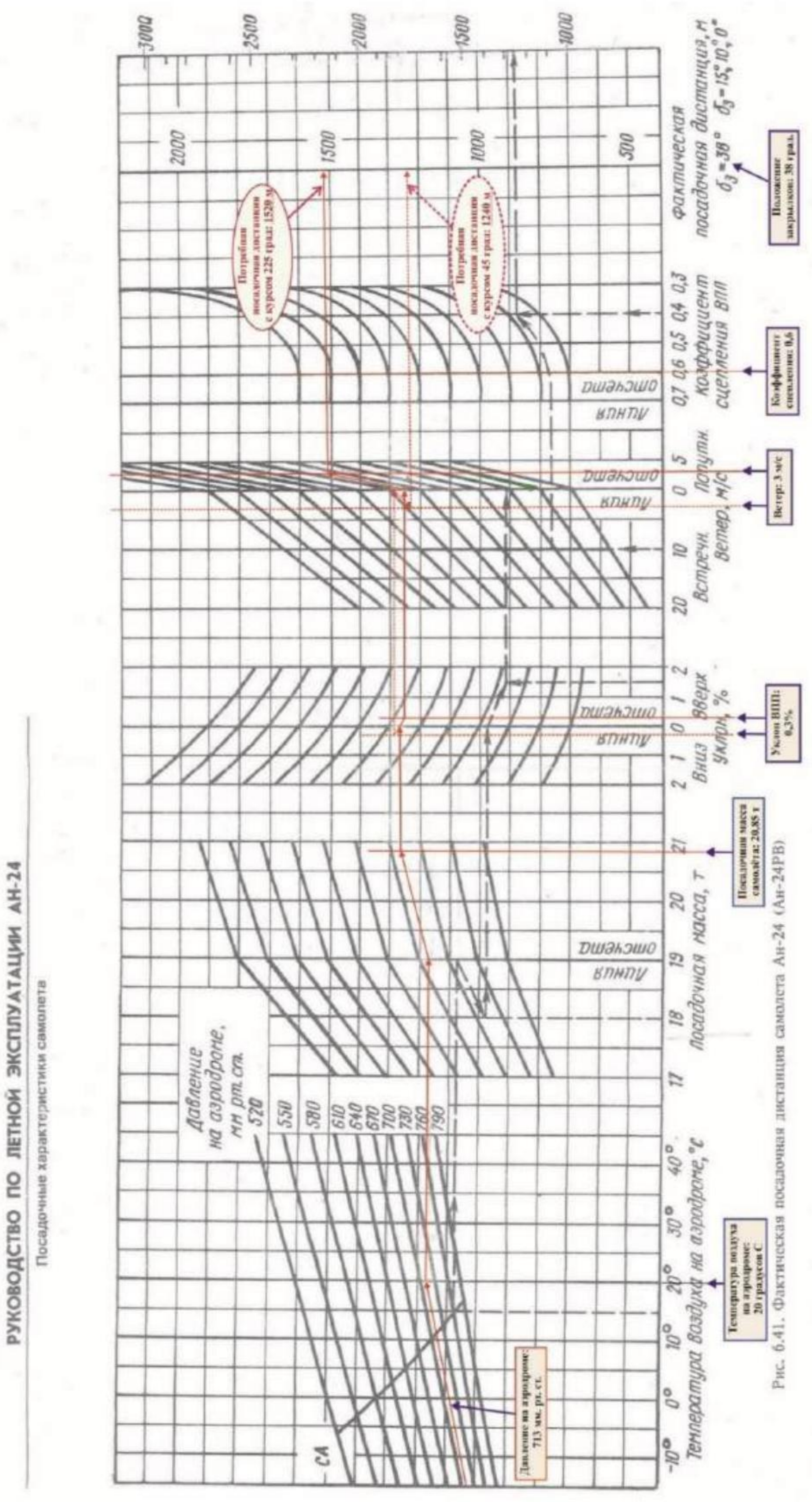


Fig. 37. Nomogram 6.41 of the An-24RV aircraft flight manual. Calculation of the actual landing distance of the aircraft

At 02:07:57 2P contacted the control tower dispatcher, reported the flight conditions and requested an approach with a heading of 225°: *“Nizhneangarsk-Tower, good morning again, 46366, flight level sixty-four...line 64, flight level one hundred seventy. Weather information I have, QFE seven one three millimeters, entry of the OPRS, allow according to the scheme with the course "twenty...two hundred twenty-five hours."*

According to 2P[20], the PIC chose MKpos = 225° due to the landing distance could have been 200 m longer, since the end of runway 05 was previously moved to this value to the center of the runway.

Note: *According to the information provided by the Armed Forces of the Ministry of Transport of the Russian Federation to the Federal Air Transport Agency, in 2001, after inspecting the Nizhneangarsk airfield for compliance requirements of the NGEA, due to the unsatisfactory condition artificial pavement, the end of runway 05 was moved by 200 m. At the moment of the accident, there was a strip free of debris behind the exit threshold of runway 23. obstacles, 250 m long with artificial surface, on the other The length of this strip was 150 m, the surface was dirt.*

It should be noted that, based on the latest meteorological information that was available the crew, when landing with MK = 225° the tailwind component did not exceed 3 m/s and did not exceeded the limits. According to the flight manual of the An-24 (An-24RV) aircraft, the maximum tailwind component during landing - up to 5 m/s.

At 02:08:19 the control tower controller instructed the crew to proceed at flight level 170, Estimated descent, approaching flight level 100 (3050 m): *“47366, good morning. Milestone 54 confirmed, identified, proceed to flight level 170, descend as calculated, on approach "take flight level 100."* The control tower controller's response did not contain information about the approach course. for boarding.

At 02:08:37 2P reported: *“366, this is the beginning, calculation of the beginning of the descent, to the drive flight level 100, and... with a course... according to pattern 225 with a landing course of 45."* The presented phrase is not gives a clear understanding of the heading at which the crew requested landing this time.

In response, the controller cleared the approach to landing on runway 23 (MKpos = 225°) and transmitted Meteorological conditions at the airfield: *“366, understood, calculate approach to runway 23. Descend to the OPRS, flight level 100. Nizhneangarsk weather at 02:00: wind 50 degrees 3, visibility over 10, slight cumulonimbus clouds at 1500, temperature 20 degrees, QFE 713 millimetres, the strip will be 23 for you, dry, grip 06".*

[20] The on-board recorder did not record any discussion of this issue. It is possible that the discussion took place without the use of SPU.

The second pilot confirmed receipt of the meteorological information and began again explain the planned approach pattern to the dispatcher: *"clear approach pattern two hundred twenty...",* but the PIC interrupted this message, giving an emotional command to the second pilot: *"I gave him two hundred… 23, 23 we'll be coming in!"*

The commission notes that when changing the landing course, the PIC additionally did not carry out pre-landing preparation, calculation of the required landing distance on runway 23 for the existing conditions, the crew also did not produce it.

Note: paragraph 3.78 of FAP-128:

> *"When changing the runway, landing course or when conditions arise that require changes to previously adopted decisions must be made by the aircraft crew additional preparation and re-testing should be carried out "completed operations."*

According to nomogram 6.41 of the An-24 aircraft flight manual, the required landing distance with the aircraft landing weight of 20,850 kg, with the flaps extended to the 38° position, and with a tailwind of 3 m/s was 1520 m (Fig. 37), that is, it slightly exceeded available landing distance (1503 m), which, according to formal criteria, is not allowed the crew to land on runway 23.

At 02:09:07 the PIC gave the command: *"Low flight."* At 02:09:10 it was switched off. autopilot, after which the left and right engine throttles were moved to the position flight idle, and the aircraft was put into descent. At 02:09:51, the flight mechanic reported: *"Autopilot power is off, flight idle throttle is set,"*

The rest of the flight was carried out with the autopilot turned off.

At 02:09:54 the PIC informed the crew: *"…we are descending, vertical 6 meters."*

At 02:09:58 the PIC informed the crew that a descent to level 100: *"Assigned level 100, altitude three thousand fifty."* The second pilot and The flight engineer confirmed this information: at 02:10:02 2P: *"Echelon one hundred, three thousand fifty",* at 02:10:05 flight mechanic: *"We are descending to a speed of three thousand."*

At 02:10:43 the co-pilot reported: *"Departure 40, track 485."*

At 02:11:15 the captain gave the command to turn on the anti-aircraft fire alarm: *"Turn on anti-aircraft fire alarm completely."*

At 02:11:27, 12 seconds after the command to turn on the POS, at an altitude of 4060 m there was the anti-icing system of the inlet guide vanes is turned on and engine air intakes, almost simultaneously with the activation of the POS, MSRP-12-96 registered a one-time command: *"Negative thrust of the left engine."*

Note: *One-time command "Left engine negative thrust"*

> *registered when a signal is received from the alarm device*

negative thrust of the left engine SDU-5-2.5 with simultaneous

the left engine's "KFL" lamp comes on.

At 02:11:33 the flight engineer moved the left engine throttle from the PMG position to position 29°, the oil pressure in the PCM system of the left engine has not changed.

At 02:11:38 the PIC addressed the crew with the question: *"Our engine is working, No?"*

The flight parameters and trajectory with radio traffic applied are shown in Fig. 38, Fig. 39.

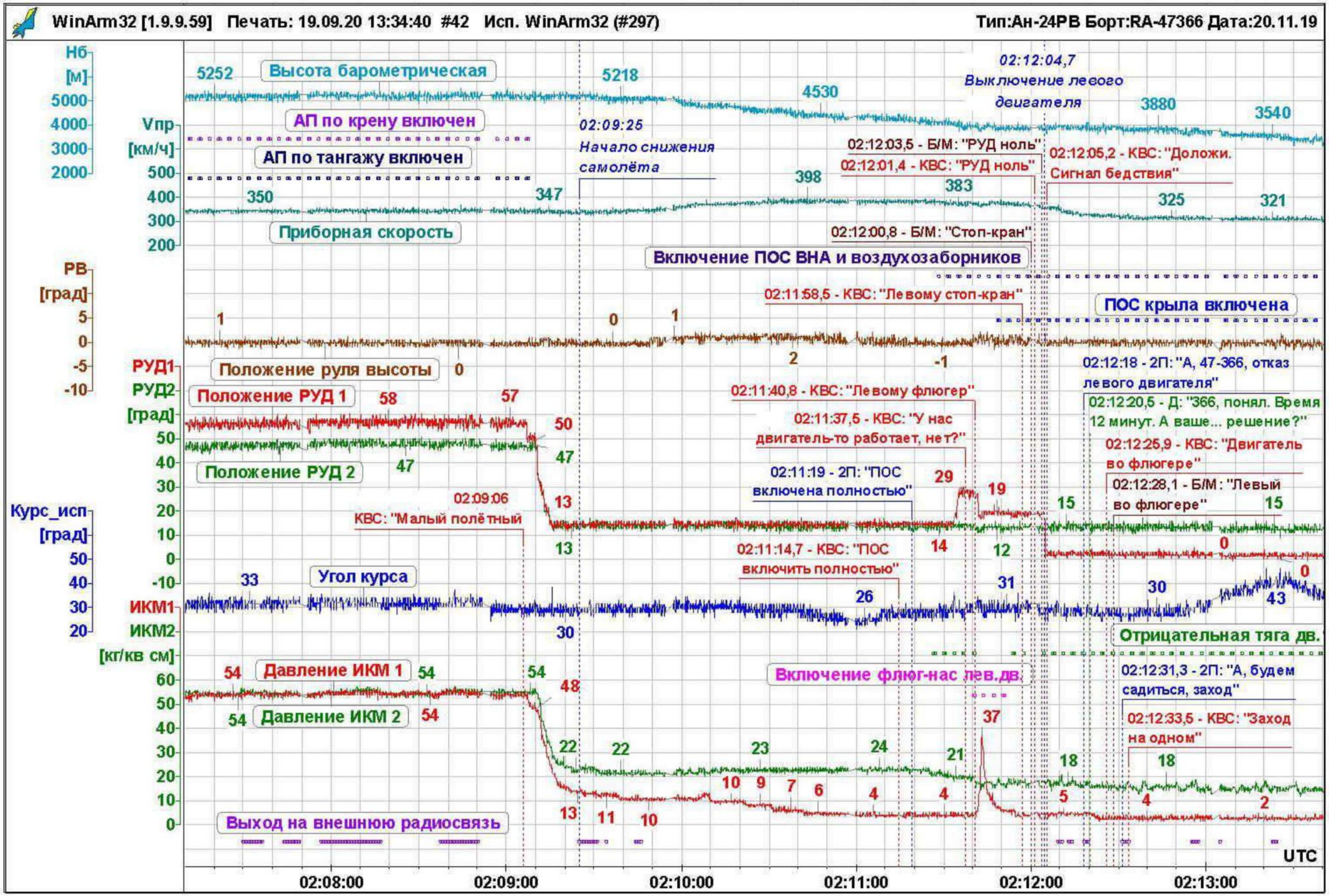


Fig. 38. Flight parameters during descent and failure of the left engine

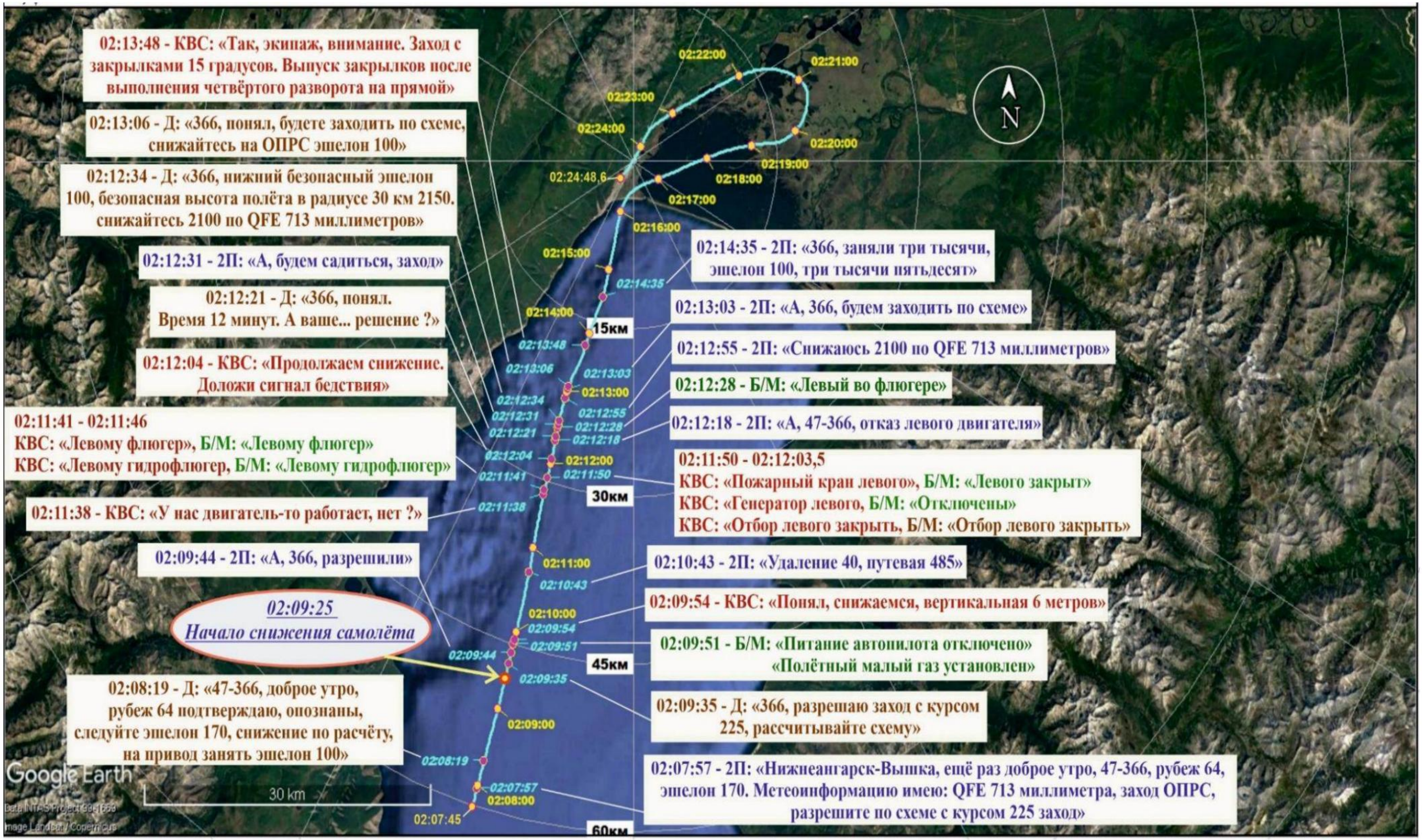


Fig. 39. Flight trajectory of the An-24RV RA-47366 aircraft from the start of descent from flight level 170

Moving the throttle to a mode greater than 28° in the presence of a signal from the alarm negative draft, led to the removal of the lock and automatic activation left engine propellant feathering systems.

Note: *RLE of An-24 aircraft (An-24RV):*

"5.1. Engine failure.

5.1.2. Crew actions in case of engine failure.

Notes: 1. In case of engine failure at modes less than (26±2)° according to the UPRT on aircraft equipped with a negative wind auto-feathering system thrust, moving the throttle of the failed engine to a position greater than 28° The UPRT will lead to a slowdown in 5–7 *s (system slowdown time) automatic feathering of the propeller from the negative pressure sensor "traction."*

At 02:11:41, automatic feathering of explosives was initiated, a one-time command was given *"Turning on the left engine vane pump."*

The crew duplicated the actions of feathering the explosives, the flight engineer reported closing the emergency stop valve and setting the left engine throttle to "0". Crew actions when In case of engine failure in flight, the provisions of the flight manual of the An-24RV aircraft were met.

After the engine stopped, the crew made no attempts to start it.

Note: *paragraph 5.1.10 of the An-24RV aircraft flight manual "Stopping and starting the engine in flight".*

"It is prohibited to start a faulty engine that has been stopped in flight automatically or forcibly by the crew."

In case of engine failure during descent, the crew, in accordance with paragraph 8.5.5. of the An-24RV aircraft flight manual, should have started the RU19A-300 engine, but this was not accomplished. The crew The possibility of launching RU19A-300 was not discussed.

Note: *p. 8.5.5. RLE aircraft An-24(An-24RV).*

"Start the RU19A-300 engine and set it to the required mode."

At 02:12:05 the PIC informed the crew: *"We continue to descend, signal distress." After the crew turned on the "Distress" signal, the control room screen a radar signal appeared near the An-24RV RA-47366 aircraft marker disasters (letters BD in red).*

At 02:12:09 2P reported to the control tower dispatcher about the engine failure: *"A, 47366, failure "engine."* The captain clarified: *"Failure of the left engine."*

At 02:12:15 the control tower dispatcher asked for clarification of the information: *"366, nep… "Repeat."*

At 02:12:18, 2P reported to the control tower: *"47366, left engine failure."*

At 02:12:21 the control tower dispatcher confirmed receipt of the information and requested a decision Crew: *"366, understood. Time is 12 minutes. And your... decision?"*

At 02:12:26 the PIC clarified with the flight engineer the fact of feathering of the left engine and decided to land at Nizhneangarsk airfield on one engine.

calculations of the required landing distances to perform a landing with a failed engine, taking into account the actual weather conditions and the landing weight of the aircraft, the crew did not produced.

It should be noted that in the flight manual of the An-24 aircraft, when calculating the required landing distances (according to the nomogram in Fig. 6.41) no distinction is made based on the number of workers engines, when landing, only the landing configuration of the aircraft is taken into account (flaps position). In this case, according to the flight manual, if one engine fails, the approach performed with flaps deflected by 15° (see also below).

Results of calculations of the required landing distance for actual conditions landing of the aircraft, landing weight of 20850 kg and flap position of 15°, performed commission and the Antonov State Enterprise showed that for runway 05 it was 1760 m, and for runway 23 – 2160 m, which did not allow the crew to land (Fig. 40). Thus, taking into account indicated above, the required landing distance did not exceed that available only when landing with attitude control = 45° with flaps extended to position 38°.

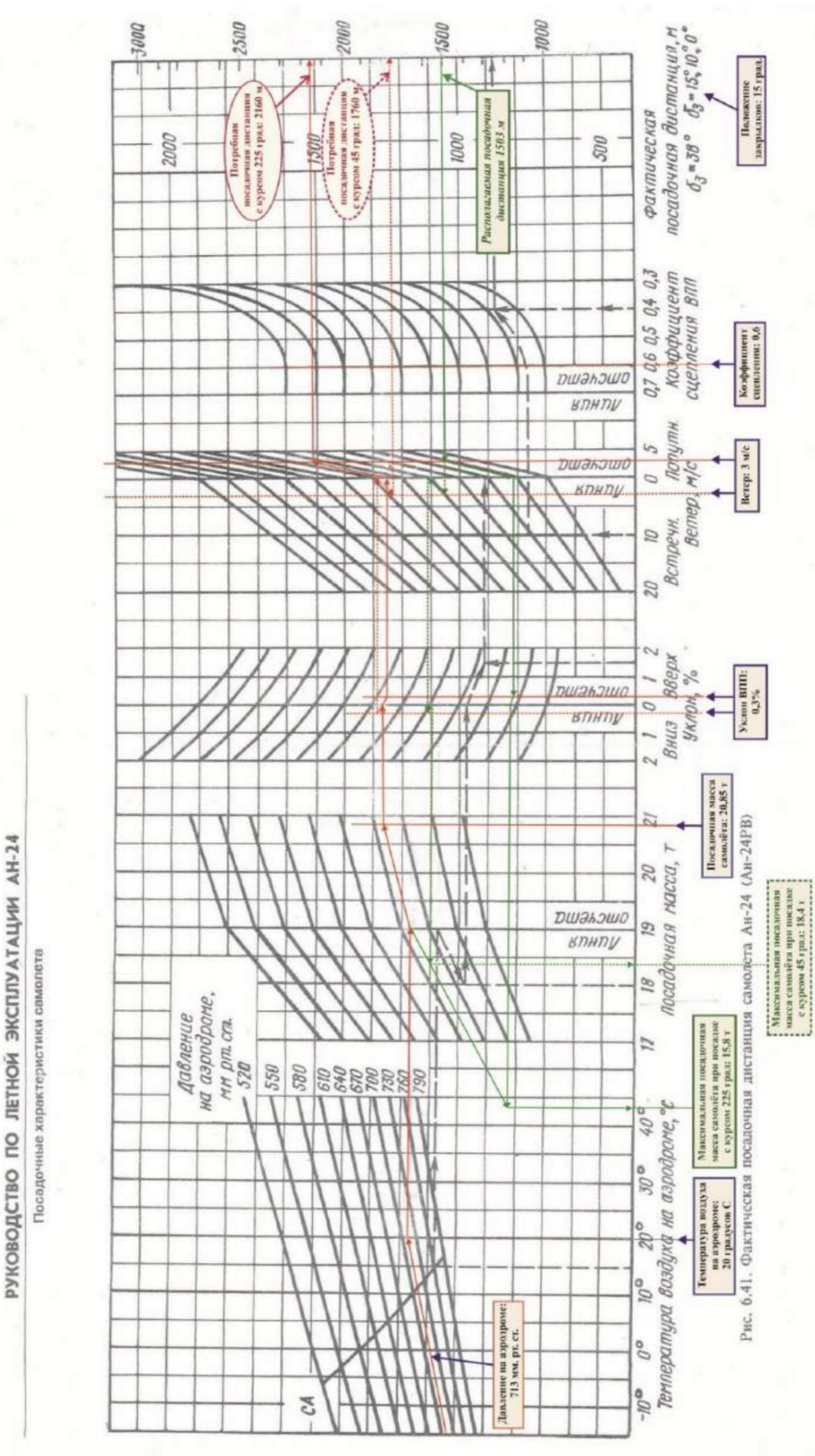


Fig. 40. Nomogram 6.41 of the An-24RV aircraft flight manual. Calculation of the required landing distance of the aircraft

In accordance with schedule 6.41 of the An-24 aircraft flight manual, with the available landing distance of the runway at the Nizhneangarsk airfield is 1503 m maximum landing the mass of the An-24RV aircraft with flaps extended to 15°, when landing at a heading of 225° (with tailwind component) should not exceed 15.8 tons, and when landing at a course of 45° (from headwind component) 18.4 t.

In this case (given the actual weight of the aircraft), the crew had to make a decision or to carry out a landing on a heading of 45° with flaps extended at 38°, or about leaving for the alternate airfield Irkutsk, located at a distance of ~ 500 km from the place where the aircraft's right engine failed.

Note: *1. FAP-128 p. 5.41.*

"Diversion to an alternate airfield must ensure the flight within 30 minutes in the holding area at an altitude of 450 m."

2. Engineering and navigation calculation of a flight to an alternate airfield Irkutsk at FL120 with the right engine running at nominal mode and with the RU19A-300 engine turned on, taking into account paragraph 5.41 of FAP-128 stated above showed that the remaining fuel after landing would be ~ 300 kg.

Discussion of the option of leaving for an alternate airfield using the onboard tape recorder is not registered.

At 02:12:33 the PIC informed the crew and the tower controller: *"Approach on one."*

At 02:12:34 the control tower controller informed the crew of the minimum safe altitude and cleared the descent to an altitude of 21 2100 m: *"366, safe altitude, lower safe flight level 100, safe flight altitude within a radius of 300 kilometers... 30 kilometers, 2150. Approach from course... descend two thousand one hundred QFE, 713 millimeters."*

At 02:12:55 2P responded to the control tower dispatcher: *"Descending 2100 on QFE "713 millimeters."*

At 02:13:00 the control tower controller announced the "Alarm" signal at the airfield.

At 02:13:03, the 2nd Flight Commander, on command from the PIC, clarified for the control tower dispatcher: *"366, we will be approaching according to the scheme."*

At 02:13:06 the controller gave permission for approach according to the pattern and descent *to the NRS level 100",* which corresponded to the standard arrival route scheme MOPON 1C (Fig. 4) The co-pilot issued a receipt: "366, according to the diagram, 210022."

[21] From here on, the QFE height values are given.
[22] See also Section 1.10, Fig. 4-9.

At 02:13:48 the PIC informed the crew: *“Okay, crew, attention. Approach from flaps 15 degrees. Release flaps after completing the fourth turn on direct".*

Note: *paragraph 5.1.7 of the An-24RV aircraft flight manual “Approach and landing with "feathered propeller of a failed engine."*

1. The landing approach is permitted to be carried out both in the direction of the operating, and a failed engine, with a bank angle of no more than 15°.

2. The PIC selects the approach method. It is recommended to perform the approach using the rectangular route method.

3. The 4th turn is performed without descending at the height of the rectangular route at a speed of 260 km/ h with retracted landing gear and flaps. After 4 turns before entering the glide path, release the landing gear and Deflect flaps by 15°. Maintain a speed of 220° on the glide path. 240 km/ h depending on flight weight.

According to Table 6.1 of the aircraft flight manual, with an aircraft landing weight of 19,000 kg maintain speed while gliding with flaps extended position 15°, before the start of alignment 230 km/ h, at landing weight 21,000 kg – 240 km/ h.

At 02:14:07 the PIC informed the crew: *“Vertical speed 4 meters… The descent speed on a straight line is 200...40 kilometers per hour."*

At 02:14:34, at a distance of about 13 km from the OPRS 2P, on command from the PIC, he reported to the control tower dispatcher about occupying FL100: *“366, we have occupied three thousand, FL100, three "thousand fifty."*

At 02:14:45 the control tower controller cleared the approach according to the NPR and descent to 2100 m: *“366, Direct bearing 218, OPR approach cleared, transition level 100, QFE 713 mm descend according to the two thousand one hundred meters pattern to the OPR."* The crew confirmed receipt information.

Note: *According to the approach chart for the runway 23 NDS (Fig. 6), the crew must was to reach the OPR at FL 100, after which in the holding area make a right turn to descend to an altitude of 2100 m by pressure aerodrome (QFE), then perform an approach according to the established scheme.*

Analysis of the trajectory of the further flight showed that the crew exited the OPR and The descent in the holding area was not carried out. The controller brought this to the crew's attention. The deviation from the landing approach pattern was not noticed. The analysis showed that other airline crews were making their way to the holding area and flying in the holding zone (section 1.18.3).

The Commission notes that such a decision by the crew at this stage did not create direct risks to flight safety, although it reduced the time margin for mitigation available energy of the aircraft (speed reduction and damping). At the same time, based on general philosophy of constructing and executing landing approach procedures for one NPR, its a span with an installed MPU is mandatory, since only at this moment time the crew can accurately determine their position on the approach pattern using the standard instrumentation of the aircraft.

At 02:14:56 the crew turned off the de-icing system of the inlet guide vanes and air intakes.

At 02:15:09, on command from the PIC, the 2nd pilot and the flight engineer installed and reported the installation The airfield altimeters show 713 millimeters of pressure.

At 02:15:18 the PIC transferred control to 2P, set the airfield pressure on his altimeter and took control again.

At 02:15:29 the PIC gave the order to complete the section of the checklist "AFTER TRANSITION TO AIRFIELD PRESSURE": *"We are descending 2100. Control by cards.*

From 02:15:37 to 02:16:03 the crew completed this section of the checklist checks in full.

At 02:16:00, on the command of the PIC, the 2nd pilot reported to the control tower: *"47366, at flight level transition, QFE 713 installed, descending, took 2100."*

At 02:16:16 the control tower controller confirmed receipt of the information and cleared the descent. 1500 m: *"366, OPR confirm, continue approach, on ... descend according to the scheme to paired 1500 meters."* At 02:16:32 2P confirmed receipt of the information: *"To "paired 1500, 366".*

At 02:16:33, at an altitude of 2100 m, approximately 3 km before reaching the landing strip, the PIC began right turn to bring the aircraft onto a course of 79°, which was announced to the crew: *"Course 79 degrees, descending 1500."*

The flight path[23] of the aircraft from the moment of approach to the NRS is shown in Fig. 41 and Fig. 42. Flight parameters are shown in Fig. 43.

[23] The aircraft flight trajectory was constructed using information recorded by MSRP-12-96.

Machine Translated by Google

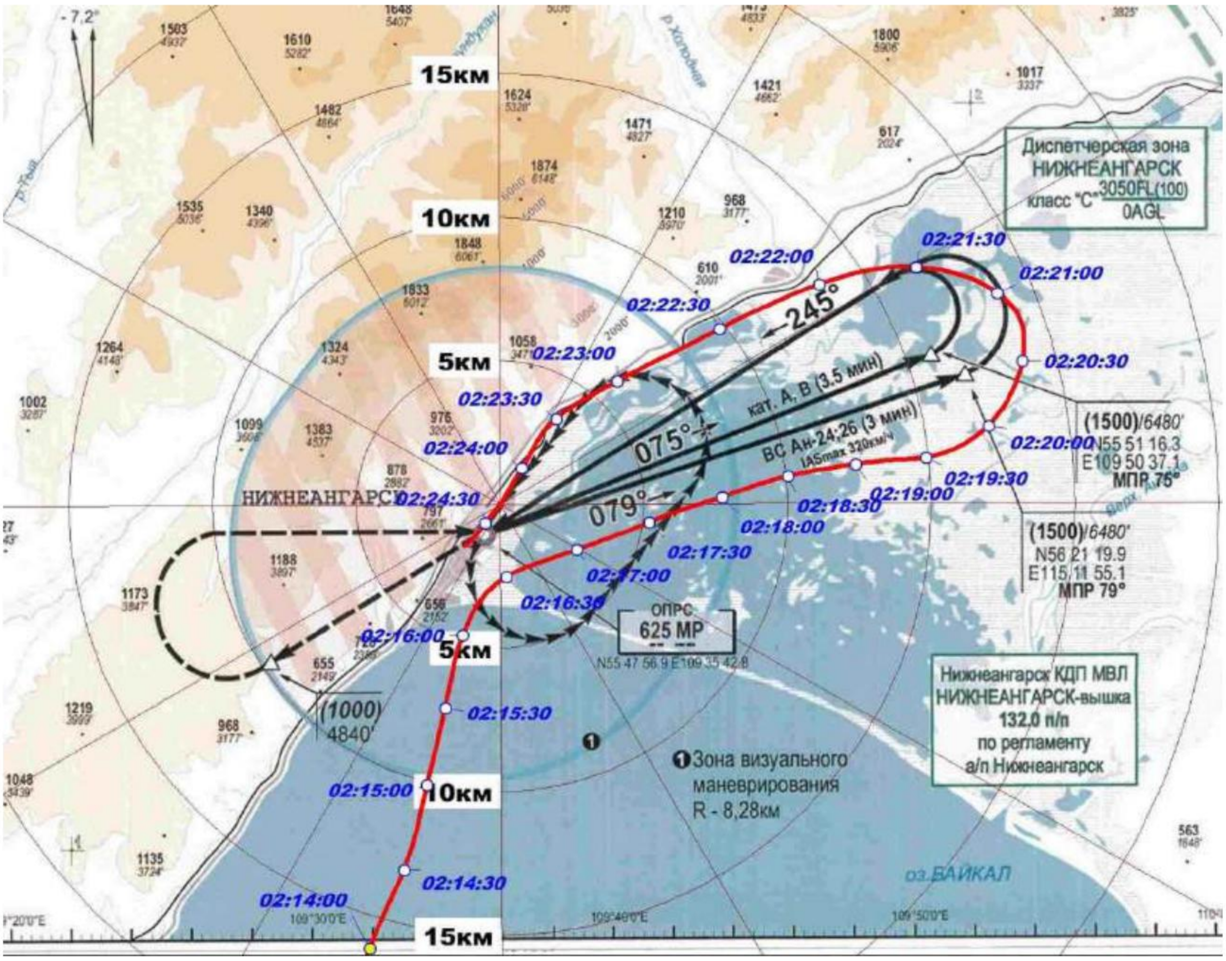


Fig. 41. Flight path of an aircraft when entering the airfield area and approaching for landing

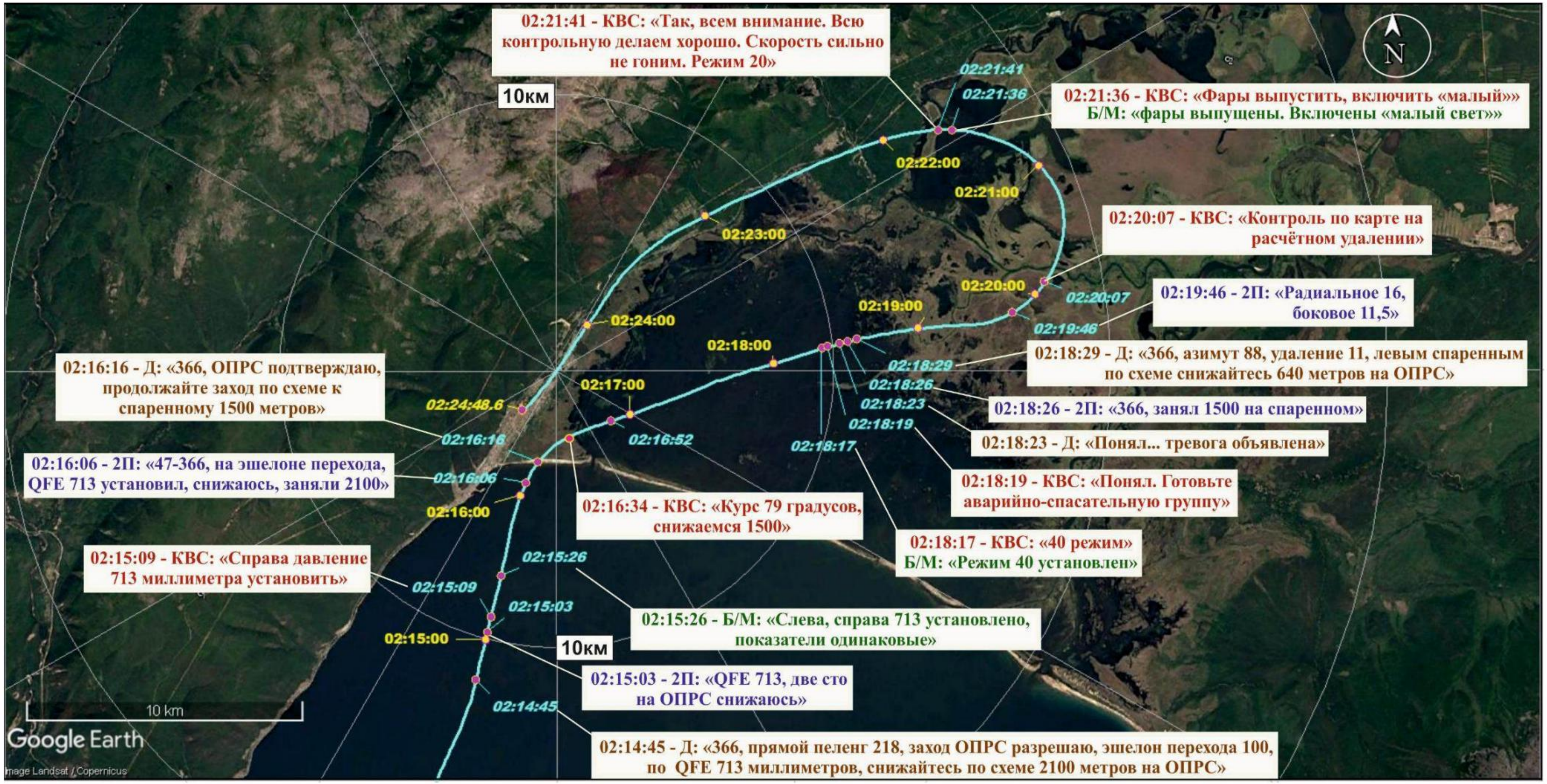


Fig. 42. Flight path of an aircraft when entering the airfield area and approaching for landing.

Machine Translated by Google

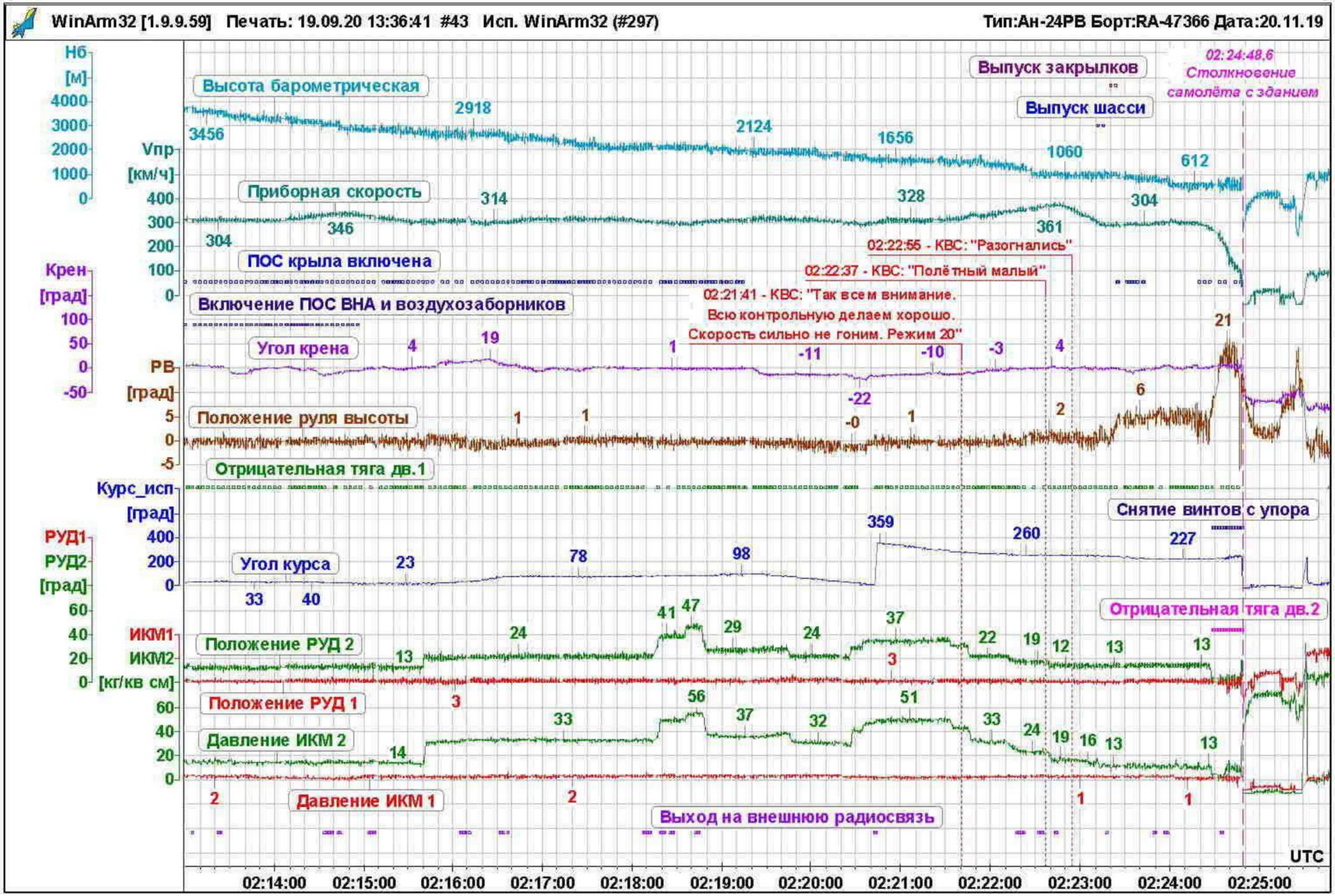


Fig. 43. Aircraft flight parameters when entering the airfield area and approaching for landing

At 02:17:03 the PIC informed the crew: *“Crew, attention. If there is Confidence in a safe landing, we will release the flaps to 30 degrees at altitude “Not less than…100 meters.”* The crew confirmed receipt of the information.

It should be noted that the flight manual for the An-24 (An-24RV) aircraft contains a warning that with flaps extended to 38°, go-around with one engine running impossible.

Note: *Flight Manual for the An-24 (An-24RV) aircraft, 5.1. Engine failure, 5.1.7. Approach landing and landing with one engine failed:*

“4. After the final decision has been made to carry out the landing and clarification of the calculation is permitted in visual flight in the area of the mid-range marker Extend the flaps to 38°. If you are unsure about the landing calculation to be carried out with the flaps deflected by 15°, since the go to the second "A circle with flaps deflected at 38° is impossible."

At 02:17:56 the control tower dispatcher asked the crew about the presence of dangerous goods on board the aircraft: *"Ah, 47366, report... dangerous goods."* At 02:18:08 the captain responded: *"47366, dangerous goods there is no… on board”.*

At 02:18:17 the PIC gave the command to the flight mechanic: *"Mode 40."* At 02:18:18 the flight mechanic reported the installation of the mode: *“Mode 40 is installed.”*

At 02:18:20 the PIC informed the control tower dispatcher: *“... Prepare the emergency rescue group."* At 02:18:23 the control tower controller responded to the announcement of a signal at the airfield "Alarm": *"Understood, Alarm has been declared."*

At 02:18:26 2P reported reaching an altitude of 1500 m (the actual altitude is almost corresponded) and about its location: *“366, took 1500, in paired.”*

At 02:18:29 the control tower controller cleared a left turn to the landing zone and a descent to altitude 640 m: *“366, azimuth 88, distance 11, descending left paired according to the pattern, descend 640 meters on the OPRS."* The crew confirmed receipt of the information.

Note: *According to the approach chart for runway 23, left paired turn on course 245° is carried out without descending, at an altitude of 1500 m. After after completing the turn, a descent is made to the altitude of the landing zone 640 m.*

At 02:18:45, the captain gave the command to the flight engineer: *"Mode 28."* The flight engineer reported: *“I’m reducing the power,”* after which the crew continued their descent.

At 02:19:14 the crew switched off the wing's anti-skid system, as evidenced by the cessation of registration of a one-time command.

At 02:19:19, at the request of the captain, 2P reported: *“Radial 14, lateral 10.”*

Machine Translated by Google

At 02:19:20 the PIC began to perform a left turn and informed the crew: *"Got it, let's go. It'll carry you 3 kilometers, the list will be small, no more than 15, with one engine."*

At 02:19:43, the captain gave the command to the flight engineer: *"Mode 20."* The flight engineer reported: *"20 installed."*

At 02:20:07 the PIC gave the command to carry out the section of the checklist "ON "ESTIMATED DISTANCE." The map section was completed in full, including: the front wheel switch is set to the *"Take-off-landing" position,* the setter The dangerous altitude on the RV is set at VPR - 50 m. The flight engineer reported that the ice missing on the wing and stabilizer.

At 02:20:25, the captain gave the command to the flight engineer: *"Mode 30."* The flight engineer reported: *"30 installed."*

During the turn, 2P periodically reported the radial direction to the PIC. distance to the entrance threshold of runway 23 and lateral deviation from the extended center line Runway.

At 02:21:36 the PIC gave the command to the flight mechanic: *"Extend headlights, turn on low."* The flight mechanic reported: *"Headlights extended. Low beams on."*

At 02:21:41 the PIC informed the crew: *"Okay, everyone's attention. The entire control* We're doing well. We're not pushing the speed too much. The flight engineer set the mode *to 20.* the running engine at 20° and reported: *"Mode 20 set."* Despite the decrease engine operating mode due to the aircraft's speed decreasing from 300-310 km/h began to grow.

The flight path of the aircraft with the radio traffic superimposed at this stage of the flight is shown in Fig. 44.

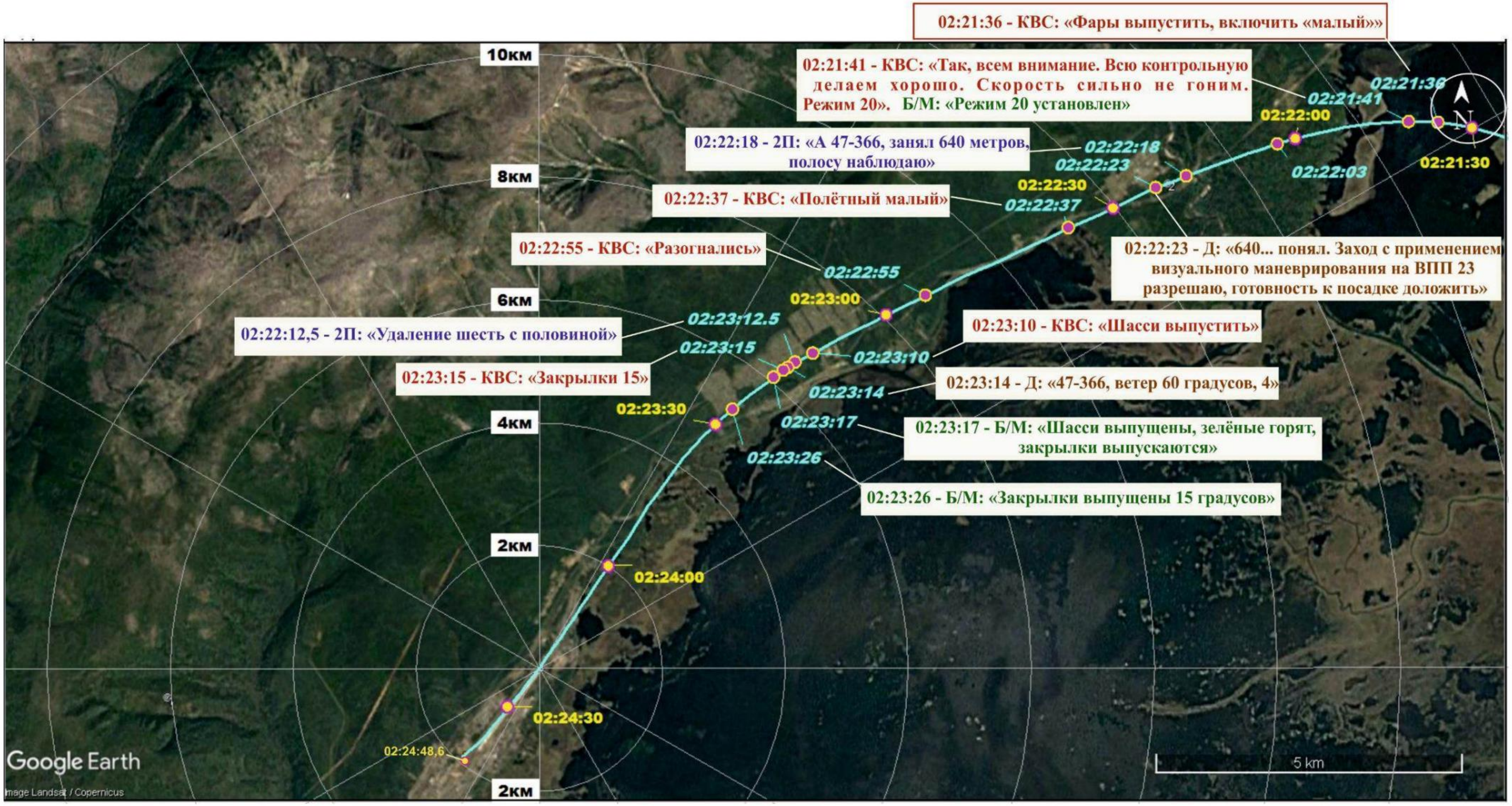


Fig. 44. Aircraft flight path during visual approach to landing.

At 02:22:09 the PIC gave the command to the flight engineer: *“Mode 20, so let it stay at 18.”* The flight engineer reported: *“Set to 18.”*

At 02:22:14, the captain gave the flight engineer the command: *"Mode 16."* The flight engineer reported: *"Installed."*

At 02:22:18, at a distance of about 11.5 km from the runway threshold, 2P, on the command of the PIC, reported to the control tower dispatcher about reaching an altitude of 640 m (at that moment the aircraft was descending at approximately 100-150 m above this altitude) and visual visibility of the runway: *“A, 47366, took “640 meters, I’m observing the strip.”*

At 02:22:23 the control tower controller cleared the crew to approach runway 23 using visual maneuvering: *"640...understood. Approach using visual "Maneuvering on runway 23 is cleared, readiness for landing, please report."*

The Commission believes that the said permission from the control tower controller concerned the exit to the visual maneuvering point (VMP) at an altitude of 640 m, followed by performing the "Circle-to-land" maneuver for landing approach. However, further development events shows that the PIC has decided to carry out a visual check from the current point landing approach without further execution of the established approach pattern (that is, without (re-entry to the NRS). The PIC did not report his intentions to the controller.

It should be noted that visual approach and approach using visual maneuvering, despite their certain similarities, primarily in terms of the presence conditions for maintaining constant visual contact with the runway and its landmarks, have fundamental differences, which, among other things, are related to the aircraft's energy management and the creation of a landing configuration. Definitions, purpose, and implementation criteria These types of approaches are given in section 1.18.2. At the same time, the flight manual of the An-24RV aircraft and The airline's flight manual does not contain a section describing the procedure for performing an approach to landing with using visual maneuvering (Circle-to-land), only the order is given performing a visual approach with two engines running. Features of performing a visual approach to landing with one working engine, associated primarily with the management of aircraft energy and moments The release of the chassis and mechanization is also not included in the RLE.

Informal surveys of pilots, including instructors and flight commanders composition, showed that a significant part of them do not see a significant difference between these two types of approaches, and therefore it is difficult to name the criteria for choosing one or the other of a different type depending on the current flight situation. The Commission notes that this The confusion most likely dates back to the Department of Aviation's order transport dated September 20, 1993 No. DV-123/38 "On the implementation of civil aviation practice visual approaches for landing on aircraft of classes 1-3", where these two types of approaches are essentially mixed.

In relation to the considered stage of the emergency flight, safe landing was only possible when performing an approach using visual maneuvering. When performing a visual approach from the current the aircraft's location made it impossible to ensure a stable approach landing and landing at the estimated speed due to the presence of significant excess energy, while the aircraft was still in flight configuration. The PIC's decision to perform a "fly-by" landing resulted in the aircraft not having time to descend before entering the "standard glide path" at the calculated speed, and a vigorous descent led to an increase in the indicated airspeed, which the crew was unable to reduce.

The commission analyzed five randomly selected landing approaches, performed by the airline crews at the Nizhneangarsk airfield on runway 23 during the period from 31.08-10.09.2021. Weather conditions were VFR for all flights. Information about information on these flights is given in section 1.18.3. From the information provided, one can draw two main conclusions:

- The airline crews used a visual approach maneuvering (circle-to-land maneuver) and visual landing approach;
- comparative analysis of the landing approach in the emergency flight and others approaches in which visual maneuvering was not performed showed that in previous approaches that ended in successful landings, the total aircraft energy (altitude and speed reserve) after exiting the paired the turnaround was significantly smaller.

At 02:22:37, at a distance of about 11 km from the runway threshold and at a speed of 370 km/h, the PIC gave command to the flight mechanic: *"Flight small",* after which the flight mechanic moved the throttle engine to position 12° (according to the MSRP-12-96 record, the throttle remained in this position before landing).

At 02:22:39 the control tower dispatcher transmitted information about the aircraft's location to the crew: *"366, bearing 56, distance 11."* 2P acknowledged receipt of the information.

At 02:22:46 the flight engineer reported: *"Altitude 400. Headlights on 'big beam'."*

At 02:22:49 2P reported: *"Radial 9, lateral 1.5."*

At 02:22:55 the PIC informed the crew: *"We've gained speed. We need to lower the landing gear, flaps",* after which, by deflecting the elevator to pitch up, he reduced the vertical rate of descent. Upon reaching a speed of 300 km/h (the maximum permissible (release of landing gear and flaps) the PIC gave the command to release the landing gear and requested removal of up to The runway, to which the co-pilot reported: *“clearance six and a half.”*

The flight path on the landing course is shown in Fig. 45, the flight parameters are shown in Fig. 46.

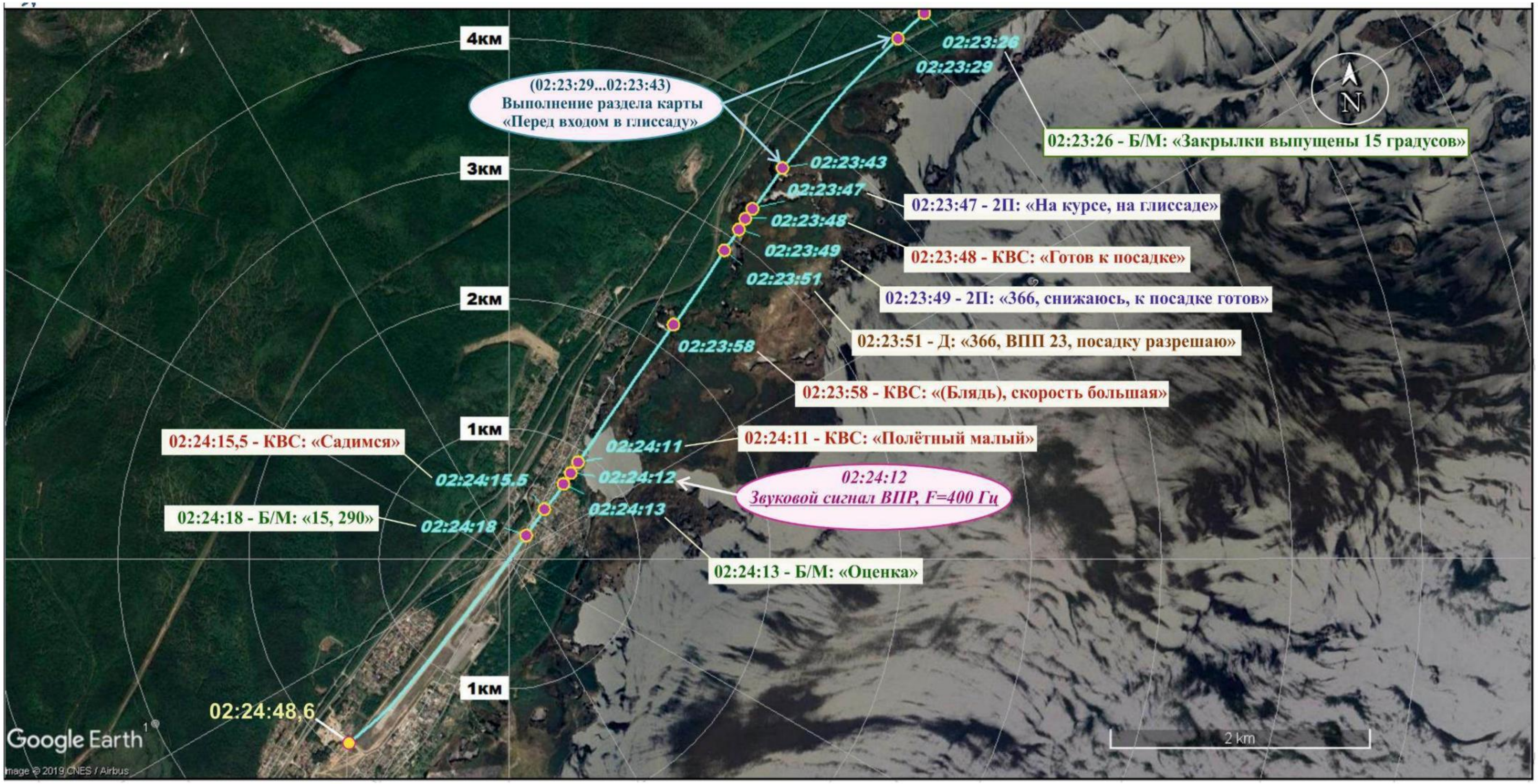


Fig. 45. Airplane flight path on landing course

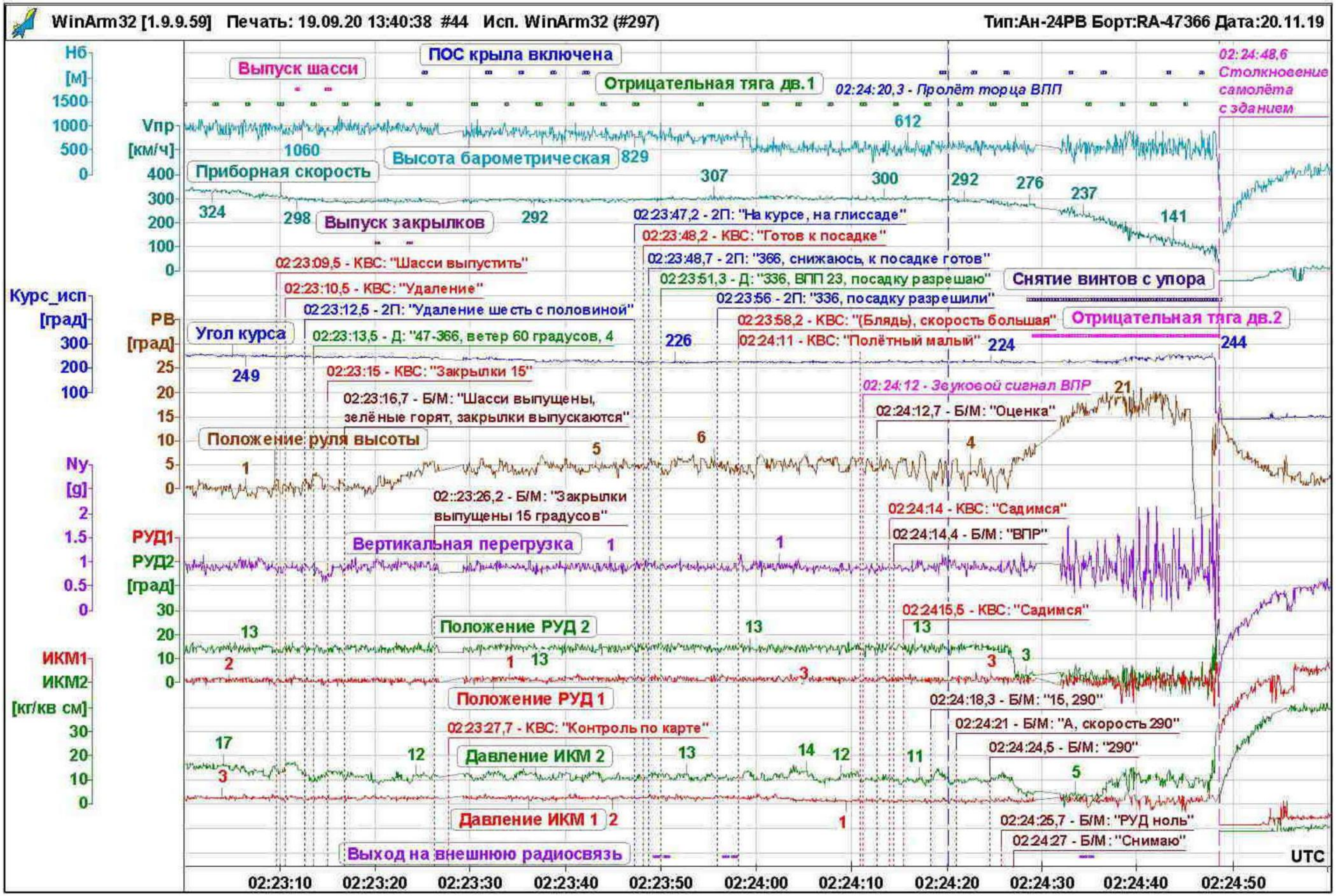


Fig. 46. Aircraft flight parameters on landing course

At 02:23:14 the control tower dispatcher transmitted information about the direction and speed to the crew landing wind: *"47366, wind 60 degrees, 4".*

After releasing the chassis, the flight mechanic, on command from the captain, began to fly at a speed of 290 km/h. to release the flaps by 15°, informing the crew about the intermediate positions of the flaps. The speed of landing gear and flaps extension was higher than recommended by paragraph 5.1.7.a3 of the Flight Manual An-24RV aircraft "LANDING WITH ONE ENGINE FAILED" more than 30 km/h (recommended speed for landing gear and flaps release according to the flight manual is 260 km/h).

After the flaps were released, at an altitude of about 400 m, the anti-icing system of the wing, as evidenced by the registered 02:23:25 by the MSRP-12-96 system a one-time command *"Wing POS is on".* The most the system was probably manually activated by the crew in order to reduce thrust, because There were no conditions for automatic activation. After 18 seconds, the POS was switched off. There was no discussion of the need and reasons for including the POS, as well as the commands for it. On/off switching by the on-board radio was not registered.

In the time interval 02:23:28…02:23:43, on the command of the PIC, the section was performed checklists "BEFORE ENTERING THE GLIDE SIDE". The section was completed in in full and completed when the aircraft was on a straight line with a course of 225° at a distance 3.8 km from the runway.

At 02:23:47 the co-pilot reported that the aircraft was on course and glide path. After The PIC reported: *"Ready for landing,"* which the second pilot reported to the dispatcher at 02:23:49: *"366, descending, ready for landing."* The controller responded to this message at 02:23:51: *"366, Runway 23, cleared for landing."*

According to the calculations carried out, after entering the landing course, at a distance of 4 km from the end of the runway, the aircraft was descending with a trajectory angle of 4.5°, which made it impossible establishing the recommended landing approach speed of 240 km/h according to the flight manual.

At 02:23:58 the PIC said: *"Fuck24, high speed",* at this time the instrument the speed was about 300 km/h. The criteria for a stabilized landing approach, The airline's established RPPs were not met.

Note: *Airline Regulations, Part B, Chapter 2:*

> ***Note:*** *The landing approach must be stabilized before:*
>
> *a) (300 m) relative to the runway threshold when landing in meteorological conditions IFR, or*
>
> *b) (150 m) when approaching for landing in visual conditions.*

---

[24] Here the commission presents the conversations of the crew members, preserving the obscene language, since in this In this case, this characterizes their psycho-emotional state.

The landing approach is considered stabilized if:

a) the aircraft is on the estimated glide path and landing course;

b) small forces are required to maintain the descent trajectory corrective movements of controls along the course and glide path;

c) the indicated airspeed corresponds to the set one entry speed, taking into account the accuracy of maintaining the rating of no less than "four" and not less than the estimated approach speed (according to rating standards: "good" – speed +15 km/ h), created the required landing configuration according to the aircraft flight manual;

d) the vertical rate of descent does not exceed 5 m/ sec.

Note: *If the final approach phase requires maintain a vertical descent rate of more than 5 m/ sec, this is must be discussed during pre-boarding preparation.*

e) the engine operating mode corresponds to the landing configuration aircraft and approach speed;

FAP-128, p. 3.90:

"The PIC is obliged to stop the descent and carry out an aborted approach to landing (go around) if:

- landing approach during commercial air transportation not stabilized according to the requirements established in the RPP reaching an altitude of 300 m above the airfield level during a flight in instrument meteorological conditions or upon reaching altitude 150 m above the aerodrome level when flying in visual meteorological conditions conditions, unless otherwise provided by the RLE."

At 02:24:11, at a distance of about 900 m from the entrance threshold of runway 23, the PIC again gave command to the flight mechanic: *"Flight small."* On the command of the captain, the engine throttle was already set to the "Flight idle" position before the landing gear was released at 02:22:38. It is possible that, being under increased psycho-emotional stress, the PIC "forgot" about this.

When approaching for landing at speeds exceeding the recommended ones by 60-80 km/h, the only correct decision would have been to go around. According to paragraph 5.1.8 of the Flight Manual An-24RV aircraft, go-around with one engine failed (the propeller of the failed engine) engine feathered) with the flaps deflected by 15° and the landing gear extended, when flight weight of no more than 20,800 kg, at a gliding speed of 220–240 km/h according to the instrument, possible from a height of at least 50 m. The speed of the aircraft was significantly higher, that is, the procedure for leaving was simplified. However, the crew had the opportunity to go to the second I didn't consider the circle.

At 02:24:12, an audible alarm sounded in the cockpit at the set altitude of 50 m. a signal indicating that the VPR had been passed, after which the flight mechanic said " Assessment", that the PIC at 02:24:16 announced the decision: *"We're landing."* Commands to extend the flaps in There was no landing position, the flaps remained at 15°.

At 02:24:19, the wing's anti-aircraft warning system was activated again. There were no commands to activate the anti-aircraft warning system. The POS remained on until the aircraft was destroyed.

At 02:24:20, at an indicated airspeed of approximately 295 km/h, the threshold of runway 23 was overshot.

According to the recordings from the flight recorders, the video footage presented, which was performed by passengers of the aircraft from the left and right sides of the aircraft, airfield records video cameras and traces on the runway from the wheels of the left and right main landing gear were the moments of the first and subsequent touches of the wheels on the runway, as well as the trajectory, were determined aircraft movements after landing. Aircraft flight parameters upon landing, braking, run and rollout are shown in Fig. 47. The trajectory of the aircraft along the runway is shown in Fig. 47, Fig. 48 and Fig. 49.

At 02:24:26, at a distance of 530 m from the threshold of runway 23, at indicated airspeed At approximately 275 km/h, the first contact with the runway occurred with the wheels of the left main landing gear chassis, while the wheels of the right landing gear did not touch the runway. At the moment of the first contact The flight engineer reported: *"Thrust zero."*

After the first touchdown, the aircraft separated from the runway and at 02:24:27.2, at a distance of 633 m from the runway entry edge, touched the runway again with the wheels of the left landing gear. At this At that moment, the flight engineer reported, *"I'm taking off."* At 02:24:27.7, the captain gave the command, *"Slow down."* The commission was unable to clearly determine to whom this command was addressed. During pre-landing preparation, the PIC determined that he was performing the braking himself.

At 02:24:28, the PIC gave the command to remove the propellers from the stop: *"Remove."* At 02:24:28 The flight engineer reported again: *"Removing."* Removing the propellers from the stop and, as a result, the occurrence of negative thrust on the right engine occurred at 02:24:28.5 and 02:24:29 accordingly, as evidenced by the registrations by the MSRP-12-96 system one-time commands *"Remove propellers from stop"* and *"Negative thrust of the right engine".*

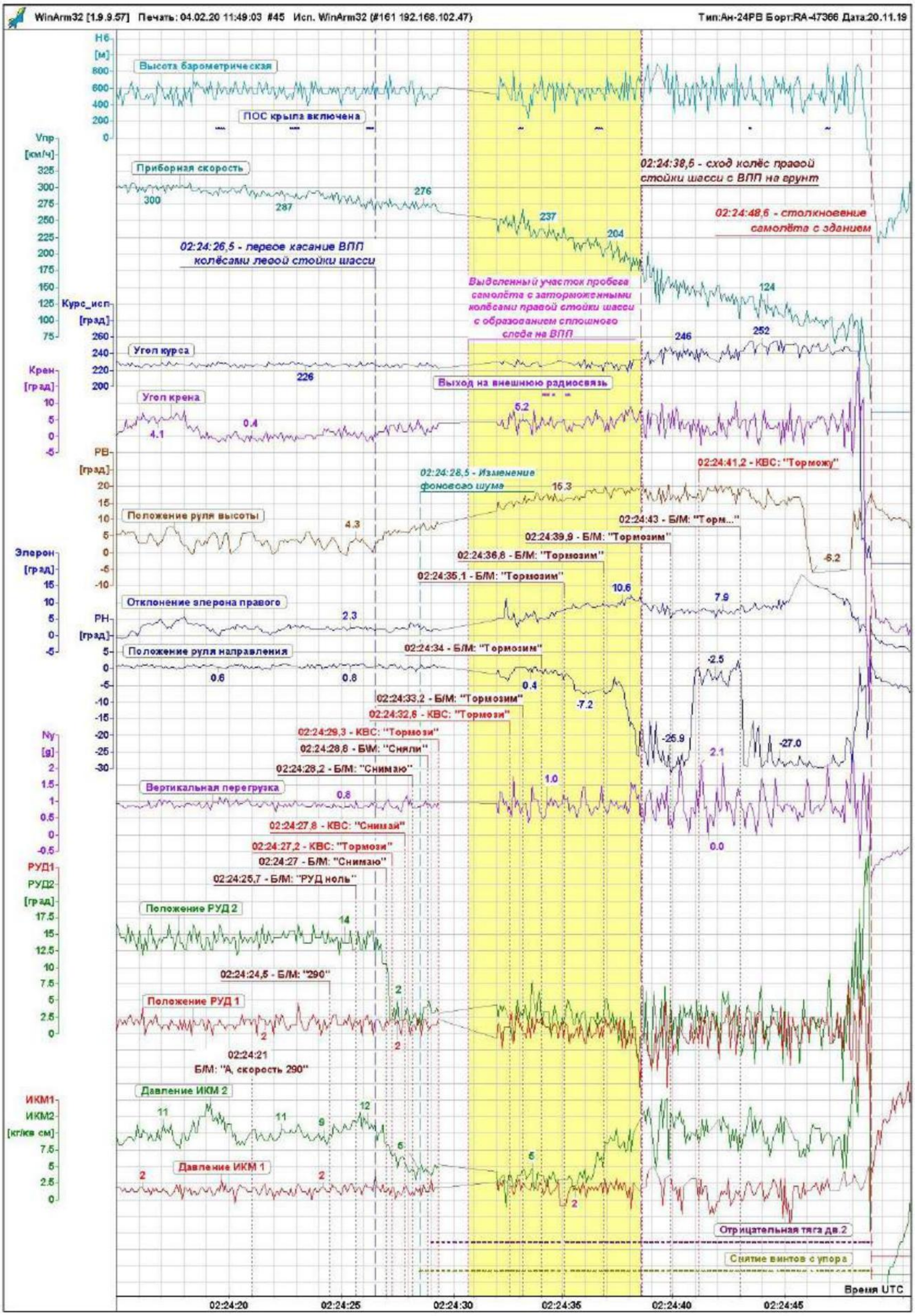


Fig. 47. Aircraft flight parameters during landing, braking, takeoff run and rollout

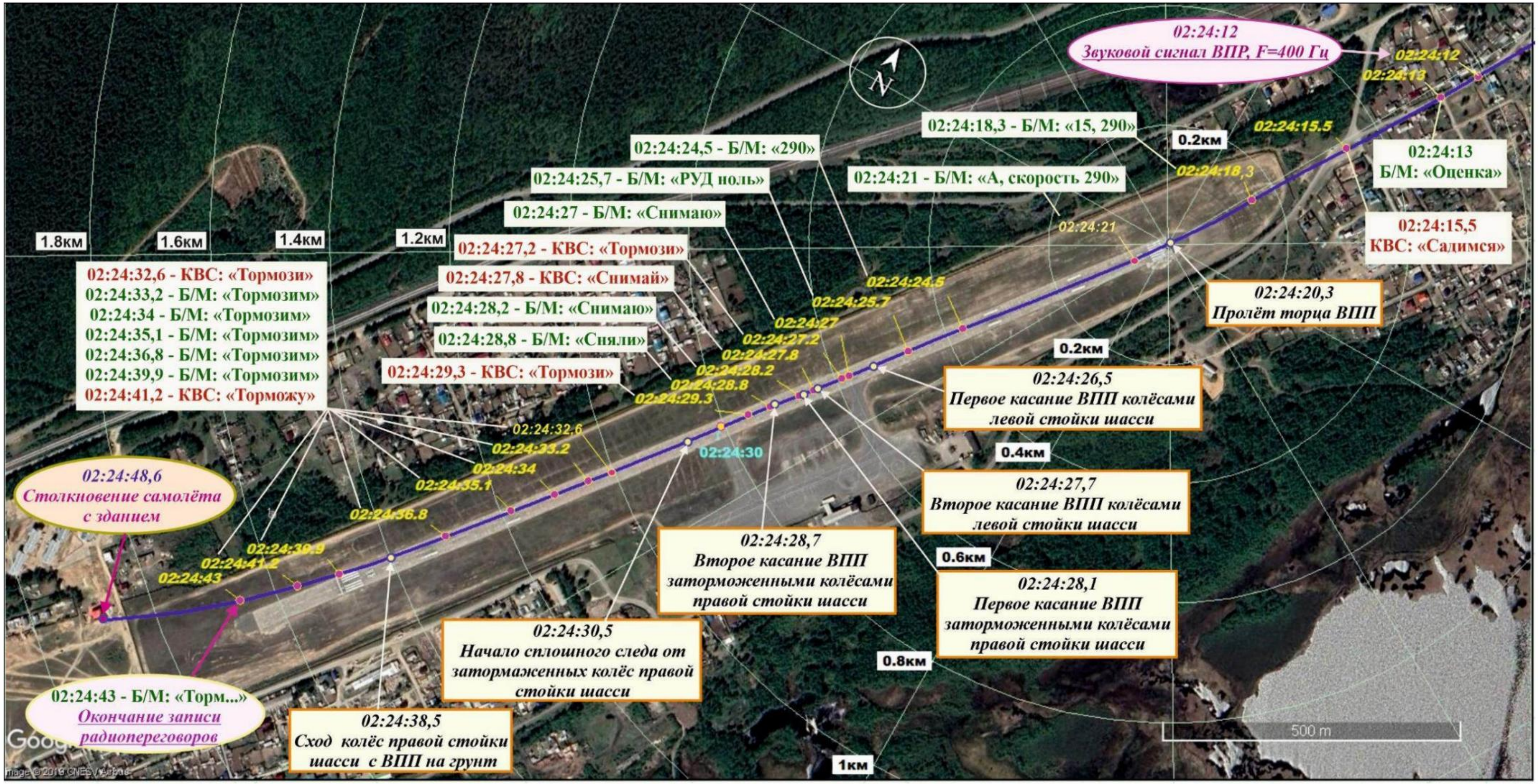


Fig. 48. Aircraft trajectory with superimposed radio traffic during landing, braking, takeoff run and rollout

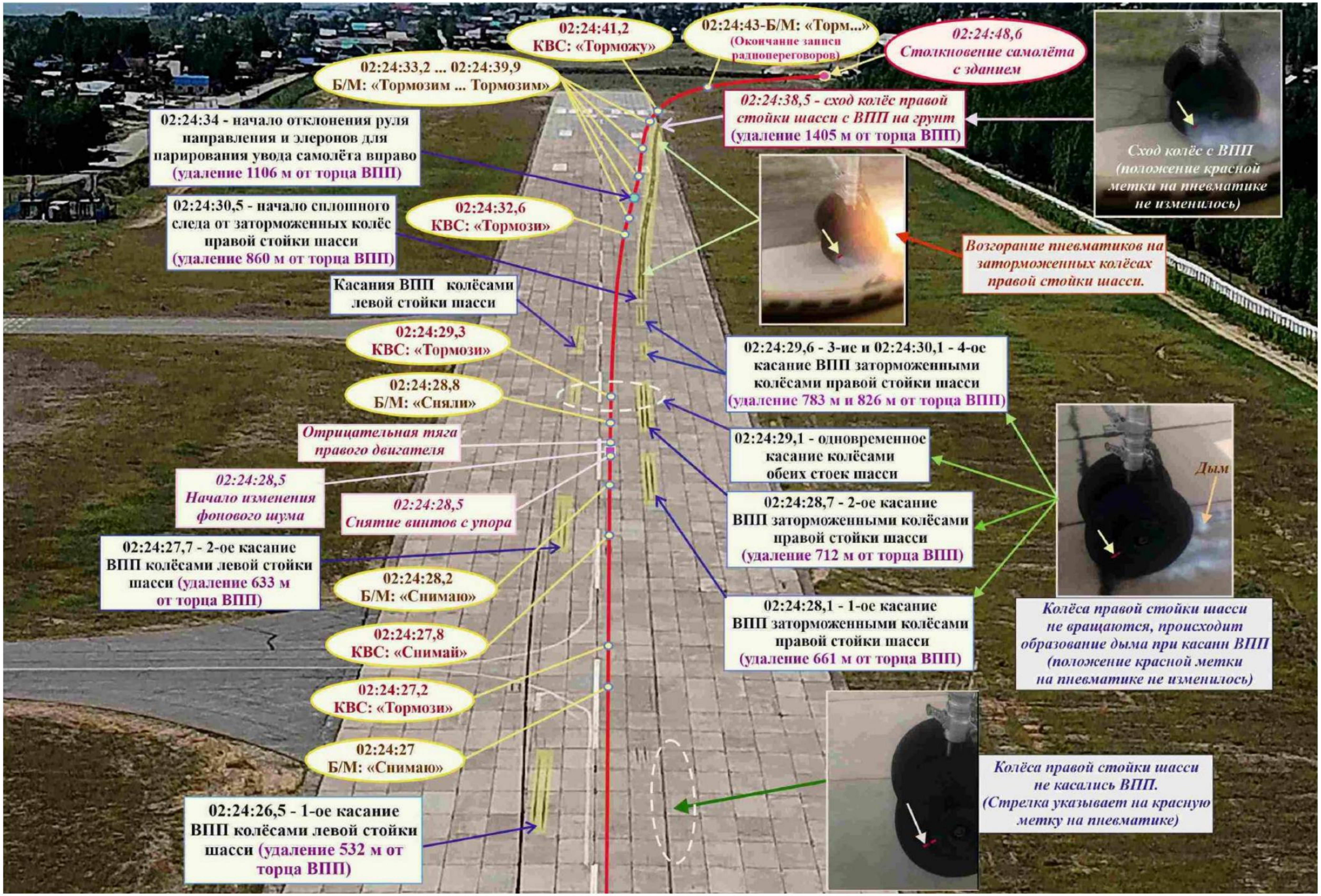


Fig. 49. Aircraft trajectory along the runway after landing

At 02:24:28.1, at a distance of 661 m from the entrance threshold of runway 23, the first contact occurred. The runway is non-rotating (braked) wheels of the right main landing gear (Fig. 49), which is confirmed by video footage and characteristic traces on the runway. Probably, that after the previous touchdowns of the runway with the wheels of the left main landing gear, the PIC "pinched" brake pedals. These actions of the PIC were contrary to the provisions of the flight manual, which determine the application of braking only after the nose landing gear has been lowered, and, were obviously caused by the desire to begin braking as quickly as possible actual significantly increased aircraft speed.

Note: *1. From the letter of the State Enterprise "Antonov" dated 03.02.2020 No. 724/1468-20:*

“2. The An-24 aircraft is not equipped with protection against landing on braked wheels, so landing with the brakes pressed pedals are not allowed, and the use of braking in accordance with The RLE is permitted only after the front wheels have been lowered onto the runway. When landing with a bank, when the wheels of one side are in contact from the runway and rotate, and the wheels of the other side do not have this contact and do not rotate, pressing both brake pedals with entry into Contact with the runway of non-rotating wheels can lead to their skidding damage."

Clause 6 of Section 5.1.7. "Approach and Landing with an Engine Failed" RLE of the An-24RV:

“After landing, smoothly lower the nose gear, set the throttle the running engine to the 0° position according to the control lever and remove the screw from the stop. The resulting turning moment to the side the running engine is countered by the rudder, controlled front support and, if necessary, wheel brakes."

2. RLE of the aircraft An-24 (An-24RV), 7. Operation of systems, 7.6. Chassis:

"7.6.3. In-flight operation.

The brake wheels are equipped with inertial sensors, which together with the automatic braking system are intended to prevent wheel skidding when braking."

Due to landing on the braked tires of the right main landing gear The chassis began to collapse and leave characteristic black marks on the runway.

Note: *From the conclusion No. 9981-AK/ 104 of the Federal Autonomous Institution “Aviation Register of the Russian Federation” dated 21.11.2019:*

"Destruction of the inner and outer aircraft tires... of the right main landing gear The landing gear of the An-24RV RA-47366 aircraft was damaged as a result of the aircraft landing. with locked wheels."

After removing the propellers from the stop, the flight engineer, on the command of the captain, applied the emergency braking, which confirms 2P.

Note: *From explanations 2P from 03.07.2019:*

"After touchdown… the PIC applied braking, there was no feeling of braking happened. Then the captain gave the command to the flight engineer: "Slow down." Received the answer was "I'm braking." Out of the corner of my eye, I saw the flight engineer apply emergency braking."

It should be noted that when applying braking from the emergency system the automatic braking system, which serves to prevent wheel skidding, does not work. When using the emergency braking system, the RLE requires that the brakes be released pedals, that is, stop using the main braking system.

Note: *Item 7 of Section 7.6.5. "Possible Malfunctions and Crew Actions" RLE of the An-24RV:*

"Signs of failure. *During taxiing, takeoff, or rollout during compression The brake pedals do not brake the wheels.*

Crew actions: *Release the brake pedals. Braking perform this using the emergency brake handles.*

Warning: *During emergency braking, the anti-skid system The automatic system does not work, to avoid skidding, press The levers should move smoothly, without any sudden jerks."*

At 02:23:34.5 the second pilot reported to the dispatcher that the landing had been completed, and, according to him, he did not take part in braking the plane.

Note: *1. From explanations 2P from 03.07.2019:*

"The plane was moving along the runway centerline. I reported the landing and assisted maintain direction. My feet were in the "feet on" position flight" (after the aircraft landed and before the collision, I did not use the brakes, I (I didn't slow down).

2. RLE of the aircraft An-24 (An-24RV), 7. Operation of systems, 7.6. Chassis:

"7.6.3. In-flight operation.

Warnings: 1. Simultaneous pressing by both pilots on "Brake pedals are not allowed during primary braking."

As the aircraft speed decreases while running along the runway with damaged pneumatics of the braked wheels of the right main landing gear, at negative the thrust of the right engine, the feathered propeller of the left engine and the rotating wheels the left main landing gear of the aircraft began to deviate to the right and at 02:24:41 it left the runway soil. According to the results of mathematical modeling (section 1.16.5), at speeds below 200 km/h maximum available total directional moment from front landing gear tires and full rudder deflection becomes insufficient to maintain the specified direction of movement along the runway.

At the moment of exiting the runway, at a distance of 1405 m from its entry threshold, the speed of the aircraft was about 160 km/h, which was due to the increased landing speed of the aircraft and possibly ineffective use of the wheel braking system due to simultaneous use by the crew of the main and emergency braking systems.

Research of main landing gear brake wheels and shuttle valves brake system of the aircraft, conducted by the Federal Autonomous Institution "Aviation Register of the Russian Federation", showed that[25]: *"The brake devices of the wheels of the left and right landing gear aircraft… are in satisfactory technical condition, signs of failure are absent.*

The shuttle valves are in their extreme positions[26], which indicates that the supply of brake pressure to the brake devices from the main system braking."

Specialists from the Antonov State Enterprise, taking into account the results of the brake research systems, analyzed the performance of the braking system (main and emergency) during the aircraft's landing run. According to their conclusion, the use of the PIC for braking the main system, and the flight mechanic of the emergency system, could lead to abnormal operation of the braking system up to a complete lack of pressure in the brakes.

Note: *1. From the response of the State Enterprise "Antonov" dated 21.02.2020 No. 724/2304-20:*

> *"When the brake pedals are fully depressed, the shuttle the UG97-7 valve is pressed against the seat of the emergency channel fitting by pressure liquid (95 ± 5) kgf/ cm² plus the force of the retainer spring. If in this in this situation, apply the same pressure to the emergency channel fitting values, then the shuttle will remain pressed against the seat of the emergency channel for account of the clamp force. If the pressure is released from both at the same time fittings, then the shuttle will remain pressed against the seat of the emergency channel.*

[25] For more details, see Section 1.16.3 of this report.
[26] This refers to the condition at the time of the study.

It is important to keep in mind the following: The shuttle valve operates when condition when pressure is applied on one side and the other is connected to drain. If this condition is not met and the drain is blocked, then the shuttle can shift from higher pressure to lower pressure equalization of the forces acting on it from both sides. In this case, Shuttle vibrations and blocking of both feed channels by the shuttle are possible brake fluid.

Thus, taking into account the explanations of the 2nd pilot, the PIC, working intensively pressed the rudder pedals until they reached their full travel brake pedals are not fully depressed, blocking the main channel drain, and BM, by pressing the emergency brake handles, created conditions, in which the shuttles of the left and right pairs of wheels could block the feed fluid in the brakes of the left or right wheels, differently, in constant or intermittent mode, which led to abnormal operation of the brake systems up to a complete lack of pressure in the brakes.

In the absence of registration of the brake pressure value and the fact pressing the brake pedals, this situation can be considered as version».

2. An-24 Aircraft. Technical Description. Book IV. Edition II. V/O AVIAEXPORT USSR Moscow:

Chapter I. Wheel braking system:

It should be remembered that simultaneous braking with the main and The emergency system does not increase the effectiveness of the brakes. Such simultaneous braking must not be allowed to avoid "Brake line shut-off with shuttle valves."

After leaving the runway, the plane continued to move along the ground, continuing to evade to the right. At a speed of about 90 km/h, the plane broke through the airfield fence and at 02:24:49 collided with the wastewater treatment plant building. As a result of the collision with the building, A fire broke out on the plane, killing two crew members.

The commission analyzed the compliance of the Nizhneangarsk airfield with the level required fire protection (RFP).

On the day of the accident, runway 23 at Nizhneangarsk airfield was assigned Category 3. according to the level of fire protection (see section 1.10.). The firefighter who was at the airfield vehicle, its fire tanks and the performance of fire pumps met the requirements for runways with category 3 UTPZ.

According to the information available to the investigation commission (section 1.15), the temporary the interval between the arrival of the fire engine to the scene of the fire and the start of extinguishing the fire complied with the requirements of regulatory documents. In fact, fire extinguishing Even liquids and foaming agents were used in the process of extinguishing the fire more than the criteria for category 3 UTPZ. All passengers were evacuated from the burning aircraft, and the death of two crew members was due to them being trapped in the cockpit, most likely unconscious due to the destruction the nose of the aircraft upon collision with the wastewater treatment plant building.

The commission of inquiry also conducted, within the framework of the available information, analysis of the feasibility of the construction of the wastewater treatment plant building, which suffered aircraft collision (see also section 1.18.1).

Provisions of Article 47 of the Air Code of the Russian Federation on the need to establish an aerodrome territories were introduced by the Federal Law of 01.07.2017 No. 135-FZ “On Amendments amendments to certain legislative acts of the Russian Federation in terms of improving the procedure establishment and use of the aerodrome area and sanitary protection zone." This Federal Law was developed and put into effect in order to ensure urban zoning of territories and elimination of approval procedures construction of each individual facility near the airfield.

For the period prior to the establishment of the aerodrome territory, part 3 of Article 4 of this The Federal Law provides for the approval of the placement of objects within the boundaries of the strips air approaches and sanitary protection zones (previously, the norm was in force coordination of construction projects).

On the date of the accident involving the An-24RV aircraft RA-47366 the aerodrome area of the Nizhneangarsk aerodrome according to the rules provided Article 47 of the RF Water Code (as amended by Federal Law No. 135-FZ of 01.07.2017) was not established, therefore the approval of the construction of the facility in the aerodrome area with was mandatory by the airport administration. Thus, in the opinion of the commission According to the investigation, the construction carried out without approval did not comply the current legislation.

On the other hand, the constructed facility (the sewage treatment plant building) structures with a height of 8.2 m) was not located in the air approach zone, its height is not exceeded the established level for structures in the aerodrome zone, that is, the commission the investigation did not reveal ***any technical reasons*** why the agreement could have been be refused when applying for it in accordance with the established procedure.

Thus, the main factors that negatively influenced the outcome
The following were involved in the emergency flight after the engine shut down:

- making a decision to land on a runway that does not provide the required landing distance. This decision was made without carrying out some calculations and proper analysis of the situation;
- incorrect selection of the type and landing approach trajectory, which led to impossibility of timely reduction of flight speed;
- failure to make a decision to go around when the aircraft is not stabilized landing approach, which resulted in landing at a significantly elevated altitude speed with the separation of the aircraft from the runway and predetermined a significant the distance required to stop the aircraft;
- improper use of the aircraft's braking systems, which resulted in destruction of tires, reduction in braking intensity, and the aircraft rolls off the runway to the right.

The said unfounded decisions were taken by the Captain of the Air Force, while none of the members the crew never once drew his attention to the violation of standard operating procedures procedures, discrepancy between the current flight parameters and the required values and the unsafety of landing under current conditions. Thus, the interaction in the crew, cross-checking and implementation of standard operating procedures were at an insufficient level.

The commission notes that the captain had extensive experience in flying aircraft. of this type. All necessary training and checks were completed on time and in full. volume. Personal indiscipline of the PIC and crew members during the emergency flight, as well as no violations of the work and rest regime were found. At the same time, the commission drew attention to the peculiarities of the psychological profile of the captain (section 1.16.6).

The commission of inquiry had at its disposal psychological data inspections of the KVS for 2017, 2018 and 2019. The results of the analysis revealed two features, each of which, individually or in combination, could lead to the above-mentioned erroneous decisions and actions of the PIC.

The first feature is associated with a noticeable decrease in a number of cognitive functions. First of all, we are talking about a decrease in the level of analytical (logical) thinking and increasing unevenness in the pace of mental activity, which negatively affects the speed and quality of information processing, decision-making and the accuracy of their execution, the ability to change wrong decisions.

The commission notes that the level of analytical (logical) thinking is “lower "average" has been identified in KVS by VLEK psychologists since at least 2017.

According to some methods applied in 2018 and 2019, this indicator was on the border between “low” and “below average” levels, and in some cases was “low”. Psychologists also determined that those at the “below average” level were: working and short-term memory.

In the conditions of an emergency flight, it is precisely this (low level of analytical (logical) thinking and uneven pace of mental activity) can The adoption by the PIC of a number of unfounded decisions listed above can be explained above, and the lack of their correction despite their awareness of the existence of this objective conditions. For example, making a decision to land with one It was decided to take the failed engine onto a runway that was clearly not suitable in size The captain alone, without making any calculations or consulting with other members crew. Alternative actions were not even considered. Also, with full understanding by the PIC of a significant excess of flight speed on the glide path and discussion this fact with the crew members, the decision to go around was not made was considered. It can be assumed that the PIC was in the in an unfavorable emotional state (dominant, make the landing as possible) rather), which further negatively impacted the decision-making process.

Guide to psychological support for selection, training and professional activities of air traffic control and flight personnel of civil aviation of the Russian Federation”, approved by the order of the Ministry of Transport of Russia dated October 31, 2000 No. 57 (hereinafter in this section referred to as the Guidelines) defines “low” level of cognitive (mental) functions includes the worst 5% of results shown by healthy people individuals, and the "below average" level is the results from the 5th to the 25th "percentile" (that is, 20%) all results of healthy individuals).

In this case, according to Section 2.4 of Part 3 of the Guidelines, the assessment of the general level cognitive functions has the following semantic interpretation:

"low" level of cognitive function preservation cannot in itself be grounds for dismissal from work, but is a factor that significantly aggravating the forecast for the safety of flight (dispatch) activities;

the level “below average” is taken into account as an unfavorable factor in the presence of other clear signs of damage to the brain or its vessels27 .

[27] According to the medical record, KVS had a diagnosis of *“cerebral vascular atherosclerosis…”*. Also, as stated in section 1.16.6, it is most likely that in 2019 the PIC experienced stress related to

Machine Translated by Google

Thus, the current wording of the Guidelines determines that the levels cognitive functions "low" and "below average" are risk factors and create a significant risk to flight safety. However, neither the Management nor FAP MO GA, regardless of the results of the studies obtained, do not allow VLEK psychologists can independently decide on the pilot's fitness/unfitness for flight operations, which reduces the level of their (psychologists') responsibility.

Leading aviation doctors and psychologists have been concerned about this for a long time state of affairs. In particular, the issues of declining mental functions due to aging The brain received much attention at the Second International Congress Association of Aviation Medicine Physicians (2019). It was particularly noted that the negative the influence of insufficient mental functions in older pilots is not can guarantee successful piloting of the aircraft, especially in emergency situations situations. The Congressional resolution states that *"aviation specialists Psychologists point out that the reliability of a pilot (ATC controller) depends not only on his professional training, but also on his level of health and psychological readiness to perform professional functions in the regular and emergency modes."* According to the Congress participants, this problem can be solved by introducing appropriate changes to the Federal Aviation Administration's Civil Aviation Regulations by supplementing the list grounds for dismissal from flight work under the clause on the general insufficient level mental functions caused by age-related aging of the brain. Resolution The Congress contains information on the preparation of a corresponding appeal to to the aviation authorities.

In the case under consideration, the final forms of psychological examinations of the KVS for 2018 and 2019, the psychologist did not indicate the integral level of preservation of the function "thinking", which had levels of "below average" and "low" (according to different methods). This is contrary to the provisions of Section 2.4 of Part 3 of the Guidelines, which states: *“When When making an overall assessment, the psychologist records not only the overall level of safety mental functions, but also necessarily indicates which functions (memory,* ***thinking,*** *attention, sensorimotor coordination, etc.) were "low" and/ or "below average" level. Therefore, the conclusion, if necessary, should be not only general,* ***but also functional in nature,*** *which is important for making expert decision depending on the specifics of the activity being performed."*

some kind of somatic disease, while the identification of such indicators goes beyond the provisions of the Guidelines and the actual qualifications of the “linear” psychologists of the VLEK.

Section 3, “Psychologist’s Report,” of Part 3 of the Guidelines also states that in The psychologist's conclusion must indicate: *"assessment of the general level of development of cognitive functions* ***with additional indication and functional interpretation of the indicators assessed at 1 and 2 points28”,*** which was also not done. Accordingly, no *recommendations of a psychologist,* as provided for in the said section of the Guide, were given *on the need for additional research (consultations), treatment rehabilitation and/ or psychocorrective measures."* No additional research to determine the causes of cognitive decline in the KVS and the need for appropriate correction was not prescribed or carried out.

The second feature is related to individual psychological characteristics. personality of the captain. With strongly expressed leadership qualities and the ability to accept The captain of the armed forces did not have the emotional stability and flexibility to make leadership decisions. thinking for their prompt correction in case of error. At the same time, the increase stress (suboptimal psycho-emotional state) caused by complication flight situation due to engine shutdown and the indicated erroneous decisions, negatively affected the organization and consistency of the actions of the captain, lengthened the time required to make subsequent decisions and provoked the adoption impulsive (irrational) decisions. Thus, the approach trajectory chosen by the PIC landing did not allow the plane to extinguish its available energy in a timely manner. Judging by negotiations, the PIC was aware of this fact while still on the double turn, however did not take the necessary measures, which naturally led to a significant excess of speed on the glide path, unstabilized approach and clearly non-landing position of the aircraft.

The manual defines the Standardized Multivariate Method personality research (SMIL), as one of the main methods of personality research during a medical examination at the Civil Aviation Authority. The results are interpreted also in accordance with the provisions of the Guidelines. With the help of SMIL, the following are identified: personality traits that are more likely to lead to negative psychological reactions, as well as accentuated character traits, increasing the likelihood of such reactions occurring in conditions of emotional tension, especially in emergency situations, inherent in the flight profession.

At the same time, the IAC investigation commissions have repeatedly noted that the recruitment psychodiagnostic methods used in GA to determine individual- personal characteristics of the pilot mainly appeals to verbal methods.

---

[28] That is, the levels are “low” and “below average”.

This high level of intelligence allows the subjects (pilots) to successfully manipulate test results in verbal methods and exclude answers, which may present them in an unfavorable light. In this regard, the profile's inclusion the personality within the normal range is not in itself a sufficient basis for conclusions on the level of adaptation/maladaptation. Based on the results of investigations by commissions repeatedly (starting in 2006, the crashes of A310 aircraft at the Irkutsk airport, Russian Federation and Tu-154 in the area of Donetsk, Ukraine) were given the corresponding recommendations, including those for finalizing the Guidelines. For the most part, the data The recommendations were not implemented.

In the case under consideration, the results of the PIC testing according to the SMIL methodology were within the established criteria of the norm (quantitative indicators obtained by test results are within the norm), his admission to flights with "from a psychological point of view, it was formally carried out in accordance with the Guidelines. In fact, the captain had significant personal characteristics that could the negative impact of which was aggravated by the growing deficit of cognitive functions. When interpreting the results of the KVS testing, psychologists did not attention is drawn to a whole range of features, in-depth qualitative (and not only quantitative) assessment of which would allow us to recommend the use additional non-verbal methods of psychological examination. At the same time, "linear" psychologists of the VLEK with the required qualifications to conduct high-quality personality profile assessments, as a rule, do not have, and the Guide does not have such assessments requires.

The Commission notes that in a significant number of cases of severe aviation incidents that have taken place in recent years, namely personal psychological characteristics of crew members that largely determine their behavior in emergency situations situations, were the “last piece of the puzzle” that translated the situation that had arisen into flight a special (abnormal) situation in an emergency or catastrophic situation. These the features were reflected in the tests they passed using the SMIL methodology, however were not identified by psychologists due to existing shortcomings in the Guidelines, insufficient the qualifications of the psychologists themselves and their heavy workload, leading to "conveyor" (formal) approach when conducting surveys.

Overall, the Commission notes that there have been no changes to the Guidelines since its publications, lack of specificity and vagueness of individual formulations, limitations in selection of methods, many of which are repeated from year to year and are available for preliminary study (memorization) on the Internet, insufficient scientific the validity of the "derivation" of integral assessments without taking into account the "weight" of one or another of applied basic and additional methods, references when choosing additional ones methods for popular literature, allow one to obtain practically any required test result[29] and do not provide the required level of psychological selection in modern conditions. Thus, the main goal of psychological selection in GA, declared in paragraph 1.2. of Part 1 of the Guidelines, *"…achieving and maintaining compliance personnel capabilities meet requirements, presented by the professional activities while maintaining the proper functional state of the body, the performance of aviation specialists, their physical and mental health",* is not achieved.

2.2. Analysis of the causes of engine failure in flight

As a result of the analysis of the performance of the left engine, the following was established.

The last repair of the left engine took place in July 2018. For the purpose of evaluation the quality of the repairs performed and further adjustments, as well as the functionality engine in operation, the Commission contacted the Engine Developer – State Enterprise "Zaporizhzhya Machine- Building Design Bureau "Progress" named after Academician A. G. Ivchenko (Ukraine) (hereinafter referred to as the State Enterprise "IVCHENKO-PROGRESS"). The response received, in particular, states:

"Based on the results of the study of the engine control test materials AI-24 2 series No. H47622024 at the last repair (section 15 of the form), as well as materials of ground testing of the engine during its installation on the aircraft and during When carrying out maintenance in operation, we inform you:

…

2. The control test of the engine after repair was carried out under the following conditions: Rn.v. = 753 mm Hg; tn.v. = + 27 °C, while the following values were obtained in takeoff mode parameters: fuel pressure ÿÿ = 75 kgf/ cm², fuel consumption ÿÿ = 654 kg/ h, pressure oil in the torque meter Rcm = 78 kgf/ cm², gas temperature for turbine t6=524 °C. That is, at a regulated pressure of 30 (the norm is not >85 kgf/ cm²) and flow rate fuel (the norm for these conditions is 667 + 30-20), we have an underestimated value of Rikm (should be ~ 84 kgf/ cm²) and the maximum permissible temperature of gases behind the turbine (maximum permissible temperature in the VZL mode is 525 °C).

[29] According to information received from the VLEK of JSC Irkutsk International Airport, at least for the last For 10 years there has not been a single case of dismissal from flight work for "psychological" reasons.
[30] This means "down-regulated" to ensure compliance with the maximum gas temperature.

We see the same thing in the nominal mode – the temperature of the gases behind the turbine t6=472 °C, and the maximum permissible temperature in NOM mode is 475 °C."

With the engine operating parameters "adjusted downwards" after the last During repairs, the temperature of the gases behind the turbine in several modes was almost equal maximum permissible. According to the conclusion of the State Enterprise "IVCHENKO-PROGRESS": *"This indicates as a result of the repairs performed, such an engine could produce for some time parameters during operation under low tÿ.ÿ. conditions, but with increasing temperature "The non-compliance of the technical specifications would become increasingly apparent under these conditions."*

Thus, with formal (according to documents) compliance of the work parameters engine after the last repair, its actual condition (quality of repair) created significant risks that during further operation these parameters would be go beyond the TU.

This conclusion is confirmed by the study of the engine test results in operation during installation on the An-24RV RA-47366 aircraft on November 19, 2018, and when performing various works on 01/22/2019 and 06/22/2019. To check compliance with the technical specifications The engine parameters during testing should be used to determine fuel consumption standards for specific conditions, and then the measured costs are compared with these standards. In its In response to the State Enterprise "IVCHENKO-PROGRESS" he notes: *"In the protocols it is always the norm for standard conditions is written as ÿÿ.ÿ. = 760 mm Hg; tÿ.ÿ. = + 15 °ÿ, although The measured consumption is significantly higher than even the upper flow limit of these standards (see Table 2).*

Table 2

Режим работы двигателя	Норма расхода Q записаная в протоколах для $P_н$ =760 мм рт.ст.; $t_{н.в.}$= +15°C	Расход Q для $P_н$ =723 мм рт.ст.; $t_{н.в.}$= -14°C (уст-ка на с-т 19.11.18г.)		Расход Q для $P_н$ =719 мм рт.ст.; $t_{н.в.}$= -8°C (период-е ТО 22.01.19г.)		Расход Q для $P_н$ =716 мм рт.ст.; $t_{н.в.}$= +18°C (ТО 22.06.19г.)	
	ТУ	ЗАМ	ТУ	ЗАМ	ТУ	ЗАМ	ТУ
0,7 Ном. (41° по УПРТ)	480^{+30}_{-20}	<u>520</u>	467^{+30}_{-20}	510	467^{+30}_{-20}	500	466^{+30}_{-20}
0,85 Ном. (52° по УПРТ)	533^{+30}_{-20}	<u>580</u>	519^{+30}_{-20}	550	519^{+30}_{-20}	560	518^{+30}_{-20}
Номинал (65° по УПРТ)	591^{+30}_{-20}	<u>620</u>	575^{+30}_{-20}	610	575^{+30}_{-20}	620	575^{+30}_{-20}
Взлетный (100° по УПРТ)	678^{+30}_{-20}	<u>700</u>	666^{+30}_{-20}	650	666^{+30}_{-20}	700	667^{+30}_{-20}

Judging by the change in measured consumption in different modes during testing, apparently Adjustments were made to reduce costs, but they still exceed the norms for specific conditions (there are no records of adjustments in the ADT-24 passport)."

From the information provided, we can conclude that already during installation on The aircraft engine did not comply with the technical specifications. Its non-compliance with the technical specifications, despite the ongoing adjustments, was confirmed during further testing. This is also confirmed conclusion of the State Enterprise Ivchenko-Progress: *"Taking into account the above, it can be said that engine parameters at low air temperatures were maintained by "increased fuel consumption, the engine did not meet the specifications."*

However, the engine continued to be used. The commission was unable to to clearly determine why the comparison of the actually obtained parameters was carried out with parameters for standard, not actual, atmospheric conditions. This it could have been due to a lack of qualifications of the technical personnel who carried out the work work, both intentionally and intentionally, to ensure continued operation.

The left engine adjustment deficiencies also became apparent during the accident flight. According to cockpit communications, during the start-up process the left engine did not immediately reached the engine speed due to the activation of the temperature limiter (system PRT). The activation of the PRT system during launch could be due to incorrect adjusted starting process (inaccurate adjustment of ADT-24).

Note: *According to the RTE of the AI-24 2 series engine, the limit control system*

temperature and speed correction (PRT) limits *gas temperature behind the turbine depending on the operating mode engine (ÿÿ) and the static pressure of the environment ÿÿ (height fields).*

The system at pressure ÿÿ=760 mmHg limits the gas temperature behind the turbine depending on the position of the engine control lever on ADT-24:

- at start-up (ÿÿ=0-7±1°) up to 720°ÿ

- in the "Nominal" mode (ÿÿ=7±1° - 72.5±1°) - 435°ÿ;

- in the "Maximum" mode (ÿÿ=72.5±1 -100°) - 495°ÿ.

The gas temperature behind the turbine is limited by the influence of the system PRT on the bypass needle of the ADT-24 unit for changing the fuel consumption the engine.

During the process of starting the right engine and entering the MG mode, according to the report B/M, the temperature limiter did not activate.

The takeoff was carried out with the engines operating in takeoff mode.

The oil pressure values in the ICM system of the left and right engines were equal 85 kgf/cm2 and 89 kgf/cm2 respectively at the same throttle position (Fig. 50).

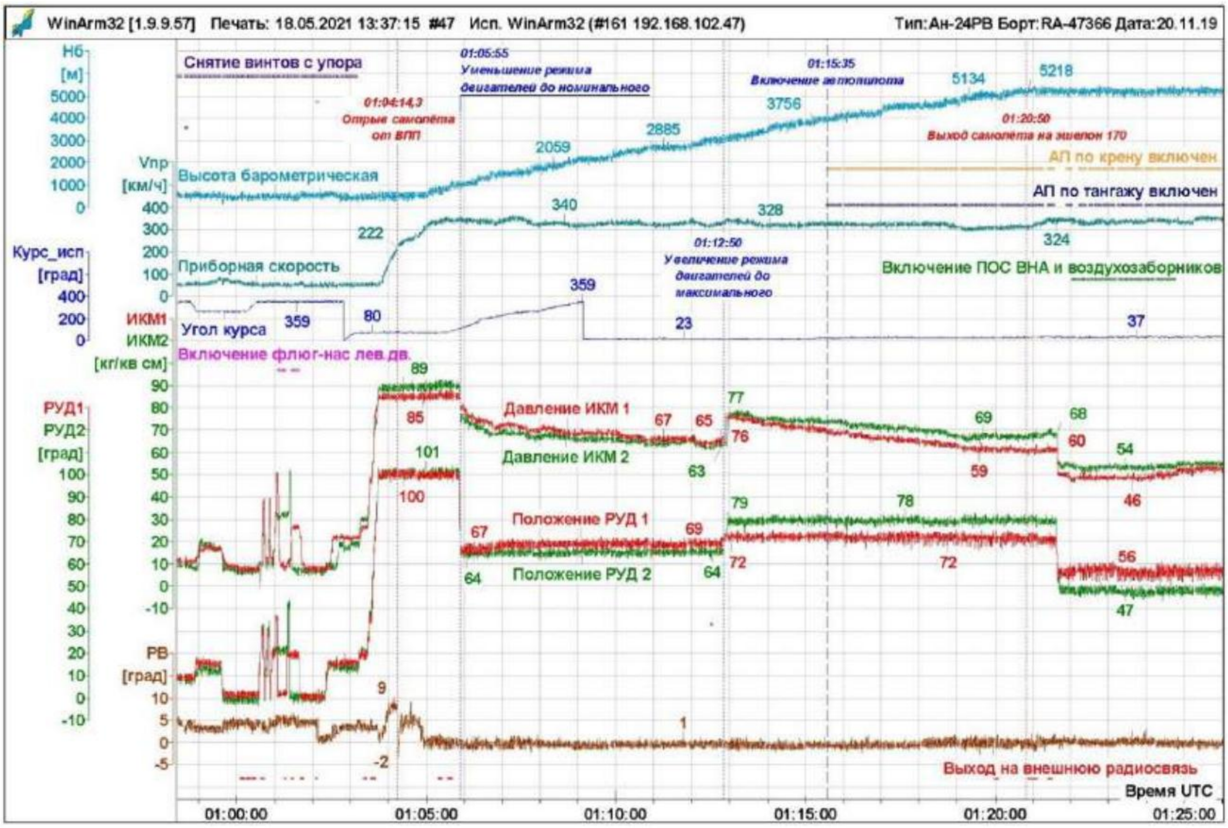


Fig. 50. Changing the engine operating mode after takeoff during the climb

The oil pressure in the left engine's PCM system was below the minimum permissible value (87 kgf/cm2) for actual takeoff conditions (Fig. 51)

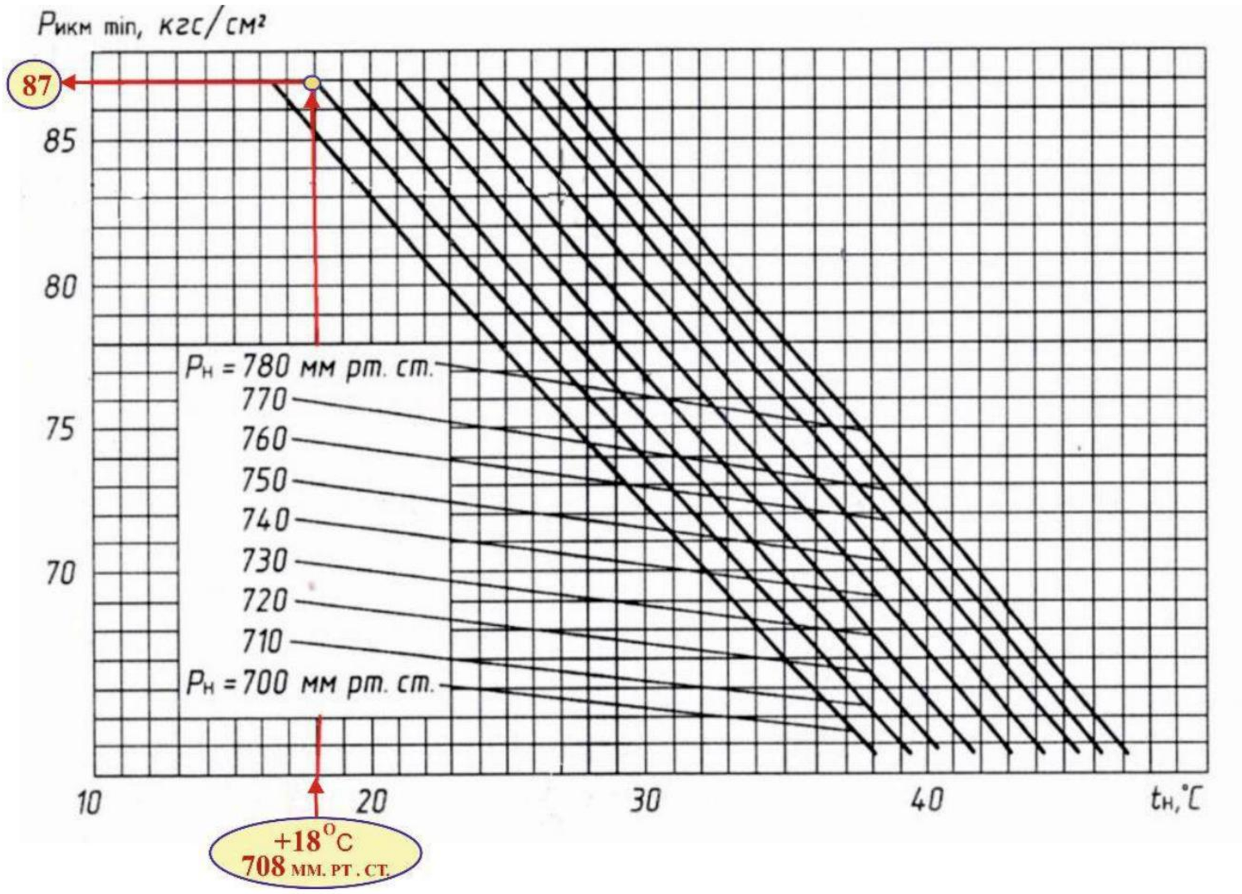


Fig. 51. Change in the minimum permissible Rim depending on the pressure (Pn) and temperature (tn) of the ambient air

The oil pressure value in the PCM of the left engine is below the norm indicate the activation of a temperature limiter, the operation of which could may be due to either external atmospheric conditions or imprecise tuning PRT systems or ADT-24 adjustments, or malfunctions of the PRT system or ADT-24.

Note: *p.7.1.2. RLE of the An-24(An-24RV):*

“When the engine is running in takeoff mode, the pressure RIC should be equal 88ÿ $^{2}_{1}$ *kgf/cm2 for AI-24 engines of the 2nd series and (92+2 - 1) kgf/cm2 for AI-24T, if there is no fuel “cut” by the PRT system (absent voltage on the voltmeter "IM-24 SHAFT POSITION").*

The PRT system comes into operation depending on the barometric air pressure and temperature according to the graph.

Zone I – operating zone of the AI-24 2nd series engine on the ICM limiter, the PRT limiter does not work;

Zone II – engine operating zone at the PRT limiter (partial drain);

Zone III is the engine operating zone on the barostat with complete drain."

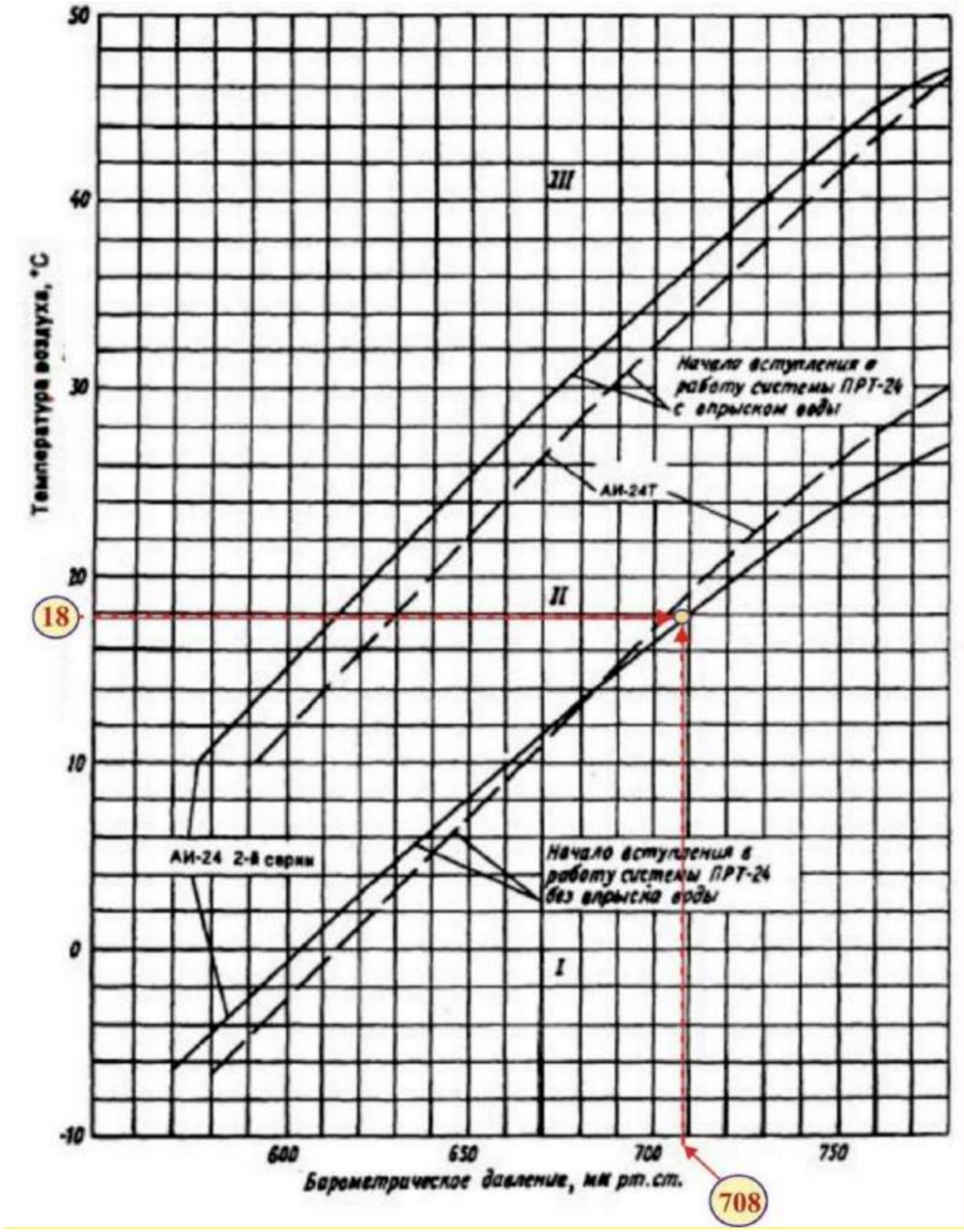


Fig. 52. Changing the moment of entry into operation of the PRT system with injection and without water injection depending on atmospheric conditions during takeoff mode (Fig. 7.4. RLE)

At the time of takeoff from the Ulan-Ude airfield, the atmospheric pressure was 708 mm Hg, outside air temperature +18°C. From Fig. 52

The graph shows that for actual takeoff conditions the engine operating mode was at the beginning of zone II (engine operating zone at the PRT limiter - partial drain, which characterized by a decrease in engine power), but in close proximity to the transition boundary from zone I (the engine operating zone at the ICM limiter, where (full engine power is provided). Due to the proximity of the above zones, to ensure the delivery of full engine power, precise adjustment must be made adjustment of its fuel regulating equipment (ADT-24 or (and) units of the system PRT).

Machine Translated by Google

It should be noted that the An-24RV aircraft are not equipped with a water injection system. AI-24 engines of the 2nd series, which are used at high temperatures or low atmospheric pressure when the takeoff power of the engines is reduced. Application water injection system restores takeoff power of engines and reduces gas temperature behind the turbine when operating at the PRT limiter (zone II on the graph in Fig. 52).

On An-24RV aircraft to create additional thrust during takeoff, climb altitude and at high outside air temperatures and low atmospheric pressure pressure, as well as to improve flight safety in the event of failure of one engine, in The right nacelle is equipped with a third turbojet engine RU19A-300. Therefore the decrease in power of the left engine, in the case under consideration, was compensated for additional thrust RU19A-300.

At a barometric altitude of approximately 3000 m (at 01:12:50 in Fig. 50) the crew increased the engine operating mode above the nominal: throttle levers of the left and right engines were 72° and 79°, respectively. The oil pressure in the left engine's ICM system was 76 kgf/cm2 and the right one - 77 kgf/cm2 .

According to the second pilot's explanatory note, the difference ("fork") between the throttle levers engines was necessary to equalize the oil pressure values in the ICM systems left and right engines.

Note: *According to the An-24 aircraft operating instructions, changing the engine operating mode occurs when the position of the control levers of the units changes ADT-24. The engine operating mode is set and controlled in in the crew cabin according to the fuel lever position indicator UPRT-2, DS-11 sensors which are installed on the ADT-24. The indicator scale calibrated in degrees of the angle of rotation of the ADT lever in the range from 0º to 115º with a division value of 2º.*

Zero degrees according to UPRT-2 corresponds to the earth's low throttle mode, and 100º - takeoff mode. Engine control sectors, mounted on the central control panel of the pilots, kinematically linked with the levers of the ADT-24 units using cable wiring, which require periodic adjustment of cable tension.

Specialists of the State Enterprise "IVCHENKO-PROGRESS" based on the results of the parametric analysis information came to the conclusion that *"signs of altitude gain are evident Malfunctions in the left engine. When retracting the throttle from the vertical position to ÿÿ = 68-70° along the UPRT The risk of RICM stabilizing (decreasing) takes longer, and with minor changes mode with ÿÿ = 68-70° to ÿÿ = 72° Rikhm sharply increases to 76 kgf/ cm2 . This value of Rikhm* *should correspond to a higher engine operating mode. With a climb*

The ricochet declines more sharply compared to the right one."

Throughout the entire flight at the flight level, the left engine throttle position (56º) corresponded to the engine operating mode slightly higher than 0.85 Nominal (52±2º), but significantly below the nominal mode (65±2º), and the right 47º corresponded to the mode 0.7 Nominal (47±2º).

Note: *1. p. 7.1.1. RLE of the An-24(An-24RV):*

Mode	*Situation RUD by UPRT, city*	*Time continuous work, min*	*Opening hours, % of resource*
Takeoff	*87-100*	*5*	*3*
Nominal	*65±2*	*60*	*25*
0,85 Nominal	*52±2*	*Not limited*	*No restrictions within resources*
0.7 Nominal	*41±2*	*Same*	
0.6 Nominal	*34±2*	*Same*	
0.4 Nominal	*22±2*	*Same*	
Earth's minor gas	*0*	*30*	

2. p. 6.4.1 RLE of An-24(RV):

“At the altitude of the given flight level, such a cruising level should be established a flight mode that will ensure the aircraft arrives at its destination assignments at a specified time, but not higher than 0.85 of the nominal mode (throttle position according to throttle control 52º)".

According to the altitude-speed characteristics of the AI-24 engine of the 2nd series, true flight speed of 120 m/s at an altitude of 5200 m and atmospheric conditions, corresponding to the MSA, in the 0.85 nominal mode (52° UPRT) the developed screw the power is approximately equal to 1500 hp (Fig. 53), which corresponds to the oil pressure in the system ICM 58 kgf/cm2 .

Note: *Oil pressure in the ICM system corresponding to the screw power 1500 hp engine, was determined from the ratio Nÿ = 25.8902×Pÿÿ, given in Section 14 of the Technical Operation Manual*

AI-24 2 series engine, where RCM is the oil pressure in the ICM system in kgf/ cm2 .

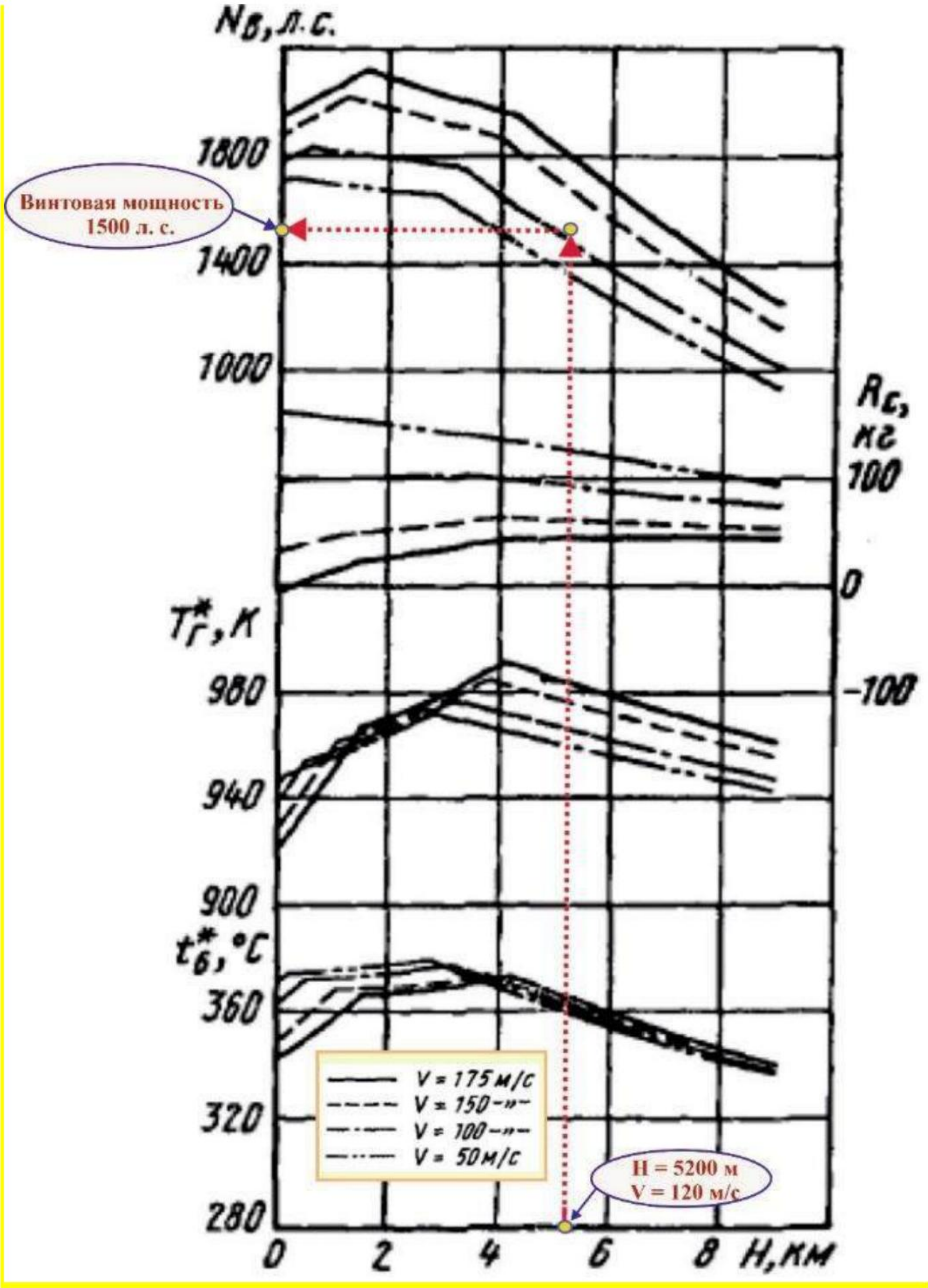


Fig. 53. Engine altitude and speed characteristics
(Operating mode – 0.85 nominal, tÿ = tÿÿÿ; ÿÿ = ÿÿÿÿ; nT = 15100 rpm).

The oil pressure values in the ICM system of the left and right engines recorded by the MSRP-12-96 on-board recorder were 53 and 55 kgf/cm2.

accordingly, they were below 0.85 of the Nominal Mode, which did not contradict provisions of the RLE.

Thus, the position of the left engine throttle corresponded to the mode, greater than 0.85 Nominal, but the propeller power of the engine was lower than this mode. Low oil pressure values in the left engine's PCM system indicate a possible malfunction of the fuel control equipment (units of the PRT system and (or) the ADT-24 unit), or about incorrect adjustment of the specified units. The MSRP-12-96 system failed to register the temperature of gases turbine, engine rotor speed and fuel consumption did not allow for full to evaluate the performance of the fuel control equipment. At the same time, Taking into account the above information, the commission believes that the most likely the reason for the deviations in the operation of the left engine was precisely the setting (regulation) fuel control equipment, which lowered the engine operating parameters in order to "fit" within the restrictions on the maximum temperature of gases behind the turbine.

At 02:09:07 the crew began to descend. At first, only the left engine's throttle was moved from position 57° to position 50° (the throttle of the right engine was in this position) in the 47° position), and then, during the next 6 seconds, the left and right throttle The engines were synchronously switched to the flight idle position. Throttle control led to a decrease in oil pressure in the ICM systems of the left and right engines to 13 kgf/cm2 and 22 kgf/cm2, respectively (Fig. 54).

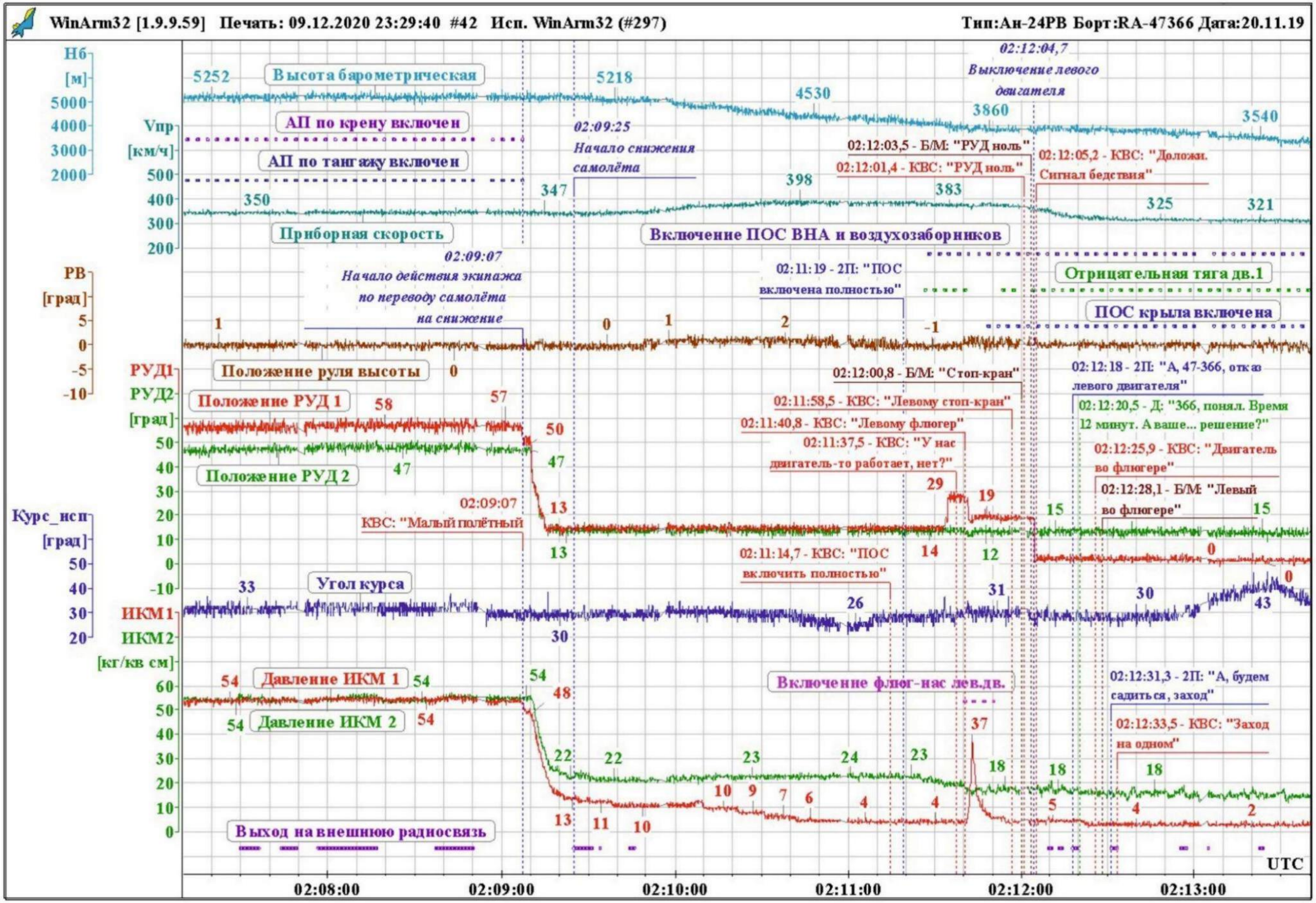


Fig. 54. Flight parameters of the An-24RV RA-47366 aircraft

An analysis of the records of previous flights performed on the aircraft was conducted An-24RV RA-47366 for the period from 01/29/2019 to 04/19/2019 showed that there is a “fork” in the values The RCM readings varied between 3 kgf/cm2 and 12 kgf/cm2 with the same throttle position in the PMG mode on almost every flight. , while the values The oil pressure in the right engine's PCM system was always higher than the pressure values oil in the left engine's PCM system.

A clear definition of what difference in the PCM readings is acceptable in the PMG mode There is no such thing in the ETD. There is only the concept: *“exceeding the pressure according to the ICM up to 15 kgf/cm2 on one of the engines compared to the normal pressure of the second engine's ICM."* What there should be “normal pressure of the engine ICM in the PMG mode” in the ETD also not defined. The main thing is that there are no signs of engine malfunction, for example - an increase or decrease in the engine rotor speed beyond the permissible limits limits, engine rotor speed variations more than ±1%, pressure drop oils in the ICM, etc.

In accordance with paragraph 5.1.9 of Section 5 “Special Flight Cases” of the An-24 aircraft flight manual "Landing with asymmetric engine thrust at flight idle power", the crew at when descending from the flight level to the altitude of the landing circle of the aerodrome, you must check the difference in pressure readings of the ICM engines when the control levers are set to the idle stop. If the pressure according to the ICM exceeds 15 kgf/cm2 on one of the engines, compared to the normal pressure of the second engine's ICM, landing should be carried out as in under normal conditions.

According to the provisions of Chapter VIII of the RTE of the AI-24 engine of the 2nd series, asymmetrical Engine thrust in the PMG mode could occur in cases where:

- hourly fuel consumption does not comply with the technical specifications in the PMG mode;
- The aircraft fine filter, jet, and propeller mesh are clogged 85 ADT-24 units;
- inaccurate adjustment or failure of one of the units of the PRT system or ADT - 24.

Clogging of fine filters is an unlikely event, since the last maintenance of the An-24RV RA-47366 aircraft, performed On May 20, 2019, the filters were washed. After technical This was the second day of flying for maintenance.

Note: *The An-24 RA-47366 aircraft did not fly from 20.04.2019 to 22.06.2019, because work was carried out to study the technical condition of the glider and its systems to continue to operate in accordance with* *requirements of the Bulletins of October 31, 2017 No. AN-24-GS001 and of 09.24.2015 No. 1520-BE-G. As part of the above works, the following was carried out: only external inspection of the power structures of the Armed Forces intended for power plant attachments, to identify signs of violation integrity of power structures.*

The series of flights under consideration on the route: Irkutsk – Nizhneangarsk - Ulan-Ude - Nizhneangarsk was the first after the end the specified works.

According to the materials presented, the measurement on the An-24RV RA-47366 aircraft hourly fuel consumption of the left engine according to the pulse and time counter (PTC) the ground in the PMG mode was performed on November 20, 2018 after replacing the left engine (form F1). After 20.11.2018, the control of hourly fuel consumption of the left engine according to the SIV was not was carried out, since in accordance with Note 4.02.42 of the Technical Regulations servicing of the An-24, An-26 aircraft, part 2, the specified measurement is necessary only in in case of replacement of the engine or the wind tunnel unit.

Note: *clause 4.02.42 of the Technical Maintenance Regulations. Form F2.*

"Measure the hourly fuel consumption of the engines using the pulse counter and time (SIV) in flight. Compare the results obtained with the standards costs and make adjustments if necessary. In case Adjustments: Measure the hourly fuel consumption on the ground and again in flight and compare them with the standards.

Note: Measurement of hourly fuel consumption on the ground and in flight produce in cases:

- *replacement of engine(s);*
- *ADT unit replacement;*
- *if the values of hourly fuel consumption during the flight do not fit into the norms;*
- *if the crew has any comments."*

01/22/2019, after replacing the ADT-24 unit of the right engine, during the aircraft maintenance according to periodic form F-28 (aircraft flight time 14,000 hours) in the volume of F2 (frequency of execution every 1,000 hours), was The hourly fuel consumption measurement according to the SIV on the ground was carried out, in flight it was indicated measurement was not performed.

Analysis of the fuel consumption measurement protocols for the left and right engines according to the SIV from 20.11.2018 and 22.01.2019 showed that the hourly fuel consumption indicated in them in the PMG mode they were slightly underestimated (the left engine had a deviation of minus 5 kg/hour, and the right one - minus 7 kg/hour), but complied with the technical specifications (no more than 10 kg/hour).

After 22.01.2019, hourly fuel consumption measurements for the left and right engines according to the SIV in the PMG mode, no operations were carried out, since there were no cases of replacing the engine (engines) or there were no wind turbine units, so it is unknown at the time of the accident, Did the hourly fuel consumption of the left (failed) engine in the PMG mode comply with the technical specifications?

During the study of operating documentation, in order to determine timeliness and completeness of the execution of scheduled maintenance work in accordance with the current ETD, it was established that in the List of Works (Appendix No. 1 of the Regulations on Technical servicing of An-24, An-26 aircraft, part 2) there is no information on the need performing a control flight (fly-by) for periodic measurement of hours fuel consumption of engines according to the SIV in case of replacement of one WT unit, and in paragraph 4.02.42. the specified Regulation does not define the type of flight (check flight or scheduled flight), at which measurements must be taken.

Lack of clear and consistent requirements for performing measurements hourly fuel consumption of engines according to the SIV in the specified Regulation may not allow the operator to promptly identify deviations in the operating parameters of AI-24 engines 2 series and in case of exhaustion of the permissible range of adjustments, remove from further operation of the failed unit or engine.

In an emergency flight, after the start of the descent from the flight level, with the aircraft remaining unchanged in the throttle position, the oil pressure in the left engine's ICM system began to decrease monotonously and after 40 seconds reached a value of 3 - 4 kgf/cm2 according to MSRP-12-96, which corresponds to the values for the engine turned off. In this case, the limit value (the boundary) of a possible fall of RCM is not specified in the RLE. The lack of registration by the system MSRP-12-96 engine rotor speed, engine oil pressure, temperature gas behind the turbine did not allow us to determine the change in these parameters. Oil pressure in The right engine's ICM system was approximately constant and amounted to 22...24 kgf/cm2 .

According to the conclusion of the State Enterprise “IVCHENKO-PROGRESS”: *“When transferring RUD engines with ÿÿ=48° on the PMG (13° on the UPRT) there was a drop in power of the left engine and its shutdown without feathering the propeller (the throttle was below the range "(switching readiness on). The engine has entered autorotation mode."*

Almost simultaneously with this, a one-time registration began on MSRP-12-96 the *"Negative left engine thrust"* command , which was generated during upon receipt of a signal from the SDU-5-2.5 with the simultaneous lighting of the indicator light engine failure in the KFL-37 button. Due to the peculiarity of registering one-time commands of the MSRP-12-96 It is not possible to determine the response time precisely, down to the second.

Approximately 6 seconds after the left power steering failure indicator comes on engine (Fig. 54) the crew moved its throttle lever from position 13° first to 19°, and then to 29°, however, the pressure in the ICM system did not change and remained at the same level (3-4 kgf/cm2 according to the MSRP data).

Moving the throttle to the mode corresponding to the value of 26±2°, if available the signal from the negative draft indicator led to the removal of the lock and automatic activation of the feathering system, which is confirmed by the beginning registration on the MSRP of a one-time command *“Turn on the left engine vane pump”* and recording the parameter RIM of the left engine, characteristic of automatic feathering the engine.

Note: *p. 5.1.2. RLE of the An-24 (An-24RV):*

“In case of engine failure at modes less than (26 ± 2) ° according to the UPRT on airplanes, equipped with a negative draft auto-weather system, translation The throttle of a failed engine will be set to a position greater than 28° according to the UPRT after 5-7 s (system slowdown time) to automatic feathering screw from the negative thrust sensor."

It should be noted that at engine operating modes below 24º according to the UPRT, the ignition is switched on The feathering system can only be operated manually using the feathering button CFL – 37.

As a result of the study of the left engine by FAU specialists The Aviation Register of Russia has established that *“signs indicating spontaneous Engine shutdown in flight, no. Damage to parts detected during engine research, were obtained as a result of an aviation accident and subsequent fire at a time when the engine was not rotating. Parts that do not have damage, are in satisfactory technical condition. Availability rotation of the drive shafts of the units and their location in their regular places "testifies to their performance."*

Significant damage to the fuel control equipment units on the left engine damage resulting from the aircraft's collision with a ground structure and subsequent ground fire did not allow their research to be carried out.

The Commission analyzed all cases available to the IAC. failures of AI-24 series 2 engines due to deviations in the operation of the fuel control system equipment, while none of the engine failures under consideration corresponded to full manifestations of engine failure in flight on the An-24RV RA-47366 aircraft.

Thus, based on the results of the analysis of all available commission of materials, it can be argued that the reason for the abnormal operation of the left the engine and its self-shutdown in flight were caused by deviations in the operation of the fuel automation. Specialists from the State Enterprise "ZMKB Ivchenko-Progress" believe that *"presumably, the engine shutdown occurred due to a drop in pressure to 0 and fuel consumption due to malfunction of control elements ADT-24".* The Commission shares this assumption and considers it the most probable, however, it is impossible to clearly determine the fact of failure of any specific unit fuel control equipment was not possible. Moreover, in any In this case, the fact of abnormal operation of the left engine (deviation of its parameters from the specifications) could be detected by both technical personnel and flight crews well before the day AP.

Machine Translated by Google

3. Conclusion [31]

The accident with the An-24RV RA-47366 aircraft occurred at performing a landing with one engine inoperative as a result of a longitudinal-lateral rolling off the runway and subsequent collision with a building outside airfield with the destruction of the aircraft structure and the outbreak of a fire. The landing was carried out on a runway with an available landing distance (1503 m), which was significantly less than required (2160 m) for actual conditions.

Most likely, the aviation accident was the result of a combination of the following factors [32]:

the decision by the PIC (pilot in command) to carry out a landing without calculation of the required landing distance;

incorrect choice by the PIC of the type and approach trajectory, which led to the impossibility of timely reduction of flight speed. Instead of visual maneuvering (circle-to-land maneuver) provided for by the landing approach pattern and agreed upon by the crew with the dispatcher, the crew performed a visual landing approach;

the absence of procedures and procedures for implementation in the airline's RPP and the aircraft's flight manual landing approach using visual maneuvering method (circle-to-land maneuver);

failure of the crew to take measures to go around in the event of a significant discrepancy between the actual flight parameters and the criteria for a stabilized approach, provided by the airline's RPP;

lack of interaction within the crew, cross-checking and failure to perform a number of standard operating procedures in terms of informing the PIC (pilot pilot) by other crew members about significant deviations of actual parameters flight from established values, as well as insufficient management of crew resources;

landing at a significantly increased speed (275 km/h instead of recommended 220 km/h), which resulted in the aircraft landing with an overshoot (530 m from the entrance runway threshold) and repeated separation from the runway;

incorrect use of the main landing gear wheel braking system by the crew chassis, which resulted in premature compression of the brake pedals (in the air), which the repeated contact resulted in landing on the braked wheels of the right main landing gear with destruction of tires and, subsequently, lateral rollout of the aircraft;

---

[31] According to Annex 13 of the Chicago Convention, "Aircraft Accident and Incident Investigation," the determination of the causes and contributing factors of an aircraft accident *"does not imply attribution of fault or the establishment of administrative, civil or criminal liability."*

[32] In accordance with the ICAO Manual of Aircraft Accident and Incident Investigation (Doc 9756 AN/965), contributing factors are listed without priority rating.

increased psycho-emotional stress of the captain against the background of his existing features of mental activity that contributed to the acceptance of unfounded solutions in the current situation.

The left engine shut down in flight due to abnormal operation. fuel control equipment, probably ADT-24. Due to the impact of the fire on elements of the fuel control equipment to clearly identify the failed unit and The cause of the failure is not possible. Abnormal adjustments of the left engine and deviations in the operation of its fuel control equipment became apparent long before the day aviation accident and could be identified by both flight and engineering technical staff.

4. Other deficiencies identified during the investigation

4.1. During the investigation, a discrepancy was found in the magnetic declination value the airfield used by the Nizhneangarsk AMSG, the actual value:

in the Instructions for meteorological support of flights at the aerodrome Nizhneangarsk, approved on March 31, 2017 by the head of the Nizhneangarsk airport, magnetic declination is specified as minus 5°;

The magnetic declination at the Nizhneangarsk airfield is indicated as minus 7.2°, ANI collection No. 122 – minus 7°.

In response to the commission's proposal to the head of Nizhneangarsk airport 07/29/2019 No. 05-11-344 on eliminating the discrepancy in magnetic declination, no response received.

4.2. The runway surface type is not indicated in the Nizhneangarsk airfield's ANP. The ANP also has errors and inaccuracies.

5. Recommendations for improving flight safety

To the aviation authorities of Russia[33]

5.1. The circumstances and causes of the crash of the An-24RV RA-47366 aircraft shall be brought to flight and engineering personnel of civil aviation.

5.2. In accordance with the provisions of Article 47 of the Air Code of the Russian Federation, establish the airfield area of the Nizhneangarsk airfield. Assess the situation with establishment of aerodrome areas at other aerodromes.

5.3. Enter aeronautical data for the Nizhneangarsk airfield into the AIP of the Russian Federation Federation.

5.4. Consider the feasibility of conducting unscheduled one-time inspections of districts airfields and aerodrome areas for the presence of objects on them, the placement of which does not comply with current regulations and/or the construction of which was carried out without the necessary approvals.

5.5. Consider the feasibility of publishing methodological materials that explain visual approach and visual maneuvering procedures (circle-to-land maneuver), paying particular attention to their differences and validity choice depending on the conditions at the landing airfield.

5.6. Establish an expert working group to conduct an analysis of the provisions Guidelines for psychological support of selection, training and professional development activities of flight and dispatch personnel of civil aviation and the practice of its application (taking into account the shortcomings identified in this and previous investigations) the subject of its compliance with modern knowledge in the field of psychology, including in part the adequacy of the methods used to identify the personal characteristics of pilots, concerning the methods of emotional response and behavior in emergency situations, as well as to analyze the programs and required level of training of VLEK psychologists.

5.7. Consider the feasibility of the position of airline psychologist as mandatory.

For An-24, 26 operators

5.8. Conduct an analysis of the data from flight recorders to identify cases the presence of a “fork” in the throttle positions and/or engine operating parameters. When the need to take appropriate corrective measures.

5.9. Conduct training sessions with flight and engineering personnel on the assessment procedure. compliance of the operation of AI-24 series 2 engines with the established technical specifications. When accepting

[33] Aviation administrations of other States Parties to the Agreement shall consider the applicability of these recommendations taking into account the actual state of affairs in the States.

engines from repair, conduct a comprehensive assessment of its quality, paying special attention attention to the mutual combination and magnitude of engine operating parameters when control checks and their compliance with technical specifications under actual atmospheric conditions.

5.10. JSC Angara Airlines

5.11. Conduct training with crews on the procedure for determining the adequacy of the runway length. for landing, performing approaches and landings with one engine failure engine and the use of aircraft braking systems. Pay special attention to crew resource management, crew interaction and standard operations operational procedures.

5.12. Taking into account the significant reserve (excess) of altitude after exiting the fourth turn when making an approach to land at Nizhneangarsk airfield in accordance with existing schemes, assess the level of risks when performing visual approaches to landing. Take corrective measures if necessary.

5.13. Include provisions on the procedure for performing visual inspection in the airline's operational manual maneuvering (circle-to-land maneuver) and conduct appropriate training for flight personnel crews.

5.14. Eliminate other deficiencies noted in this report.

OJSC "Airports of Local Air Lines of Buryatia"

5.15. Conduct verification of the positions of the Nizhneangarsk airfield ANP.

5.16. In the Instructions for meteorological support of flights at the aerodrome Nizhneangarsk magnetic declination must be brought into line with the airfield's ANP.

To the developers of the aircraft and engine, Federal State Unitary Enterprise GosNII GA

5.17. State Enterprise "ANTONOV", ZMKB "PROGRESS" together with the Federal State Unitary Enterprise "GosNII GA" to clarify clause 4.02.42 of the Regulations for the technical maintenance of An-24, An-26 aircraft in part determining the types of flights in which periodic measurements should be performed hourly fuel consumption of engines using the SIV, and also to resolve issues, associated with the need to perform the specified measurement in cases of replacement engine(s) or unit(s) of the ADT-24.

State Enterprise "ANTONOV"

5.18. Consider the feasibility of introducing the procedure for performing the following into the An-24, 26 flight manual: visual maneuvering (circle-to-land maneuver), as well as the specifics of execution visual approach with one engine running.